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Inventor(s)	Desjarlais; John R. et al.

Heterodimeric antibodies that bind prostate specific membrane antigen (PSMA) and CD3

Abstract

Provided herein are novel antigen binding domains and antibodies (e.g., heterodimeric antibodies) that bind Prostate Specific Membrane Antigen (PSMA). In exemplary embodiments, the anti-PSMA antibodies also bind CD3. Such antibodies that bind PSMA and CD3 are useful, for example in the treatment of PSMA-related cancer.

Inventors: Desjarlais; John R. (Pasadena, CA), Nisthal; Alex (Monrovia, CA), Hedvat; Michael (Encino, CA), Dragovich; Matthew Adam (Monrovia, CA)

Applicant: Xencor, Inc. (Pasadena, CA)

Family ID: 76306039

Assignee: Xencor, Inc. (Pasadena, CA)

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Primary Examiner: Lucas; Zachariah

Assistant Examiner: Alsomairy; Sarah A

Attorney, Agent or Firm: Morgan, Lewis & Bockius, LLP

Background/Summary

PRIORITY (1) This application is a continuation of U.S. patent application Ser. No. 17/321,325, filed May 14, 2021, which claims the benefit of U.S. Provisional Patent Application Nos. 63/025,082, filed May 14, 2020, and 63/042,315, filed Jun. 22, 2020, which are hereby incorporated by reference in their entireties.

SEQUENCE LISTING

(1) The instant application contains a Sequence Listing which has been submitted electronically in XML format and is hereby incorporated by reference in its entirety. Said XML copy, created on Jan. 17, 2024, is named 067461-5269-US01 SL.xml and is 1,199,689 bytes in size.

BACKGROUND

(2) Antibody-based therapeutics have been used successfully to treat a variety of diseases, including cancer. An increasingly prevalent avenue being explored is the engineering of single immunoglobulin molecules that co-engage two different antigens. Such alternate antibody formats that engage two different antigens are often referred to as bispecific antibodies. Because the considerable diversity of the antibody variable region (Fv) makes it possible to produce an Fv that recognizes virtually any molecule, the typical approach to bispecific antibody generation is the introduction of new variable regions into the antibody.

(3) A particularly useful approach for bispecific antibodies is to engineer a first binding domain which engages CD3 and a second binding domain which engages an antigen associated with or upregulated on cancer cells so that the bispecific antibody redirects CD3.sup.+ T cells to destroy the cancer cells. Prostate cancer (PC) is one of the most prevalent cancers in men, and end stage (castration-resistant prostate cancer) has no curative treatment option. Prostate Specific Membrane Antigen (PSMA), a type II transmembrane protein with a large extracellular domain, has long

generated interest as a therapeutic target. It is highly overexpressed in PC compared to normal tissue, and its expression has been shown to correlate with malignancy. Previous attempts to target PSMA include antibody-based radiotherapy and antibody drug conjugates, which have shown some success but can be hampered by the inherent toxicity of the modality. Thus, there is a need for additional anti-PSMA antibodies for the treatment of PSMA-related cancers include, for example, prostate cancer.

BRIEF SUMMARY

(4) Provided herein are novel bispecific antibodies to CD3 and PSMA that are capable of localizing CD3^{sup}+ effector T cells to PSMA expressing tumors such as in prostate cancer. The anti-PSMA antibodies provided herein include PSMA binding domain with binding affinities and valencies that allow for the advantageous selectivity for cells expressing high levels of PSMA while minimizing reactivity on low PSMA expressing cells. In some embodiments, such anti-PSMA antibodies include CD3 binding domains with binding affinity that further contribute to selective targeting of high-PSMA expressing cells lines. Such PSMA antibodies are useful, for example, for cancers that express high levels of PSMA including, for example, prostate cancer.

(5) In one aspect, provided herein is a composition that includes a Prostate Specific Membrane Antigen (PSMA) binding domain. The PSMA binding domain includes: a) a variable heavy domain that includes the variable heavy complementary determining regions 1-3 (vhCDR1-3) of PSMA-H variable heavy domain H1 (FIG. 17); and b) a variable light domain that includes the variable light complementary determining regions (vlCDR1-3) of a PSMA-H variable light domain selected from PSMA-H variable light domains L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E). In some embodiments, the vhCDR1-3 and vlCDR1-3 are selected from the vhCDR1-3 and vlCDR1-3 sequences of any one of the anti-PSMA binding domains provided in FIGS. 19A-19X.

(6) Also provided herein are nucleic acid compositions that include polynucleotide(s) encoding the subject PSMA binding domains, expression vectors that include such polynucleotides and host cells that include such expression vectors. Also provided herein are methods of making such PSMA binding domains.

(7) In another aspect, provided herein is composition that includes a Prostate Specific Membrane Antigen (PSMA) binding domain. The PSMA binding domain includes: a) a variable heavy domain, wherein the variable heavy domain is the PSMA-H variable heavy domain H1 (FIG. 17); and b) a variable light domain selected from PSMA-H variable light domains L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E).

(8) In another aspect, provided herein is an anti-PSMA antibody that includes a Prostate Specific Membrane Antigen (PSMA) binding domain. The PSMA binding domain includes: a) a variable heavy domain that includes the variable heavy complementary determining regions 1-3 (vhCDR1-3) of PSMA-H variable heavy domain H1 (FIG. 17); and b) a variable light domain that includes the variable light complementary determining regions (vlCDR1-3) of a PSMA-H variable light domain selected from PSMA-H variable light domains L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E). In some embodiments, the vhCDR1-3 and vlCDR1-3 are selected from the vhCDR1-3 and vlCDR1-3 of selected from any one of the anti-PSMA binding domains provided in FIGS. 19A-19X.

(9) In one aspect, provided herein is a Prostate Specific Membrane Antigen (PSMA) binding domain. The PSMA binding domain includes: a) a variable heavy domain, wherein the variable heavy domain is the PSMA-H variable heavy domain H1 (FIG. 17); and b) a variable light domain selected from PSMA-H variable light domains L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E). In some embodiments, the antibody includes: a) a first monomer that includes a first antigen binding domain and a first constant domain; and b) a second monomer that includes a second antigen binding domain and a second constant domain, wherein either of the first antigen binding domain or second antigen binding domain is the PSMA binding domain. In certain embodiments, the first antigen binding domain and the second antigen binding domain bind different antigens.

(10) In some embodiments, the first antigen binding domain is the PSMA binding domain and the second antigen binding domain is a CD3 binding domain. In particular embodiments, the CD3 binding domain includes the vhCDR1-3, and vlCDR1-3 of any of the following CD3 binding domains: H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. **10A-10F**). In certain embodiments, the vhCDR1-3 and vlCDR1-3 of the CD3 binding domain are selected from the vhCDR1-3 and vlCDR1-3 in FIGS. **10A-10F**. In some embodiments, the CD3 binding domain includes the variable heavy domain and variable light domain of any of the following CD3 binding domains: H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. **10A-10F**). In certain embodiments, the CD3 binding domain is an anti-CD3 scFv. In some embodiments, the scFv includes a charged scFv linker. In some embodiments, the first and second constant domains each include CH2-CH3. In certain embodiments, the first and second constant domains each include CH1-hinge-CH2-CH3.

(11) In particular embodiments, the first and second constant domains each are a variant constant domain. In some embodiments, the first and second monomers include a set of heterodimerization variants selected from the group consisting of those depicted in FIGS. **1A-1E**. In exemplary embodiments, the set of heterodimerization variants selected is from the group consisting of S364K/E357Q:L368D/K370S; S364K:368D/K370S; S364K:L368E/K370S; D401K: T411E/K360E/Q362E; and T366W:T366S/L368A/Y407V. In some embodiments, the first and second monomers each further include an ablation variant. In exemplary embodiments, the ablation variant is E233P/L234V/L235A/G236del/S267K.

(12) In certain embodiments, at least one of the first or second monomer further includes one or more pI variants. In exemplary embodiments, the one or more pI variants is N208D/Q295E/N384D/Q418E/N421D.

(13) In another aspect, provided is a heterodimeric antibody that includes: a) a first monomer, b) a second monomer; and c) a light chain. The first monomer includes: i) an anti-CD3 scFv that includes a first variable heavy domain, an scFv linker and a first variable light domain; and ii) a first Fc domain, wherein the scFv is covalently attached to the N-terminus of the first Fc domain using a domain linker. The second monomer includes a VH2-CH1-hinge-CH2-CH3 monomer, wherein VH is a second variable heavy domain and CH2-CH3 is a second Fc domain; and the light chain includes a second variable light domain. The second variable heavy domain and the second variable light domain form an PSMA binding domain.

(14) In some embodiments of the heterodimeric antibody, the second variable heavy domain includes the variable heavy complementary determining regions 1-3 (vhCDR1-3) of PSMA-H variable heavy domain H1 (FIG. **17**); and the second variable light domain includes the variable light complementary determining regions (vlCDR1-3) of a PSMA-H variable light domain selected from PSMA-H variable light domains L1 and L1.1-L1.84 (FIGS. **17** and **18A-18E**). In exemplary embodiments, the vhCDR1-3 of the second variable heavy domain and the vlCDR1-3 of the second variable light domain are selected from any one of the anti-PSMA binding domains provided in FIGS. **19A-19X**. In some embodiments, the second heavy variable domain is PSMA-H variable heavy domain H1 (FIG. **17**); and the second variable light domain is selected from PSMA-H variable light domains L1 and L1.1-L1.84 (FIGS. **17** and **18A-18E**).

(15) In certain embodiments, the anti-CD3 scFv includes the vhCDR1-3 and the vlCDR1-3 of any of the following CD3 binding domains: H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. **10A-10F**). In some embodiments, the vhCDR1-3 and vlCDR1-3 of the anti-CD3 scFv are selected from the vhCDR1-3 and vlCDR1-3 in FIGS. **10A-10F**. In exemplary embodiments, the first variable heavy domain and the first variable

light domain are the variable heavy domain and variable light domain, respectively, of any of the following CD3 binding domains: H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. 10A-10F).

(16) In one embodiment, the first variable light domain is covalently attached to the N-terminus of the first Fc domain using a domain linker. In some embodiments, the first variable heavy domain is covalently attached to the N-terminus of the first Fc domain using a domain linker.

(17) In some embodiments of the heterodimeric antibody, the scFv linker is a charged scFv linker. In exemplary embodiments, the scFv linker is a charged scFv linker having the amino acid sequence (GKPGS).sub.4 (SEQ ID NO: 1).

(18) In certain embodiments, the first and second Fc domains are variant Fc domains. In some embodiments, the first and second monomers include a set of heterodimerization variants selected from the group consisting of those depicted in FIGS. 1A-1E. In exemplary embodiments, the set of heterodimerization variants selected is from the group consisting of S364K/E357Q:L368D/K370S; S364K:L368D/K370S; S364K:L368E/K370S; D401K: T411E/K360E/Q362E; and T366W:T366S/L368A/Y407V, wherein numbering is according to EU numbering. In some embodiments, the first and second monomers further includes an ablation variant. In exemplary embodiments, the ablation variant is E233P/L234V/L235A/G236del/S267K, wherein numbering is according to EU numbering.

(19) In some embodiments, one of the first or second monomer includes one or more pI variants. In particular embodiments, the one or more pI variants are N208D/Q295E/N384D/Q418E/N421D, wherein numbering is according to EU numbering.

(20) In exemplary embodiments of the heterodimeric antibody, the first monomer includes amino acid variants S364K/E357Q/E233P/L234V/L235A/G236del/S267K, the second monomer includes amino acid variants L368D/K370S/N208D/Q295E/N384D/Q418E/N421D/E233P/L234V/L235A/G236del/S267K, and the numbering is according to EU numbering.

(21) In certain embodiments of the heterodimeric antibody, the first and second monomers each further include amino acid variants 428/434S.

(22) In exemplary embodiments, the heterodimeric antibody is one of the following heterodimeric antibodies: XENP14484, XENP33755, XENP33756, XENP33757, XENP33758, XENP33759, XENP33760, XENP33761, XENP33762, XENP34234, XENP34235, XENP34236, XENP16873, XENP16874, and XENP19722.

(23) In another aspect, provided herein is a heterodimeric antibody that includes: a) a first monomer; b) a second monomer; and c) a light chain. The first monomer includes, from N-terminus to C-terminus, a scFv-linker-CH2-CH3, wherein scFv is an anti-CD3 scFv and CH2-CH3 is a first Fc domain. The second monomer includes, from N-terminus to C-terminus, a VH-CH1-hinge-CH2-CH3, wherein CH2-CH3 is a second Fc domain. The light chain includes a VL-CL. The first variant Fc domain includes amino acid variants S364K/E357Q, the second variant Fc domain includes amino acid variants L368D/K370S, the first and second variant Fc domains each include amino acid variants E233P/L234V/L235A/G236del/S267K, and the hinge-CH2-CH3 of the second monomer includes amino acid variants N208D/Q295E/N384D/Q418E/N421D. The VH and VL form an PSMA binding domain that includes the variable heavy domain and the variable light domain, respectively, of an PSMA binding domain selected from PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; and PSMA-H H1_L1.13. Further, the anti-CD3 scFv includes the variable heavy domain and the variable light domain of a CD3 binding domain selected from H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and

L1.47_H1.31. In such heterodimeric antibodies, the numbering of the amino acid variants is according to EU numbering.

(24) In some embodiments of this heterodimeric antibody, the scFv includes a charged scFv linker having the amino acid sequence (GKPGS).sub.4 (SEQ ID NO: 1). In certain embodiments, the first and second variant Fc domains each further include amino acid variants 428/434S, wherein numbering is according to EU numbering.

(25) In another aspect, provided herein is a heterodimeric antibody that includes: a) a first monomer; b) a second monomer; and c) a common light chain. The first monomer includes, from N-terminus to C-terminus, a VH1-CH1-linker 1-scFv-linker 2-CH2-CH3, wherein VH1 is a first variable heavy domain, scFv is an anti-CD3 scFv, linker 1 and linker 2 are a first domain linker and second domain linker, respectively, and CH2-CH3 is a first Fc domain. The second monomer includes, from N-terminus to C-terminus, a VH2-CH1-hinge-CH2-CH3, wherein VH2 is a second variable heavy domain and CH2-CH3 is a second Fc domain. The common light chain includes a variable light domain. The first variable heavy domain and the variable light domain form a first PSMA binding domain, and the second variable heavy domain and the variable light domain form a second PSMA binding domain. In some embodiments, the first and second PSMA binding domains each include the variable heavy complementary determining regions 1-3 (vhCDR1-3) of PSMA-H variable heavy domain H1 (FIG. 17); and the variable light complementary determining regions (vlCDR1-3) of a PSMA-H variable light domain selected from PSMA-H variable light domains L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E). In exemplary embodiments, the vhCDR1-3 and vlCDR1-3 of the first and second PSMA binding domains are selected from the vhCDR1-3 and vlCDR1-3 provided in FIGS. 17 and 18A-18E. In some embodiments, the first and second variable heavy domain each is a PSMA-H variable heavy domain H1 (FIG. 17), and the variable light domain of the common light chain is selected from PSMA-H variable light domains L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E).

(26) In certain embodiments, the scFv includes the vhCDR1-3 and the vlCDR1-3 of any of the following CD3 binding domains: H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. 10A-10F). In exemplary embodiments, the vhCDR1-3 and vlCDR1-3 of the scFv are selected from the vhCDR1-3 and vlCDR1-3 in FIGS. 10A-10F. In some embodiments, the scFv includes the variable heavy domain and variable light domain of any of the following CD3 binding domains: H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. 10A-10F).

(27) In certain embodiments, the scFv includes an scFv variable heavy domain, an scFv variable light domain and an scFv linker that connects the scFv variable heavy domain and the scFv variable light domain. In some embodiments, the scFv variable heavy domain is attached to the C-terminus of the CH1 of the first monomer using the first domain linker and the scFv variable light domain is covalently attached to the N-terminus of the first Fc domain using the second domain linker. In other embodiments, the scFv variable light domain is attached to the C-terminus of the CH1 of the first monomer using the first domain linker and the scFv variable heavy domain is covalently attached to the N-terminus of the first Fc domain using the second domain linker. In certain embodiments of this heterodimeric antibody, the scFv linker is a charged scFv linker. In exemplary embodiments, the scFv linker is a charged scFv linker having the amino acid sequence (GKPGS).sub.4 (SEQ ID NO: 1).

(28) In certain embodiments of this heterodimeric antibody, the first and second Fc domains are variant Fc domains. In some embodiments, the first and second monomers include a set of heterodimerization variants selected from the group consisting of those depicted in FIGS. 1A-1E. In exemplary embodiments, the set of heterodimerization variants selected is from the group consisting of S364K/E357Q:L368D/K370S; S364K:L368D/K370S; S364K:L368E/K370S;

D401K; T411E/K360E/Q362E; and T366W:T366S/L368A/Y407V, wherein numbering is according to EU numbering. In some embodiments, the first and second monomers further include an ablation variant. In exemplary embodiments, the ablation variant is E233P/L234V/L235A/G236del/S267K, wherein numbering is according to EU numbering.

(29) In some embodiments, one of the first or second monomer includes one or more pI variants. In particular embodiments, the one or more pI variants are N208D/Q295E/N384D/Q418E/N421D, wherein numbering is according to EU numbering.

(30) In exemplary embodiments of the heterodimeric antibody, the first monomer includes amino acid variants S364K/E357Q/E233P/L234V/L235A/G236del/S267K, the second monomer includes amino acid variants L368D/K370S/N208D/Q295E/N384D/Q418E/N421D/E233P/L234V/L235A/G236del/S267K, and the numbering is according to EU numbering.

(31) In certain embodiments of the heterodimeric antibody, the first and second monomers each further include amino acid variants 428/434S.

(32) In exemplary embodiments, this heterodimeric antibody is one of the following: XENP31602, XENP31603, XENP31855, XENP32218, XENP32219, XENP32220, XENP32221, XENP32222, XENP32223, XENP32224, XENP32225, XENP32226, XENP34237, XENP34238, XENP34239, XENP34625, XENP34626, XENP34627, XENP34628, XENP31853, XENP31856, XENP33063, XENP33064, XENP33065, XENP33066, XENP33067, XENP33068, XENP33069, XENP33070, XENP33071, XENP34240, XENP34241, XENP34242, XENP34629, XENP34630, XENP34631, XENP34632, XENP31854, and XENP31857.

(33) In yet another aspect, provided herein is a heterodimeric antibody that includes: a) a first monomer; b) a second monomer; and c) a common light chain. The first monomer includes, from N-terminus to C-terminus, a VH-CH1-linker 1-scFv-linker 2-CH2-CH3, wherein scFv is an anti-CD3 scFv and CH2-CH3 is a first Fc domain. The b) a second monomer includes, from N-terminus to C-terminus a VH-CH1-hinge-CH2-CH3, wherein CH2-CH3 is a second Fc domain. The common light chain includes VL-CL. The first variant Fc domain includes amino acid variants S364K/E357Q, the second variant Fc domain includes amino acid variants L368D/K370S. The first and second variant Fc domains each include amino acid variants E233P/L234V/L235A/G236del/S267K, the hinge-CH2-CH3 of the second monomer includes amino acid variants N208D/Q295E/N384D/Q418E/N421D. The VH of this heterodimeric antibody is PSMA-H variable heavy domain H1 (FIG. 17), and the VL is a variable light domain selected from PSMA-H variable light domains L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E). Further, the anti-CD3 scFv includes the variable heavy domain and the variable light domain of a CD3 binding domain selected from H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. 10A-10F). In such heterodimeric antibodies, the numbering of the amino acid variants is according to EU numbering.

(34) In some embodiments, the scFv includes a charged scFv linker having the amino acid sequence (GKPGS).sub.4 (SEQ ID NO: 1). In certain embodiments, the first and second variant Fc domains each further include amino acid variants 428/434S.

(35) In another aspect, provided herein are heterodimeric anti-PSMA×anti-CD3 antibodies XENP14484, XENP33755, XENP33756, XENP33757, XENP33758, XENP33759, XENP33760, XENP33761, XENP33762, XENP34234, XENP34235, XENP34236, XENP16873, XENP16874, and XENP19722.

(36) In yet another aspect, provided herein are heterodimeric anti-PSMA×anti-CD3 antibodies XENP31602, XENP31603, XENP31855, XENP32218, XENP32219, XENP32220, XENP32221, XENP32222, XENP32223, XENP32224, XENP32225, XENP32226, XENP34237, XENP34238, XENP34239, XENP34625, XENP34626, XENP34627, XENP34628, XENP31853, XENP31856, XENP33063, XENP33064, XENP33065, XENP33066, XENP33067, XENP33068, XENP33069,

XENP33070, XENP33071, XENP34240, XENP34241, XENP34242, XENP34629, XENP34630, XENP34631, XENP34632, XENP31854, and XENP31857.

(37) Also provided herein are nucleic acid compositions that include polynucleotide(s) encoding the subject anti-PSMA antibodies, expression vectors that include such polynucleotides and host cells that include such expression vectors. Further provided herein are methods of making such anti-PSMA antibodies, wherein a subject host cell is cultured under conditions wherein the anti-PSMA antibody is expressed, and recovering the anti-PSMA antibody.

(38) In another aspect, provided herein is a method of treating a cancer that includes administering to a patient in need thereof any one of the anti-PSMA antibody described herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A-1E depict useful pairs of Fc heterodimerization variant sets (including skew and pI variants). There are variants for which there are no corresponding “monomer 2” variants; these are pI variants which can be used alone on either monomer.

(2) FIG. 2 depicts a list of isosteric variant antibody constant regions and their respective substitutions. pI₋(-) indicates lower pI variants, while pI₊(+) indicates higher pI variants. These can be optionally and independently combined with other heterodimerization variants of the antibodies described herein (and other variant types as well, as outlined herein).

(3) FIG. 3 depicts useful ablation variants that ablate FcγR binding (sometimes referred to as “knock outs” or “KO” variants). Generally, ablation variants are found on both monomers, although in some cases they may be on only one monomer.

(4) FIG. 4 depicts particularly useful embodiments of “non-Fv” components of the antibodies described herein.

(5) FIG. 5 depicts a number of charged scFv linkers that find use in increasing or decreasing the pI of the subject heterodimeric bsAbs that utilize one or more scFv as a component, as described herein. The (+H) positive linker finds particular use herein, particularly with anti-CD3 V.sub.L and V.sub.H sequences shown herein. A single prior art scFv linker with a single charge is referenced as “Whitlow”, from Whitlow et al., Protein Engineering 6(8):989-995 (1993). It should be noted that this linker was used for reducing aggregation and enhancing proteolytic stability in scFvs. Such charged scFv linkers can be used in any of the subject antibody formats disclosed herein that include scFvs (e.g., 1+1 Fab-scFv-Fc and 2+1 Fab.sub.2-scFv-Fc formats).

(6) FIG. 6 depicts a number of exemplary domain linkers. In some embodiments, these linkers find use linking a single-chain Fv to an Fc chain. In some embodiments, these linkers may be combined. For example, a GGGGS linker (SEQ ID NO: 2) may be combined with a “half hinge” linker.

(7) FIGS. 7A-7D depict the sequences of several useful 1+1 Fab-scFv-Fc bispecific antibody format heavy chain backbones based on human IgG1, without the Fv sequences (e.g. the scFv and the VH for the Fab side). Backbone 1 is based on human IgG1 (356E/358M allotype), and includes the S364K/E357Q:L368D/K370S skew variants, C220S on the chain with the S364K/E357Q skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 2 is based on human IgG1 (356E/358M allotype), and includes S364K:L368D/K370S skew variants, C220S on the chain with the S364K skew variant, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants, and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 3 is based on human IgG1 (356E/358M allotype), and includes S364K:L368E/K370S skew variants, C220S on the chain with the S364K skew variant, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368E/K370S skew variants and the

E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 4 is based on human IgG1 (356E/358M allotype), and includes D401K:K360E/Q362E/T411E skew variants, C220S on the chain with the D401K skew variant, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with K360E/Q362E/T411E skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 5 is based on human IgG1 (356D/358L allotype), and includes S364K/E357Q:L368D/K370S skew variants, C220S on the chain with the S364K/E357Q skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 6 is based on human IgG1 (356E/358M allotype), and includes S364K/E357Q:L368D/K370S skew variants, C220S on the chain with the S364K/E357Q skew variants, N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains, as well as an N297A variant on both chains. Backbone 7 is identical to 6 except the mutation is N297S. Backbone 8 is based on human IgG4, and includes the S364K/E357Q:L368D/K370S skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants, as well as a S228P (EU numbering, this is S241P in Kabat) variant on both chains that ablates Fab arm exchange as is known in the art. Backbone 9 is based on human IgG2, and includes the S364K/E357Q:L368D/K370S skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants. Backbone 10 is based on human IgG2, and includes the S364K/E357Q:L368D/K370S skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants as well as a S267K variant on both chains. Backbone 11 is identical to backbone 1, except it includes M428L/N434S Xtend mutations. Backbone 12 is based on human IgG1 (356E/358M allotype), and includes S364K/E357Q:L368D/K370S skew variants, C220S and the P217R/P229R/N276K pI variants on the chain with S364K/E357Q skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Included within each of these backbones are sequences that are 90, 95, 98 and 99% identical (as defined herein) to the recited sequences, and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 additional amino acid substitutions (as compared to the “parent” of the Figure, which, as will be appreciated by those in the art, already contain a number of amino acid modifications as compared to the parental human IgG1 (or IgG2 or IgG4, depending on the backbone). That is, the recited backbones may contain additional amino acid modifications (generally amino acid substitutions) in addition to the skew, pI and ablation variants contained within the backbones of this figure.

(8) FIGS. 8A-8C depict the sequences of several useful 2+1 Fab.sub.2-scFv-Fc bispecific antibody format heavy chain backbones based on human IgG1, without the Fv sequences (e.g. the scFv and the VH for the Fab side). Backbone 1 is based on human IgG1 (356E/358M allotype), and includes the S364K/E357Q:L368D/K370S skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 2 is based on human IgG1 (356E/358M allotype), and includes S364K:L368D/K370S skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants, and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 3 is based on human IgG1 (356E/358M allotype), and includes S364K:L368E/K370S skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368E/K370S skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 4 is based on human IgG1 (356E/358M allotype), and includes D401K:K360E/Q362E/T411E skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with K360E/Q362E/T411E skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 5 is based on human IgG1 (356D/358L allotype), and includes

S364K/E357Q:L368D/K370S skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 6 is based on human IgG1 (356E/358M allotype), and includes S364K/E357Q:L368D/K370S skew variants, N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains, as well as an N297A variant on both chains. Backbone 7 is identical to 6 except the mutation is N297S. Backbone 8 is identical to backbone 1, except it includes M428L/N434S Xtend mutations. Backbone 9 is based on human IgG1 (356E/358M allotype), and includes S364K/E357Q:L368D/K370S skew variants, the P217R/P229R/N276K pI variants on the chain with S364K/E357Q skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Included within each of these backbones are sequences that are 90, 95, 98 and 99% identical (as defined herein) to the recited sequences, and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 additional amino acid substitutions (as compared to the “parent” of the Figure, which, as will be appreciated by those in the art, already contain a number of amino acid modifications as compared to the parental human IgG1 (or IgG2 or IgG4, depending on the backbone). That is, the recited backbones may contain additional amino acid modifications (generally amino acid substitutions) in addition to the skew, pI and ablation variants contained within the backbones of this figure.

(9) FIG. 9 depicts the sequences of several useful constant light domain backbones based on human IgG1, without the Fv sequences (e.g. the scFv or the Fab). Included herein are constant light backbone sequences that are 90, 95, 98 and 99% identical (as defined herein) to the recited sequences, and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 additional amino acid modifications.

(10) FIGS. 10A-10F depict sequences for exemplary anti-CD3 scFvs suitable for use in the bispecific antibodies described herein. The CDRs are underlined, the scFv linker is double underlined (in the sequences, the scFv linker is a positively charged scFv (GKPGS).sub.4 linker (SEQ ID NO: 1), although as will be appreciated by those in the art, this linker can be replaced by other linkers, including uncharged or negatively charged linkers, some of which are depicted in FIG. 5), and the slashes indicate the border(s) of the variable domains. In addition, the naming convention illustrates the orientation of the scFv from N- to C-terminus. As noted herein and is true for every sequence herein containing CDRs, the exact identification of the CDR locations may be slightly different depending on the numbering used as is shown in Table 2, and thus included herein are not only the CDRs that are underlined but also CDRs included within the VH and V.sub.L domains using other numbering systems. Furthermore, as for all the sequences in the Figures, these VH and V.sub.L sequences can be used either in a scFv format or in a Fab format.

(11) FIGS. 11A and 11B depict the antigen sequences for PSMA, including human, mouse and cyno, to facilitate the development of antigen binding domains that bind to both for ease of clinical development.

(12) FIG. 12 depicts illustrative IHC of biopsy cores of prostate cancer and adjacent normal tissue showing PSMA expression.

(13) FIG. 13 depicts breakdown of IHC scores of 192 biopsy cores showing PSMA expression.

(14) FIG. 14 depicts antigen density (determined using QuickCal protocol) on cancer cell lines LnCAP, 22Rv1, Huh-7, A549, ASPC-1, HT29, and SKOV3 as well as PSMA-transfected PC3 cells.

(15) FIG. 15 depicts illustrative IHC of cancer cell lines and PSMA-transfected PC3 cells showing PSMA expression.

(16) FIG. 16 depicts PSMA expression on cancer cell lines and PSMA-transfected PC3 cell lines as determined by IHC and flow cytometry.

(17) FIG. 17 depicts the variable heavy and variable light chain sequences for an exemplary humanized PSMA binding domain referred to herein as PSMA-H, as well as the sequences for XENP31858 and XENP31604, anti-PSMA mAbs based on PSMA-H and IgG1 backbone. CDRs

are underlined and slashes indicate the border(s) between the variable regions and constant domain. As noted herein and is true for every sequence herein containing CDRs, the exact identification of the CDR locations may be slightly different depending on the numbering used as is shown in Table 2, and thus included herein are not only the CDRs that are underlined but also CDRs included within the V.sub.H and V.sub.L domains using other numbering systems. Furthermore, as for all the sequences in the Figures, these V.sub.H and V.sub.L sequences can be used either in a scFv format or in a Fab format.

(18) FIGS. **18A-18E** depict the variable light chain sequences for PSMA-H variants engineered with the aim to tune binding affinity for human PSMA. CDRs are underlined and slashes indicate the border(s) between the variable regions and constant domain. As noted herein and is true for every sequence herein containing CDRs, the exact identification of the CDR locations may be slightly different depending on the numbering used as is shown in Table 2, and thus included herein are not only the CDRs that are underlined but also CDRs included within V.sub.L domains using other numbering systems. Further, as for all the sequences in the Figures, these V.sub.L sequences can be used either in a scFv format or in a Fab format. Each of the variable light domains depicted herein can be paired with any other α PSMA variable heavy domain.

(19) FIGS. **19A-19Y** depict the amino acid sequences for PSMA-H variants engineered with the aim to tune binding affinity for human PSMA formatted as bivalent anti-PSMA mAbs. CDRs are underlined and slashes indicate the border(s) between the variable regions and constant domain. As noted herein and is true for every sequence herein containing CDRs, the exact identification of the CDR locations may be slightly different depending on the numbering used as is shown in Table 2, and thus included herein are not only the CDRs that are underlined but also CDRs included within the V.sub.H and V.sub.L domains using other numbering systems.

(20) FIG. **20A-20B** depicts BLI-response, apparent dissociation constant ($K_{sub.Dapp}$), association rate ($k_{sub.a}$), and dissociation rate ($k_{sub.d}$) of affinity-engineered PSMA-H variants (in bivalent IgG1 format) as determined by Octet. Substitutions in variable light regions are based on Kabat numbering.

(21) FIG. **21A-21B** depicts a couple of formats of the present invention. FIG. **21A** depicts the “1+1 Fab-scFv-Fc” format, with a first Fab arm binding PSMA and a second scFv arm binding CD3. FIG. **21B** depicts the “2+1 Fab.sub.2-scFv-Fc” format, with a first Fab arm binding PSMA and a second Fab-scFv arm, wherein the Fab binds PSMA and the scFv binds CD3.

(22) FIGS. **22A-22F** depict the sequences for illustrative α PSMA $\times\alpha$ CD3 bsAbs in the 1+1 Fab-scFv-Fc format and comprising a H1.30_L1.47 anti-CD3 scFv (a.k.a. CD3 High[VHVL]). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α PSMA $\times\alpha$ CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

(23) FIG. **23** depicts the sequences for illustrative α PSMA $\times\alpha$ CD3 bsAbs in the 1+1 Fab-scFv-Fc format and comprising a H1.33_L1.47 anti-CD3 scFv (a.k.a. CD3 Intermediate[VHVL]). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α PSMA $\times\alpha$ CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

(24) FIG. **24** depicts the sequences for illustrative α PSMA $\times\alpha$ CD3 bsAbs in the 1+1 Fab-scFv-Fc format and comprising a H1.31_L1.47 anti-CD3 scFv (a.k.a. CD3 High-Int[VHVL]). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain

components (e.g. constant region and domain linkers). It should be noted that the α PSMA $\times\alpha$ CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

(25) FIG. 25 depicts the sequences for illustrative α PSMA $\times\alpha$ CD3 bsAbs in the 1+1 Fab-scFv-Fc format and comprising a H1.32_L1.47 anti-CD3 scFv (a.k.a. CD3 High-Int #1[VHVL]). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α PSMA $\times\alpha$ CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

(26) FIG. 26 depicts the sequences for illustrative α PSMA $\times\alpha$ CD3 bsAbs in the 2+1 Fab.sub.2-scFv-Fc format and comprising a H1.30_L1.47 anti-CD3 scFv (a.k.a. CD3 High [VHVL]). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α PSMA $\times\alpha$ CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

(27) FIG. 27 depicts the sequences for illustrative α PSMA $\times\alpha$ CD3 bsAbs in the 2+1 Fab.sub.2-scFv-Fc format and comprising a H1.32_L1.47 anti-CD3 scFv (a.k.a. CD3 High-Int #1[VHVL]). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α PSMA $\times\alpha$ CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

(28) FIGS. 28A-28K depict the sequences for illustrative α PSMA $\times\alpha$ CD3 bsAbs in the 2+1 Fab.sub.2-scFv-Fc format and comprising a L1.47_H1.32 anti-CD3 scFv (a.k.a. CD3 High-Int #1[VLVH]). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α PSMA $\times\alpha$ CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

(29) FIG. 29 depicts the sequences for illustrative α PSMA $\times\alpha$ CD3 bsAbs in the 2+1 Fab.sub.2-scFv-Fc format and comprising a H1.89_L1.47 anti-CD3 scFv (a.k.a. CD3 High-Int #2[VHVL]). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α PSMA $\times\alpha$ CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

(30) FIGS. 30A-30K depicts the sequences for illustrative α PSMA $\times\alpha$ CD3 bsAbs in the 2+1 Fab.sub.2-scFv-Fc format and comprising a L1.47_H1.89 anti-CD3 scFv (a.k.a. CD3 High-Int

#2[VLVH]). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α PSMA $\times\alpha$ CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

(31) FIG. 31 depicts the sequences for illustrative α PSMA $\times\alpha$ CD3 bsAbs in the 2+1 Fab-scFv-Fc format and comprising a H1.33_L1.47 anti-CD3 scFv (a.k.a. CD3 Intermediate[VHVL]). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α PSMA $\times\alpha$ CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

(32) FIG. 32 depicts the sequences for illustrative α PSMA $\times\alpha$ CD3 bsAbs in the 2+1 Fab-scFv-Fc format and comprising a L1.47_H1.33 anti-CD3 scFv (a.k.a. CD3 Intermediate[VLVH]). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α PSMA $\times\alpha$ CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

(33) FIG. 33 depicts the sequences for illustrative prototypic α PSMA $\times\alpha$ CD3 bsAbs in alternative formats.

(34) FIG. 34 depicts BLI-response, dissociation constant ($K_{sub.D}$), association rate ($k_{sub.a}$), and dissociation rate ($k_{sub.d}$) of affinity-engineered PSMA-H variants (in the context of PSMA $\times$ CD3 bispecifics in the 1+1 Fab-scFv-Fc format with a H1.30_L1.47 anti-CD3 scFv) for human PSMA as determined by Octet. Substitutions in variable light regions are based on Kabat numbering.

(35) FIG. 35 depicts BLI-response, dissociation constant ($K_{sub.D}$), association rate ($k_{sub.a}$), and dissociation rate ($k_{sub.d}$) of affinity-engineered PSMA-H variants (in the context of PSMA $\times$ CD3 bispecifics in the 1+1 Fab-scFv-Fc format with a H1.30_L1.47 anti-CD3 scFv) for cynomolgus PSMA as determined by Octet. Substitutions in variable light regions are based on Kabat numbering. A couple of the bsAbs are listed with NA due to odd sensorgrams.

(36) FIG. 36 depicts induction of RTCC on luciferase-transduced PC3 cells with varying surface PSMA densities by A) XENP34282 and B) XENP14484. The data show that the two prototypic 1+1 anti-PSMA $\times$ anti-CD3 induced RTCC of cell lines expressing high and low PSMA levels, including PC3 (~3K) which represent normal tissues, with similar potency.

(37) FIG. 37 depicts induction of T cell proliferation (as indicated by percentage T cells expressing Ki67) by A) XENP34282 and B) XENP14484 in the presence of PC3 cells with varying surface PSMA densities. The data show that the two prototypic 1+1 anti-PSMA $\times$ anti-CD3 induced T cell proliferation in the presence of cell lines expressing high and low PSMA levels, including PC3 (~3K) which represent normal tissues, with similar potency.

(38) FIG. 38 depicts the binding to PSMA-transfected PC3 (~32K) cells (representative of low PSMA-expressing on-target cells) by A) 1+1 Fab-scFv-Fc bispecific antibodies XENP14484, XENP33756, XENP33757, XENP33761, and XENP337652 and by B) 2+1 Fab.sub.2-scFv-Fc bispecific antibodies XENP31620, XENP32218, XENP32220, XENP32224, and XENP32225. The data show that as monovalent PSMA binding affinity is decreased in the 1+1 Fab-scFv-Fc bispecific antibodies, their binding to PC3 (~32K) cells is drastically reduced. Notably, as monovalent PSMA binding affinity is decreased in the 2+1 Fab.sub.2-scFv-Fc bispecific

antibodies, their binding to PC3 (~32K) cells is retained.

(39) FIG. **39** depicts induction of RTCC on transduced PC3 luciferase-transfected cancer cells with varying surface PSMA densities by A) XENP32218, B) XENP32220, and C) XENP32224.

(40) FIG. **40** depicts induction of T cell proliferation (as indicated by percentage T cells expressing Ki67) by A) XENP32218, B) XENP32220, and C) XENP32224 in the presence of cancer cells with varying surface PSMA densities.

(41) FIG. **41** depicts induction of T cell degranulation (as indicated by percentage T cells expressing CD107a) by A) XENP32218, B) XENP32220, and C) XENP32224 in the presence of cancer cells with varying surface PSMA densities.

(42) FIG. **42** depicts induction of RTCC on luciferase-transduced PC3 cancer cells with varying surface PSMA densities by XENP31855 (1 nM KD PSMA+CD3 High-Int #1(VLVH) in 2+1 Fab-scFv-Fc format).

(43) FIG. **43** depicts induction of T cell degranulation (as indicated by percentage T cells expressing CD107a) by XENP31856 (1 nM K.sub.D PSMA+CD3 High-Int #2(VLVH)) in the presence of PC3 cancer cells transfected with varying surface PSMA densities. The data show that XENP31856 was highly selective for high PSMA expressing PC3 (~100K) cell line; however, XENP31856 induced little to no degranulation in the presence of all the low PSMA expressing PC3 cell lines.

(44) FIG. **44** depicts induction of RTCC on luciferase-transduced PC3 cancer cells with varying surface PSMA densities by XENP33063 (7 nM K.sub.D PSMA+CD3 High-Int #2(VLVH)). The data show that XENP33063 was highly selective for high PSMA expressing PC3 (~100K) cell line; however, XENP31856 induced little to no killing on all the low PSMA expressing PC3 cell lines.

(45) FIG. **45** depicts induction of RTCC on luciferase-transduced LNCaP cancer cells and 22Rv1 cancer cells by XENP32220 (38 nM K.sub.D PSMA+CD3 High-Int #1(VLVH)). The data show that XENP32220 was able to induce cell kill on both LNCaP and 22Rv1 cancer cells.

(46) FIG. **46** depicts induction of T cell degranulation (as indicated by percentage T cells expressing CD107a) by A) XENP14484, B) XENP34282, and C) XENP34283 in the presence of cancer cells with varying surface PSMA densities. The data show that the three prototypic 1+1 anti-PSMA×anti-CD3 induced T cell degranulation in the presence of cell lines expressing higher and lower PSMA levels with similar potency.

(47) FIG. **47** depicts tumor volume on Day 19 in PSMA-transfected PC3 (~100K) and huPBMC-engrafted mice following first dose with PBS, bivalent anti-PD1 mAb, XENP32218, XENP32220, or XENP32224. Each of the tuned PSMA×CD3 bispecific antibodies significantly enhanced ($p < 0.05$ vs. PBS or α PD-1 mAb) anti-tumor activity (as indicated by tumor volume; statistics performed on baseline corrected data using unpaired t-test).

(48) FIG. **48** depicts the change in tumor volume (as determined by caliper measurements) over time in PSMA-transfected PC3 (~100K) and huPBMC-engrafted mice dosed with PBS, bivalent anti-PD1 mAb, XENP32218, XENP32220, or XENP32224.

(49) FIGS. **49A-49C** depict the pharmacokinetic data from a study in which each healthy male cynomolgus was administered by IV either a 1× dose, 10× dose, or 60× dose of the indicated test article. Xtend variants XENP34262, XENP34267, and XENP34268 showed improved pharmacokinetics over non-Xtend variants XENP32218, XENP32220, and XENP32224. All test articles were tolerated at each dose level.

(50) FIGS. **50A-50K** depict several formats for use in the anti-PSMA×anti-CD3 bispecific antibodies disclosed herein. The first is the “1+1 Fab-scFv-Fc” format (also referred to as the “bottle opener” or “Triple F” format), with a first antigen binding domain that is a Fab domain and a second anti-antigen binding domain that is an scFv domain (FIG. 1A). Additionally, “mAb-Fv,” “mAb-scFv,” “2+1 Fab2-scFv-Fc” (also referred to as the “central scFv” or “central-scFv” format), “central-Fv,” “one armed central-scFv,” “one scFv-mAb,” “scFv-mAb,” “dual scFv,” “trident,” and non-heterodimeric bispecific formats are all shown. The scFv domains depicted in

FIGS. 10A-10F can be either, from N- to C-terminus orientation: variable heavy-(optional linker)-variable light, or variable light-(optional linker)-variable heavy. In addition, for the one armed scFv-mAb, the scFv can be attached either to the N-terminus of a heavy chain monomer or to the N-terminus of the light chain. In certain embodiments, “Anti-antigen 1” in FIG. 50 refers to a PSMA binding domain. In certain embodiments, “Anti-antigen 1” in FIG. 50 refers to a CD3 binding domain. In certain embodiments, “Anti-antigen 2” in FIG. 50 refers to a PSMA binding domain. In certain embodiments “Anti-antigen 2” in FIG. 50 refers to a CD3 binding domain. In some embodiments, “Anti-antigen 1” in FIG. 50 refers to a PSMA binding domain and “Anti-antigen 2” in FIG. 50 refers to a CD3 binding domain. In some embodiments, “Anti-antigen 1” in FIG. 50 refers to a CD3 binding domain and “Anti-antigen 2” in FIG. 50 refers to a PSMA binding domain. Any of the PSMA binding domains and CD3 binding domains disclosed can be included in the bispecific formats of FIG. 50.

DETAILED DESCRIPTION

I. Overview

(51) Provided herein are novel anti-CD3× anti-PSMA (also referred to as anti-PSMA×anti-CD3, α CD3× α PSMA, or α PSMA× α CD3) heterodimeric bispecific antibodies and methods of using such antibodies for the treatment of cancers. In particular, provided herein are anti-CD3, anti-PSMA bispecific antibodies in a variety of formats such as those depicted in FIGS. 21A and 21B. These bispecific antibodies are useful for the treatment of cancers, particularly those with increased PSMA expression such as prostate cancers. Such antibodies are used to direct CD3+ effector T cells to PSMA+ tumors, thereby allowing the CD3+ effector T cells to attack and lyse the PSMA+ tumors.

(52) The anti-PSMA antibodies provided herein include PSMA binding domain with binding affinities and valencies that allow for the advantageous selectivity for cells expressing high levels of PSMA while minimizing reactivity on low PSMA expressing cells. Such PSMA antibodies are useful, for example, for cancers that express high levels of PSMA including, for example, prostate cancer.

(53) In some embodiments, such anti-PSMA antibodies include CD3 binding domains with binding affinity that further contribute to selective targeting of high-PSMA expressing cells lines. Such bispecific antibodies that have different binding affinities to human CD3 that can alter or reduce the potential side effects of anti-CD3 therapy. That is, in some embodiments the antibodies described herein provide antibody constructs comprising anti-CD3 antigen binding domains that are “strong” or “high affinity” binders to CD3 (e.g. one example are heavy and light variable domains depicted as H1.30_L1.47 (optionally including a charged linker as appropriate)) and also bind to PSMA. In other embodiments, the antibodies described herein provide antibody constructs comprising anti-CD3 antigen binding domains that are “lite” or “lower affinity” binders to CD3. Additional embodiments provides antibody constructs comprising anti-CD3 antigen binding domains that have intermediate or “medium” affinity to CD3 that also bind to PSMA. While a very large number of anti-CD3 antigen binding domains (ABDs) can be used, particularly useful embodiments use 6 different anti-CD3 ABDs, although they can be used in two scFv orientations as discussed herein. Affinity is generally measured using a Biacore assay.

(54) It should be appreciated that the “high, medium, low” anti-CD3 sequences provided herein can be used in a variety of heterodimerization formats as depicted in FIGS. 21A, 21B. In general, due to the potential side effects of T cell recruitment, exemplary embodiments utilize formats that only bind CD3 monovalently, such as depicted in FIGS. 21A and 21B, and in the formats depicted herein, it is the CD3 ABD that is a scFv as more fully described herein. In contrast, the subject bispecific antibodies can bind PSMA either monovalently (e.g. FIG. 21A) or bivalently (e.g. FIG. 21B).

(55) Accordingly, in one aspect, provided herein are heterodimeric antibodies that bind to two different antigens, e.g. the antibodies are “bispecific”, in that they bind two different target

antigens, generally PSMA and CD3 as described herein. These heterodimeric antibodies can bind these target antigens either monovalently (e.g. there is a single antigen binding domain such as a variable heavy and variable light domain pair) or bivalently (there are two antigen binding domains that each independently bind the antigen). In some embodiments, the heterodimeric antibody provided herein includes one CD3 binding domain and one PSMA binding domain (e.g., heterodimeric antibodies in the “1+1 Fab-scFv-Fc” format described herein). In other embodiments, the heterodimeric antibody provided herein includes one CD3 binding domain and two PSMA binding domains (e.g., heterodimeric antibodies in the “2+1 Fab.sub.2-scFv-Fc” formats described herein). The heterodimeric antibodies provided herein are based on the use of different monomers which contain amino acid substitutions that “skew” formation of heterodimers over homodimers, as is more fully outlined below, coupled with “pI variants” that allow simple purification of the heterodimers away from the homodimers, as is similarly outlined below. The heterodimeric bispecific antibodies provided generally rely on the use of engineered or variant Fc domains that can self-assemble in production cells to produce heterodimeric proteins, and methods to generate and purify such heterodimeric proteins.

II. Nomenclature

(56) The antibodies provided herein are listed in several different formats. In some instances, each monomer of a particular antibody is given a unique “XENP” number, although as will be appreciated in the art, a longer sequence might contain a shorter one. For example, a “scFv-Fc” monomer of a 1+1 Fab-scFv-Fc format antibody may have a first XENP number, while the scFv domain itself will have a different XENP number. Some molecules have three polypeptides, so the XENP number, with the components, is used as a name. Thus, the molecule XENP31602, which is in 2+1 Fab.sub.2-scFv-Fc format, comprises three sequences (see FIG. 26): 1) a “Fab-Fc Heavy Chain” monomer; 2) a “Fab-scFv-Fc Heavy Chain” monomer; and 3) a “Light Chain” monomer or equivalents, although one of skill in the art would be able to identify these easily through sequence alignment. These XENP numbers are in the sequence listing as well as identifiers, and used in the Figures. In addition, one molecule, comprising the three components, gives rise to multiple sequence identifiers. For example, the listing of the Fab includes, the full heavy chain sequence, the variable heavy domain sequence and the three CDRs of the variable heavy domain sequence, the full light chain sequence, a variable light domain sequence and the three CDRs of the variable light domain sequence. A Fab-scFv-Fc monomer includes a full length sequence, a variable heavy domain sequence, 3 heavy CDR sequences, and an scFv sequence (include scFv variable heavy domain sequence, scFv variable light domain sequence and scFv linker). Note that some molecules herein with a scFv domain use a single charged scFv linker (+H), although others can be used. In addition, the naming nomenclature of particular antigen binding domains (e.g., PSMA and CD3 binding domains) use a “Hx.xx_Ly.yy” type of format, with the numbers being unique identifiers to particular variable chain sequences. Thus, the PSMA binding domain PSMA-H H1_L1 (e.g., FIG. 17A) is “H1_L1”, which indicates that the variable heavy domain, H1, was combined with the light domain L1. In the case that these sequences are used as scFvs, the designation “H1_L1”, indicates that the variable heavy domain, H1 is combined with the light domain, L1, and is in VH-linker-VL orientation, from N- to C-terminus. This molecule with the identical sequences of the heavy and light variable domains but in the reverse order (VL-linker-VH orientation, from N- to C-terminus) would be designated “L1_H1.1”. Similarly, different constructs may “mix and match” the heavy and light chains as will be evident from the sequence listing and the figures.

III. Definitions

(57) In order that the application may be more completely understood, several definitions are set forth below. Such definitions are meant to encompass grammatical equivalents.

(58) By “PSMA” or “Prostate Specific Membrane Antigen” (e.g., Genbank Accession Number NP 005012.2) herein is meant a type II transmembrane protein that is expressed in all prostatic tissues, including primary prostate adenocarcinomas, metastatic prostate cancer, and in the tumor

neovasculature of many solid tumors. In prostate cancer (PCa), PSMA is highly expressed in poorly differentiated, highly metastatic prostatic cells and in castrate-resistant models. PSMA sequences are disclosed in FIGS. 11A and 11B.

(59) By “ablation” herein is meant a decrease or removal of activity. Thus for example, “ablating FcγR binding” means the Fc region amino acid variant has less than 50% starting binding as compared to an Fc region not containing the specific variant, with more than 70-80-90-95-98% loss of activity being preferred, and in general, with the activity being below the level of detectable binding in a Biacore, SPR or BLI assay. Of particular use in the ablation of FcγR binding are those shown in FIG. 5, which generally are added to both monomers.

(60) By “ADCC” or “antibody dependent cell-mediated cytotoxicity” as used herein is meant the cell-mediated reaction wherein nonspecific cytotoxic cells that express FcγRs recognize bound antibody on a target cell and subsequently cause lysis of the target cell. ADCC is correlated with binding to FcγRIIIa; increased binding to FcγRIIIa leads to an increase in ADCC activity.

(61) By “ADCP” or antibody dependent cell-mediated phagocytosis as used herein is meant the cell-mediated reaction wherein nonspecific phagocytic cells that express FcγRs recognize bound antibody on a target cell and subsequently cause phagocytosis of the target cell.

(62) As used herein, term “antibody” is used generally. Antibodies described herein can take on a number of formats as described herein, including traditional antibodies as well as antibody derivatives, fragments and mimetics, described herein.

(63) Traditional immunoglobulin (Ig) antibodies are “Y” shaped tetramers. Each tetramer is typically composed of two identical pairs of polypeptide chains, each pair having one “light chain” monomer (typically having a molecular weight of about 25 kDa) and one “heavy chain” monomer (typically having a molecular weight of about 50-70 kDa).

(64) Other useful antibody formats include, but are not limited to, the 1+1 Fab-scFv-Fc format and 2+1 Fab.sub.2-scFv-Fc antibody formats described herein, as well as “mAb-Fv,” “mAb-scFv,” “central-Fv,” “one armed scFv-mAb,” “scFv-mAb,” “dual scFv,” and “trident” format antibodies, as shown in FIG. 50.

(65) Antibody heavy chains typically include a variable heavy (VH) domain, which includes vhCDR1-3, and an Fc domain, which includes a CH2-CH3 monomer. In some embodiments, antibody heavy chains include a hinge and CH1 domain. Traditional antibody heavy chains are monomers that are organized, from N- to C-terminus: VH-CH1-hinge-CH2-CH3. The CH1-hinge-CH2-CH3 is collectively referred to as the heavy chain “constant domain” or “constant region” of the antibody, of which there are five different categories or “isotypes”: IgA, IgD, IgG, IgE and IgM. Thus, “isotype” as used herein is meant any of the subclasses of immunoglobulins defined by the chemical and antigenic characteristics of their constant regions. It should be understood that therapeutic antibodies can also comprise hybrids of isotypes and/or subclasses. For example, as shown in US Publication 2009/0163699, incorporated by reference, the antibodies described herein include the use of human IgG1/G2 hybrids.

(66) In some embodiments, the antibodies provided herein include IgG isotype constant domains, which has several subclasses, including, but not limited to IgG1, IgG2, IgG3, and IgG4. In the IgG subclass of immunoglobulins, there are several immunoglobulin domains in the heavy chain. By “immunoglobulin (Ig) domain” herein is meant a region of an immunoglobulin having a distinct tertiary structure. Of interest in the antibodies described herein are the heavy chain domains, including, the constant heavy (CH) domains and the hinge domains. In the context of IgG antibodies, the IgG isotypes each have three CH regions. Accordingly, “CH” domains in the context of IgG are as follows: “CH1” refers to positions 118-220 according to the EU index as in Kabat. “CH2” refers to positions 237-340 according to the EU index as in Kabat, and “CH3” refers to positions 341-447 according to the EU index as in Kabat. As shown herein and described below, the pI variants can be in one or more of the CH regions, as well as the hinge region, discussed below.

(67) It should be noted that IgG1 has different allotypes with polymorphisms at 356 (D or E) and 358 (L or M). The sequences depicted herein use the 356D/358M allotype, however the other allotype is included herein. That is, any sequence inclusive of an IgG1 Fc domain included herein can have 356E/358L replacing the 356D/358M allotype. It should be understood that therapeutic antibodies can also comprise hybrids of isotypes and/or subclasses. For example, as shown in US Publication 2009/0163699, incorporated by reference, the present antibodies, in some embodiments, include IgG1/IgG2 hybrids.

(68) By “Fc” or “Fc region” or “Fc domain” as used herein is meant the polypeptide comprising the constant region of an antibody, in some instances, excluding all of the first constant region immunoglobulin domain (e.g., CH1) or a portion thereof, and in some cases, optionally including all or part of the hinge. For IgG, the Fc domain comprises immunoglobulin domains CH2 and CH3 (Cy2 and Cy3), and optionally all or a portion of the hinge region between CH1 (Cy1) and CH2 (Cy2). Thus, in some cases, the Fc domain includes, from N- to C-terminal, CH2-CH3 and hinge-CH2-CH3. In some embodiments, the Fc domain is that from IgG1, IgG2, IgG3 or IgG4, with IgG1 hinge-CH2-CH3 and IgG4 hinge-CH2-CH3 finding particular use in many embodiments. Additionally, in the case of human IgG1 Fc domains, frequently the hinge includes a C220S amino acid substitution. Furthermore, in the case of human IgG4 Fc domains, frequently the hinge includes a S228P amino acid substitution. Although the boundaries of the Fc region may vary, the human IgG heavy chain Fc region is usually defined to include residues E216, C226, or A231 to its carboxyl-terminal, wherein the numbering is according to the EU index as in Kabat. In some embodiments, as is more fully described below, amino acid modifications are made to the Fc region, for example to alter binding to one or more FcγR or to the FcRn.

(69) By “heavy chain constant region” herein is meant the CH1-hinge-CH2-CH3 portion of an antibody (or fragments thereof), excluding the variable heavy domain; in EU numbering of human IgG1 this is amino acids 118-447 By “heavy chain constant region fragment” herein is meant a heavy chain constant region that contains fewer amino acids from either or both of the N- and C-termini but still retains the ability to form a dimer with another heavy chain constant region.

(70) Another type of Ig domain of the heavy chain is the hinge region. By “hinge” or “hinge region” or “antibody hinge region” or “hinge domain” herein is meant the flexible polypeptide comprising the amino acids between the first and second constant domains of an antibody. Structurally, the IgG CH1 domain ends at EU position 215, and the IgG CH2 domain begins at residue EU position 231. Thus for IgG the antibody hinge is herein defined to include positions 216 (E216 in IgG1) to 230 (p230 in IgG1), wherein the numbering is according to the EU index as in Kabat. In some cases, a “hinge fragment” is used, which contains fewer amino acids at either or both of the N- and C-termini of the hinge domain. As noted herein, pI variants can be made in the hinge region as well. Many of the antibodies herein have at least one the cysteines at position 220 according to EU numbering (hinge region) replaced by a serine. Generally, this modification is on the “scFv monomer” side for most of the sequences depicted herein, although it can also be on the “Fab monomer” side, or both, to reduce disulfide formation. Specifically included within the sequences herein are one or both of these cysteines replaced (C220S).

(71) As will be appreciated by those in the art, the exact numbering and placement of the heavy constant region domains can be different among different numbering systems. A useful comparison of heavy constant region numbering according to EU and Kabat is as below, see Edelman et al., 1969, Proc Natl Acad Sci USA 63:78-85 and Kabat et al., 1991, Sequences of Proteins of Immunological Interest, 5th Ed., United States Public Health Service, National Institutes of Health, Bethesda, entirely incorporated by reference.

(72) TABLE-US-00001 TABLE 1 EU Numbering Kabat Numbering CH1 118-215 114-223 Hinge 216-230 226-243 CH2 231-340 244-360 CH3 341-447 361-478

(73) The antibody light chain generally comprises two domains: the variable light domain (VL), which includes light chain CDRs vlCDR1-3, and a constant light chain region (often referred to as

CL or Ck). The antibody light chain is typically organized from N- to C-terminus: VL-CL.

(74) By “antigen binding domain” or “ABD” herein is meant a set of six Complementary Determining Regions (CDRs) that, when present as part of a polypeptide sequence, specifically binds a target antigen (e.g., PSMA or CD3) as discussed herein. As is known in the art, these CDRs are generally present as a first set of variable heavy CDRs (vhCDRs or VHCDRs) and a second set of variable light CDRs (vlCDRs or VLCDRs), each comprising three CDRs: vhCDR1, vhCDR2, vhCDR3 variable heavy CDRs and vlCDR1, vlCDR2 and vlCDR3 variable light CDRs. The CDRs are present in the variable heavy domain (vhCDR1-3) and variable light domain (vlCDR1-3). The variable heavy domain and variable light domain from an Fv region.

(75) The antibodies described herein provide a large number of different CDR sets. In this case, a “full CDR set” comprises the three variable light and three variable heavy CDRs, e.g., a vlCDR1, vlCDR2, vlCDR3, vhCDR1, vhCDR2 and vhCDR3. These can be part of a larger variable light or variable heavy domain, respectfully. In addition, as more fully outlined herein, the variable heavy and variable light domains can be on separate polypeptide chains, when a heavy and light chain is used (for example when Fabs are used), or on a single polypeptide chain in the case of scFv sequences.

(76) As will be appreciated by those in the art, the exact numbering and placement of the CDRs can be different among different numbering systems. However, it should be understood that the disclosure of a variable heavy and/or variable light sequence includes the disclosure of the associated (inherent) CDRs. Accordingly, the disclosure of each variable heavy region is a disclosure of the vhCDRs (e.g., vhCDR1, vhCDR2 and vhCDR3) and the disclosure of each variable light region is a disclosure of the vlCDRs (e.g., vlCDR1, vlCDR2 and vlCDR3). A useful comparison of CDR numbering is as below, see Lafranc et al., *Dev. Comp. Immunol.* 27(1):55-77 (2003):

(77) TABLE-US-00002 TABLE 2 Kabat + Chothia IMGT Kabat AbM Chothia Contact Xencor

vhCDR1	26-35	27-38	31-35	26-35	26-32	30-35	27-35	vhCDR2	50-65	56-65	50-65
	50-58	52-56	47-58	54-61	vhCDR3	95-102	105-117	95-102	95-102	95-102	93-101
vlCDR1	24-34	27-38	24-34	24-34	24-34	30-36	27-38	vlCDR2	50-56	56-65	50-56
	50-56	50-56	46-55	56-62	vlCDR3	89-97	105-117	89-97	89-97	89-97	89-96
											97-105

(78) Throughout the present specification, the Kabat numbering system is generally used when referring to a residue in the variable domain (approximately, residues 1-107 of the light chain variable region and residues 1-113 of the heavy chain variable region) and the EU numbering system for Fc regions (e.g., Kabat et al., *supra* (1991)).

(79) The CDRs contribute to the formation of the antigen-binding, or more specifically, epitope binding site of the antigen binding domains and antibodies. “Epitope” refers to a determinant that interacts with a specific antigen binding site in the variable region of an antibody molecule known as a paratope. Epitopes are groupings of molecules such as amino acids or sugar side chains and usually have specific structural characteristics, as well as specific charge characteristics. A single antigen may have more than one epitope.

(80) The epitope may comprise amino acid residues directly involved in the binding (also called immunodominant component of the epitope) and other amino acid residues, which are not directly involved in the binding, such as amino acid residues which are effectively blocked by the specifically antigen binding peptide; in other words, the amino acid residue is within the footprint of the specifically antigen binding peptide.

(81) Epitopes may be either conformational or linear. A conformational epitope is produced by spatially juxtaposed amino acids from different segments of the linear polypeptide chain. A linear epitope is one produced by adjacent amino acid residues in a polypeptide chain. Conformational and nonconformational epitopes may be distinguished in that the binding to the former but not the latter is lost in the presence of denaturing solvents.

(82) An epitope typically includes at least 3, and more usually, at least 5 or 8-10 amino acids in a

unique spatial conformation. Antibodies that recognize the same epitope can be verified in a simple immunoassay showing the ability of one antibody to block the binding of another antibody to a target antigen, for example “binning.” As outlined below, the disclosure not only includes the enumerated antigen binding domains and antibodies herein, but those that compete for binding with the epitopes bound by the enumerated antigen binding domains.

(83) In some embodiments, the six CDRs of the antigen binding domain are contributed by a variable heavy and a variable light domain. In a “Fab” format, the set of 6 CDRs are contributed by two different polypeptide sequences, the variable heavy domain (vh or VH; containing the vhCDR1, vhCDR2 and vhCDR3) and the variable light domain (vl or VL; containing the vlCDR1, vlCDR2 and vlCDR3), with the C-terminus of the vh domain being attached to the N-terminus of the CH1 domain of the heavy chain and the C-terminus of the vl domain being attached to the N-terminus of the constant light domain (and thus forming the light chain). In a scFv format, the vh and vl domains are covalently attached, generally through the use of a linker (a “scFv linker”) as outlined herein, into a single polypeptide sequence, which can be either (starting from the N-terminus) vh-linker-vl or vl-linker-vh, with the former being generally preferred (including optional domain linkers on each side, depending on the format used (e.g., from FIG. 1). In general, the C-terminus of the scFv domain is attached to the N-terminus of the hinge in the second monomer.

(84) By “variable region” or “variable domain” as used herein is meant the region of an immunoglobulin that comprises one or more Ig domains substantially encoded by any of the V κ , V δ , and/or VH genes that make up the kappa, lambda, and heavy chain immunoglobulin genetic loci respectively, and contains the CDRs that confer antigen specificity. Thus, a “variable heavy domain” pairs with a “variable light domain” to form an antigen binding domain (“ABD”). In addition, each variable domain comprises three hypervariable regions (“complementary determining regions,” “CDRs”) (VHCDR1, VHCDR2 and VHCDR3 for the variable heavy domain and VLCDR1, VLCDR2 and VLCDR3 for the variable light domain) and four framework (FR) regions, arranged from amino-terminus to carboxy-terminus in the following order: FR1-CDR1-FR2-CDR2-FR3-CDR3-FR4. The hypervariable region generally encompasses amino acid residues from about amino acid residues 24-34 (LCDR1; “L” denotes light chain), 50-56 (LCDR2) and 89-97 (LCDR3) in the light chain variable region and around about 31-35B (HCDR1; “H” denotes heavy chain), 50-65 (HCDR2), and 95-102 (HCDR3) in the heavy chain variable region; Kabat et al., SEQUENCES OF PROTEINS OF IMMUNOLOGICAL INTEREST, 5th Ed. Public Health Service, National Institutes of Health, Bethesda, Md. (1991) and/or those residues forming a hypervariable loop (e.g. residues 26-32 (LCDR1), 50-52 (LCDR2) and 91-96 (LCDR3) in the light chain variable region and 26-32 (HCDR1), 53-55 (HCDR2) and 96-101 (HCDR3) in the heavy chain variable region; Chothia and Lesk (1987) J. Mol. Biol. 196:901-917. Specific CDRs of the invention are described in Table 2.

(85) By “Fab” or “Fab region” as used herein is meant the polypeptide that comprises the VH, CH1, VL, and CL immunoglobulin domains, generally on two different polypeptide chains (e.g. VH-CH1 on one chain and VL-CL on the other). Fab may refer to this region in isolation, or this region in the context of a bispecific antibody described herein. In the context of a Fab, the Fab comprises an Fv region in addition to the CH1 and CL domains.

(86) By “Fv” or “Fv fragment” or “Fv region” as used herein is meant a polypeptide that comprises the VL and VH domains of an ABD. Fv regions can be formatted as both Fabs (as discussed above, generally two different polypeptides that also include the constant regions as outlined above) and scFvs, where the VL and VH domains are combined (generally with a linker as discussed herein) to form an scFv.

(87) By “single chain Fv” or “scFv” herein is meant a variable heavy domain covalently attached to a variable light domain, generally using a scFv linker as discussed herein, to form a scFv or scFv domain. A scFv domain can be in either orientation from N- to C-terminus (VH-linker-VL or VL-linker-VH). In the sequences depicted in the sequence listing and in the figures, the order of the VH

and VL domain is indicated in the name, e.g. H.X_L.Y means N- to C-terminal is VH-linker-VL, and L.Y_H.X is VL-linker-VH.

(88) Some embodiments of the subject antibodies provided herein comprise at least one scFv domain, which, while not naturally occurring, generally includes a variable heavy domain and a variable light domain, linked together by a scFv linker. As outlined herein, while the scFv domain is generally from N- to C-terminus oriented as VH-scFv linker-VL, this can be reversed for any of the scFv domains (or those constructed using vh and vl sequences from Fabs), to VL-scFv linker-VH, with optional linkers at one or both ends depending on the format.

(89) By “modification” herein is meant an amino acid substitution, insertion, and/or deletion in a polypeptide sequence or an alteration to a moiety chemically linked to a protein. For example, a modification may be an altered carbohydrate or PEG structure attached to a protein. By “amino acid modification” herein is meant an amino acid substitution, insertion, and/or deletion in a polypeptide sequence. For clarity, unless otherwise noted, the amino acid modification is always to an amino acid coded for by DNA, e.g. the 20 amino acids that have codons in DNA and RNA.

(90) By “amino acid substitution” or “substitution” herein is meant the replacement of an amino acid at a particular position in a parent polypeptide sequence with a different amino acid. In particular, in some embodiments, the substitution is to an amino acid that is not naturally occurring at the particular position, either not naturally occurring within the organism or in any organism. For example, the substitution E272Y refers to a variant polypeptide, in this case an Fc variant, in which the glutamic acid at position 272 is replaced with tyrosine. For clarity, a protein which has been engineered to change the nucleic acid coding sequence but not change the starting amino acid (for example exchanging CGG (encoding arginine) to CGA (still encoding arginine) to increase host organism expression levels) is not an “amino acid substitution”; that is, despite the creation of a new gene encoding the same protein, if the protein has the same amino acid at the particular position that it started with, it is not an amino acid substitution.

(91) By “amino acid insertion” or “insertion” as used herein is meant the addition of an amino acid sequence at a particular position in a parent polypeptide sequence. For example, -233E or 233E designates an insertion of glutamic acid after position 233 and before position 234. Additionally, -233ADE or A233ADE designates an insertion of AlaAspGlu after position 233 and before position 234.

(92) By “amino acid deletion” or “deletion” as used herein is meant the removal of an amino acid sequence at a particular position in a parent polypeptide sequence. For example, E233- or E233 #, E233() or E233del designates a deletion of glutamic acid at position 233. Additionally, EDA233- or EDA233 # designates a deletion of the sequence GluAspAla that begins at position 233.

(93) By “variant protein” or “protein variant”, or “variant” as used herein is meant a protein that differs from that of a parent protein by virtue of at least one amino acid modification. The protein variant has at least one amino acid modification compared to the parent protein, yet not so many that the variant protein will not align with the parental protein using an alignment program such as that described below. In general, variant proteins (such as variant Fc domains, etc., outlined herein, are generally at least 75, 80, 85, 90, 91, 92, 93, 94, 95, 96, 97, 98 or 99% identical to the parent protein, using the alignment programs described below, such as BLAST. “Variant” as used herein also refers to particular amino acid modifications that confer particular function (e.g., a “heterodimerization variant,” “pI variant,” “ablation variant,” etc.).

(94) As described below, in some embodiments the parent polypeptide, for example an Fc parent polypeptide, is a human wild type sequence, such as the heavy constant domain or Fc region from IgG1, IgG2, IgG3 or IgG4, although human sequences with variants can also serve as “parent polypeptides”, for example the IgG1/2 hybrid of US Publication 2006/0134105 can be included. The protein variant sequence herein will preferably possess at least about 80% identity with a parent protein sequence, and most preferably at least about 90% identity, more preferably at least about 95-98-99% identity. Accordingly, by “antibody variant” or “variant antibody” as used herein

is meant that modification that differs from a parent antibody by virtue of at least one amino acid modification, “IgG variant” or “variant IgG” as used herein is meant an antibody that differs from a parent IgG (again, in many cases, from a human IgG sequence) by virtue of at least one amino acid modification, and “immunoglobulin variant” or “variant immunoglobulin” as used herein is meant an immunoglobulin sequence that differs from that of a parent immunoglobulin sequence by virtue of at least one amino acid modification. “Fc variant” or “variant Fc” as used herein is meant a protein comprising an amino acid modification in an Fc domain as compared to an Fc domain of human IgG1, IgG2 or IgG4.

(95) “Fc variant” or “variant Fc” as used herein is meant a protein comprising an amino acid modification in an Fc domain. The modification can be an addition, deletion, or substitution. The Fc variants are defined according to the amino acid modifications that compose them. Thus, for example, N434S or 434S is an Fc variant with the substitution for serine at position 434 relative to the parent Fc polypeptide, wherein the numbering is according to the EU index. Likewise, M428L/N434S defines an Fc variant with the substitutions M428L and N434S relative to the parent Fc polypeptide. The identity of the WT amino acid may be unspecified, in which case the aforementioned variant is referred to as 428L/434S. It is noted that the order in which substitutions are provided is arbitrary, that is to say that, for example, 428L/434S is the same Fc variant as 434S/428L, and so on. For all positions discussed herein that relate to antibodies or derivatives and fragments thereof (e.g., Fc domains), unless otherwise noted, amino acid position numbering is according to the EU index. The “EU index” or “EU index as in Kabat” or “EU numbering” scheme refers to the numbering of the EU antibody (Edelman et al., 1969, Proc Natl Acad Sci USA 63:78-85, hereby entirely incorporated by reference).

(96) In general, variant Fc domains have at least about 80, 85, 90, 95, 97, 98 or 99 percent identity to the corresponding parental human IgG Fc domain (using the identity algorithms discussed below, with one embodiment utilizing the BLAST algorithm as is known in the art, using default parameters). Alternatively, the variant Fc domains can have from 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 amino acid modifications as compared to the parental Fc domain. Alternatively, the variant Fc domains can have up to 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 amino acid modifications as compared to the parental Fc domain. Additionally, as discussed herein, the variant Fc domains described herein still retain the ability to form a dimer with another Fc domain as measured using known techniques as described herein, such as non-denaturing gel electrophoresis.

(97) By “protein” herein is meant at least two covalently attached amino acids, which includes proteins, polypeptides, oligopeptides and peptides. In addition, polypeptides that make up the antibodies described herein may include synthetic derivatization of one or more side chains or termini, glycosylation, PEGylation, circular permutation, cyclization, linkers to other molecules, fusion to proteins or protein domains, and addition of peptide tags or labels.

(98) By “residue” as used herein is meant a position in a protein and its associated amino acid identity. For example, Asparagine 297 (also referred to as Asn297 or N297) is a residue at position 297 in the human antibody IgG1.

(99) By “IgG subclass modification” or “isotype modification” as used herein is meant an amino acid modification that converts one amino acid of one IgG isotype to the corresponding amino acid in a different, aligned IgG isotype. For example, because IgG1 comprises a tyrosine and IgG2 a phenylalanine at EU position 296, a F296Y substitution in IgG2 is considered an IgG subclass modification.

(100) By “non-naturally occurring modification” as used herein is meant an amino acid modification that is not isotypic. For example, because none of the human IgGs comprise a serine at position 434, the substitution 434S in IgG1, IgG2, IgG3, or IgG4 (or hybrids thereof) is considered a non-naturally occurring modification.

(101) By “amino acid” and “amino acid identity” as used herein is meant one of the 20 naturally

occurring amino acids that are coded for by DNA and RNA.

(102) By “effector function” as used herein is meant a biochemical event that results from the interaction of an antibody Fc region with an Fc receptor or ligand. Effector functions include but are not limited to ADCC, ADCP, and CDC.

(103) By “IgG Fc ligand” as used herein is meant a molecule, preferably a polypeptide, from any organism that binds to the Fc region of an IgG antibody to form an Fc/Fc ligand complex. Fc ligands include but are not limited to FcγRIs, FcγRIIs, FcγRIIs, FcRn, C1q, C3, mannan binding lectin, mannose receptor, staphylococcal protein A, streptococcal protein G, and viral FcγR. Fc ligands also include Fc receptor homologs (FcRH), which are a family of Fc receptors that are homologous to the FcγRs (Davis et al., 2002, Immunological Reviews 190:123-136, entirely incorporated by reference). Fc ligands may include undiscovered molecules that bind Fc. Particular IgG Fc ligands are FcRn and Fc gamma receptors. By “Fc ligand” as used herein is meant a molecule, preferably a polypeptide, from any organism that binds to the Fc region of an antibody to form an Fc/Fc ligand complex.

(104) By “Fc gamma receptor”, “FcγR” or “FcγgammaR” as used herein is meant any member of the family of proteins that bind the IgG antibody Fc region and is encoded by an FcγR gene. In humans this family includes but is not limited to FcγRI (CD64), including isoforms FcγRIa, FcγRIb, and FcγRIc; FcγRII (CD32), including isoforms FcγRIIa (including allotypes H131 and R131), FcγRIIb (including FcγRIIb-1 and FcγRIIb-2), and FcγRIIc; and FcγRIII (CD16), including isoforms FcγRIIIa (including allotypes V158 and F158) and FcγRIIIb (including allotypes FcγRIIb-NA1 and FcγRIIb-NA2) (Jefferis et al., 2002, Immunol Lett 82:57-65, entirely incorporated by reference), as well as any undiscovered human FcγRs or FcγR isoforms or allotypes. An FcγR may be from any organism, including but not limited to humans, mice, rats, rabbits, and monkeys. Mouse FcγRs include but are not limited to FcγRI (CD64), FcγRII (CD32), FcγRIII (CD16), and FcγRIII-2 (CD16-2), as well as any undiscovered mouse FcγRs or FcγR isoforms or allotypes.

(105) By “FcRn” or “neonatal Fe Receptor” as used herein is meant a protein that binds the IgG antibody Fc region and is encoded at least in part by an FcRn gene. The FcRn may be from any organism, including but not limited to humans, mice, rats, rabbits, and monkeys. As is known in the art, the functional FcRn protein comprises two polypeptides, often referred to as the heavy chain and light chain. The light chain is beta-2-microglobulin and the heavy chain is encoded by the FcRn gene. Unless otherwise noted herein, FcRn or an FcRn protein refers to the complex of FcRn heavy chain with beta-2-microglobulin. A variety of FcRn variants used to increase binding to the FcRn receptor, and in some cases, to increase serum half-life. An “FcRn variant” is one that increases binding to the FcRn receptor, and suitable FcRn variants are shown below.

(106) By “parent polypeptide” as used herein is meant a starting polypeptide that is subsequently modified to generate a variant. The parent polypeptide may be a naturally occurring polypeptide, or a variant or engineered version of a naturally occurring polypeptide. Accordingly, by “parent immunoglobulin” as used herein is meant an unmodified immunoglobulin polypeptide that is modified to generate a variant, and by “parent antibody” as used herein is meant an unmodified antibody that is modified to generate a variant antibody. It should be noted that “parent antibody” includes known commercial, recombinantly produced antibodies as outlined below. In this context, a “parent Fc domain” will be relative to the recited variant; thus, a “variant human IgG1 Fc domain” is compared to the parent Fc domain of human IgG1, a “variant human IgG4 Fc domain” is compared to the parent Fc domain human IgG4, etc.

(107) By “position” as used herein is meant a location in the sequence of a protein. Positions may be numbered sequentially, or according to an established format, for example the EU index for antibody numbering.

(108) By “target antigen” as used herein is meant the molecule that is bound specifically by the antigen binding domain comprising the variable regions of a given antibody.

(109) By “strandedness” in the context of the monomers of the heterodimeric antibodies described herein is meant that, similar to the two strands of DNA that “match”, heterodimerization variants are incorporated into each monomer so as to preserve the ability to “match” to form heterodimers. For example, if some pI variants are engineered into monomer A (e.g. making the pI higher) then steric variants that are “charge pairs” that can be utilized as well do not interfere with the pI variants, e.g. the charge variants that make a pI higher are put on the same “strand” or “monomer” to preserve both functionalities. Similarly, for “skew” variants that come in pairs of a set as more fully outlined below, the skilled artisan will consider pI in deciding into which strand or monomer one set of the pair will go, such that pI separation is maximized using the pI of the skews as well.

(110) By “target cell” as used herein is meant a cell that expresses a target antigen.

(111) By “host cell” in the context of producing a bispecific antibody according to the antibodies described herein is meant a cell that contains the exogenous nucleic acids encoding the components of the bispecific antibody and is capable of expressing the bispecific antibody under suitable conditions. Suitable host cells are discussed below.

(112) By “wild type or WT” herein is meant an amino acid sequence or a nucleotide sequence that is found in nature, including allelic variations. A WT protein has an amino acid sequence or a nucleotide sequence that has not been intentionally modified.

(113) Provided herein are a number of antibody domains that have sequence identity to human antibody domains. Sequence identity between two similar sequences (e.g., antibody variable domains) can be measured by algorithms such as that of Smith, T. F. & Waterman, M. S. (1981) “Comparison Of Biosequences,” Adv. Appl. Math. 2:482 [local homology algorithm]; Needleman, S. B. & Wunsch, CD. (1970) “A General Method Applicable To The Search For Similarities In The Amino Acid Sequence Of Two Proteins,” J. Mol. Biol. 48:443 [homology alignment algorithm], Pearson, W. R. & Lipman, D. J. (1988) “Improved Tools For Biological Sequence Comparison,” Proc. Natl. Acad. Sci. (U.S.A.) 85:2444 [search for similarity method]; or Altschul, S. F. et al, (1990) “Basic Local Alignment Search Tool,” J. Mol. Biol. 215:403-10, the “BLAST” algorithm, see <https://blast.ncbi.nlm.nih.gov/Blast.cgi>. When using any of the aforementioned algorithms, the default parameters (for Window length, gap penalty, etc) are used. In one embodiment, sequence identity is done using the BLAST algorithm, using default parameters

(114) The antibodies described herein are generally isolated or recombinant. “Isolated,” when used to describe the various polypeptides disclosed herein, means a polypeptide that has been identified and separated and/or recovered from a cell or cell culture from which it was expressed. Ordinarily, an isolated polypeptide will be prepared by at least one purification step. An “isolated antibody,” refers to an antibody that is substantially free of other antibodies having different antigenic specificities. “Recombinant” means the antibodies are generated using recombinant nucleic acid techniques in exogeneous host cells, and they can be isolated as well.

(115) “Specific binding” or “specifically binds to” or is “specific for” a particular antigen or an epitope means binding that is measurably different from a non-specific interaction. Specific binding can be measured, for example, by determining binding of a molecule compared to binding of a control molecule, which generally is a molecule of similar structure that does not have binding activity. For example, specific binding can be determined by competition with a control molecule that is similar to the target.

(116) Specific binding for a particular antigen or an epitope can be exhibited, for example, by an antibody having a KD for an antigen or epitope of at least about 10^{-4} M, at least about 10^{-5} M, at least about 10^{-6} M, at least about 10^{-7} M, at least about 10^{-8} M, at least about 10^{-9} M, alternatively at least about 10^{-10} M, at least about 10^{-11} M, at least about 10^{-12} M, or greater, where KD refers to a dissociation rate of a particular antibody-antigen interaction. Typically, an antibody that specifically binds an antigen will have a KD that is 20-, 50-, 100-, 500-, 1000-, 5,000-, 10,000- or more times greater for a control molecule relative to the antigen or epitope.

(117) Also, specific binding for a particular antigen or an epitope can be exhibited, for example, by an antibody having a K_A or K_a for an antigen or epitope of at least 20-, 50-, 100-, 500-, 1000-, 5,000-, 10,000- or more times greater for the epitope relative to a control, where K_A or K_a refers to an association rate of a particular antibody-antigen interaction. Binding affinity is generally measured using a Biacore, SPR or BLI assay.

IV. PSMA Binding Domains

(118) In one aspect, provided herein are PSMA antigen binding domains (ABDs) and compositions that include such PSMA antigen binding domains (ABDs), including anti-PSMA antibodies.

Subject antibodies that include such PSMA antigen binding domains (e.g., anti-PSMA $\times$ anti-CD3 bispecific antibodies) advantageously target cells that express high levels of PSMA over those that express levels of PSMA (e.g., normal cells). Such PSMA binding domains and related antibodies find use, for example, in the treatment of PSMA associated cancers, such as prostate cancer.

(119) As will be appreciated by those in the art, suitable PSMA binding domains can comprise a set of 6 CDRs as depicted in the sequence listing and FIGS. 17-19, either as the CDRs are underlined or, in the case where a different numbering scheme is used as described herein and as shown in Table 2, as the CDRs that are identified using other alignments within the variable heavy (VH) domain and variable light domain (VL) sequences of those depicted in FIGS. 17-19 and the Sequence Listing (see Table 2). Suitable PSMA ABDs can also include the entire VH and VL sequences as depicted in these sequences and figures, used as scFvs or as Fab domains.

(120) In one embodiment, the PSMA antigen binding domain includes the 6 CDRs (i.e., vhCDR1-3 and vlCDR1-3) of a PSMA ABD described herein, including the Figures and sequence listing. In some embodiments, the PSMA ABD includes the vhCDR1-3 of PSMA-H H1 (FIG. 17) and the vlCDR1-3 of a PSMA variable light domain selected from PSMA-H L1 (FIG. 17) and L1.1-L1.84 (FIG. 18A-E). In exemplary embodiments, the PSMA ABD is one of the following PSMA ABDs: PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; and PSMA-H H1_L1.13.

(121) In addition to the parental CDR sets disclosed in the figures and sequence listing that form an ABD to PSMA, provided herein are variant PSMA ABDs having CDRs that include at least one modification of the PSMA ABD CDRs disclosed herein. In one embodiment, the PSMA ABD includes a set of 6 CDRs with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 amino acid modifications as compared to the 6 CDRs of a PSMA ABD described herein, including the figures and sequence listing. In exemplary embodiments, the PSMA ABD includes a set of 6 CDRs with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 amino acid modifications as compared to the 6 CDRs of one of the following PSMA ABDs: PSMA ABDs: PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; and PSMA-H H1_L1.13. In certain embodiments, the variant PSMA ABD is capable of binding PSMA antigen, as measured by at least one of a Biacore, surface plasmon resonance (SPR) and/or BLI (biolayer interferometry, e.g., Octet assay) assay, with the latter finding particular use in many embodiments. In particular embodiments, the PSMA ABD is capable of binding human PSMA antigen (see Example 5).

(122) In one embodiment, the PSMA ABD includes 6 CDRs that are at least 90, 95, 97, 98 or 99% identical to the 6 CDRs of a PSMA ABD as described herein, including the figures and sequence listing. In exemplary embodiments, the PSMA ABD includes 6 CDRs that are at least 90, 95, 97, 98 or 99% identical to the 6 CDRs of one of the following PSMA ABDs: PSMA ABDs: PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; and PSMA-H H1_L1.13. In certain embodiments, the PSMA ABD is capable of binding to PSMA antigen, as measured by at least one of a Biacore, surface plasmon resonance (SPR) and/or BLI (biolayer interferometry, e.g., Octet

assay) assay, with the latter finding particular use in many embodiments. In particular embodiments, the PSMA ABD is capable of binding human PSMA antigen (see FIG. 2).

(123) In another exemplary embodiment, the PSMA ABD include the variable heavy (VH) domain and variable light (VL) domain of any one of the PSMA ABDs described herein, including the figures and sequence listing. In some embodiments, the PSMA ABD includes the PSMA-H H1 variable heavy domain (FIG. 17) and a variable light domain selected from PSMA-H L1 (FIG. 17) and L1.1-L1.84 (FIG. 18A-E). In exemplary embodiments, the PSMA ABD is one of the following PSMA ABDs: PSMA-H H1_L1 and PSMA-H H1_L1.1-L1.84 (FIGS. 17-19). In exemplary embodiments, the PSMA ABD is PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; or PSMA-H H1_L1.13 (FIGS. 17-19).

(124) In addition to the parental PSMA variable heavy and variable light domains disclosed herein, provided herein are PSMA ABDs that include a variable heavy domain and/or a variable light domain that are variants of a PSMA ABD VH and VL domain disclosed herein. In one embodiment, the variant VH domain and/or VL domain has from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid changes from a VH and/or VL domain of a PSMA ABD described herein, including the figures and sequence listing. In exemplary embodiments, the variant VH domain and/or VL domain has from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid changes from a VH and/or VL domain of one of the following PSMA-H H1_L1 and PSMA-H H1_L1.1-L1.84 (FIGS. 17-19). In exemplary embodiments, the PSMA ABD is PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; or PSMA-H H1_L1.13 (FIGS. 17-19). In certain embodiments, the PSMA ABD is capable of binding to PSMA, as measured at least one of a Biacore, surface plasmon resonance (SPR) and/or BLI (biolayer interferometry, e.g., Octet assay) assay, with the latter finding particular use in many embodiments. In particular embodiments, the PSMA ABD is capable of binding human PSMA antigen (see Example 5).

(125) In one embodiment, the variant VH and/or VL domain is at least 90, 95, 97, 98 or 99% identical to the VH and/or VL of a PSMA ABD as described herein, including the figures and sequence listing. In exemplary embodiments, the variant VH and/or VL domain is at least 90, 95, 97, 98 or 99% identical to the VH and/or VL of one of the following PSMA ABDs: PSMA-H H1_L1 and PSMA-H H1_L1.1-L1.84 (FIGS. 17-19). In exemplary embodiments, the PSMA ABD is PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; or PSMA-H H1_L1.13 (FIGS. 17-19). In certain embodiments, the PSMA ABD is capable of binding to the PSMA, as measured by at least one of a Biacore, surface plasmon resonance (SPR) and/or BLI (biolayer interferometry, e.g., Octet assay) assay, with the latter finding particular use in many embodiments. In particular embodiments, the PSMA ABD is capable of binding human PSMA antigen (see Example 5).

V. Antibodies

(126) In one aspect, provided herein are antibodies that bind to PSMA (e.g., anti-PSMA antibodies). In certain embodiments, the antibody binds to human PSMA (FIG. 11A). Subject anti-PSMA antibodies include monospecific PSMA antibodies, as well as multi-specific (e.g., bispecific) anti-PSMA antibodies. In certain embodiments, the anti-PSMA antibody has a format according to any one of the antibody formats depicted in FIGS. 21A and 21B.

(127) In some embodiments, the subject compositions include a PSMA binding domain. In some embodiments, the composition includes an antibody having a PSMA binding domain. Antibodies provided herein include one, two, three, four, and five or more PSMA binding domains. In certain embodiments, the PSMA binding domain includes any one of the vhCDR1, vhCDR2, vhCDR3,

vLCDR1, vLCDR2 and vLCDR3 sequences of an PSMA binding domain selected from those depicted in FIGS. 17-19. In some embodiments, the PSMA binding domain includes the underlined vhCDR1, vhCDR2, vhCDR3, vLCDR1, vLCDR2 and vLCDR3 sequences of a PSMA binding domain selected from those depicted in FIGS. 17-19. In some embodiments, the PSMA binding domain includes the variable heavy domain and variable light domain of a PSMA binding domain selected from those depicted in FIGS. 17-19. PSMA binding domains depicted in FIGS. 12, 13A-13B, and 14A-14I include: PSMA-H H1_L1 and PSMA-H H1_L1.1-L1.84.

(128) In one aspect, provided herein are bispecific antibodies that bind to PSMA and CD3, in various formats as outlined below, and generally depicted in FIGS. 21A and 21B. These bispecific, heterodimeric antibodies include a PSMA binding domain. In certain embodiments, the PSMA binding domain includes the VHCDR1, VHCDR2, VHCDR3, VLCDR1, VLCDR2 and VLCDR3 sequences of an PSMA binding domain selected from the group consisting of those depicted in FIGS. 17-19. In some embodiments, the PSMA binding domain includes the underlined VHCDR1, VHCDR2, VHCDR3, VLCDR1, VLCDR2 and VLCDR3 sequences of an PSMA binding domain selected from those depicted in FIGS. 17-19.

(129) These bispecific heterodimeric antibodies bind PSMA and CD3. Such antibodies include a CD3 binding domain and at least one PSMA binding domain. Any suitable PSMA binding domain can be included in the anti-PSMA×anti-CD3 bispecific antibody. In some embodiments, the anti-PSMA×anti-CD3 bispecific antibody includes one, two, three, four or more PSMA binding domains, including but not limited to those depicted in FIGS. 17-19. In certain embodiments, the anti-PSMA×anti-CD3 antibody includes an PSMA binding domain that includes the VHCDR1, VHCDR2, VHCDR3, VLCDR1, VLCDR2 and VLCDR3 sequences of an PSMA binding domain selected from the group consisting of PSMA-H H1_L1 and PSMA-H H1_L1.1-L1.84 (FIGS. 17-19). In some embodiments, the anti-PSMA×anti-CD3 antibody includes a PSMA binding domain that includes the underlined VHCDR1, VHCDR2, VHCDR3, VLCDR1, VLCDR2 and VLCDR3 sequences of an PSMA binding domain selected from the group consisting of PSMA-H H1_L1 and PSMA-H H1_L1.1-L1.84 (FIGS. 17-19). In some embodiments, the anti-PSMA×anti-CD3 antibody includes a PSMA binding domain that includes the variable heavy domain and variable light domain of an PSMA binding domain selected from the group consisting of PSMA-H H1_L1 and PSMA-H H1_L1.1-L1.84 (FIGS. 17-19).

(130) The anti-PSMA×anti-CD3 antibody provided herein can include any suitable CD3 binding domain. In certain embodiments, the anti-PSMA×anti-CD3 antibody includes a CD3 binding domain that includes the VHCDR1, VHCDR2, VHCDR3, VLCDR1, VLCDR2 and VLCDR3 sequences of a CD3 binding domain selected from the group consisting of those depicted in FIG. 10A-F. In some embodiments, the anti-PSMA×anti-CD3 antibody includes a CD3 binding domain that includes the underlined VHCDR1, VHCDR2, VHCDR3, VLCDR1, VLCDR2 and VLCDR3 sequences of a CD3 binding domain selected from the group consisting of those depicted in FIG. 10A-10F. In some embodiments, the anti-PSMA×anti-CD3 antibody includes a CD3 binding domain that includes the variable heavy domain and variable light domain of a CD3 binding domain selected from the group consisting of those depicted in FIG. 10A-10F. In some embodiments, the CD3 binding domain is selected from H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. 10A-10F). As outlined herein, these anti-CD3 antigen binding domains (CD3-ABDs) can be used in scFv formats in either orientation (e.g. from N- to C-terminal, VH-scFv linker-VL or VL-scFv linker-VH).

(131) The antibodies provided herein include different antibody domains. As described herein and known in the art, the antibodies described herein include different domains within the heavy and light chains, which can be overlapping as well. These domains include, but are not limited to, the Fc domain, the CH1 domain, the CH2 domain, the CH3 domain, the hinge domain, the heavy constant domain (CH1-hinge-Fc domain or CH1-hinge-CH2-CH3), the variable heavy domain, the

variable light domain, the light constant domain, Fab domains and scFv domains.

(132) As shown herein, there are a number of suitable linkers (for use as either domain linkers or scFv linkers) that can be used to covalently attach the recited domains (e.g., scFvs, Fabs, Fc domains, etc.), including traditional peptide bonds, generated by recombinant techniques.

Exemplary linkers to attach domains of the subject antibody to each other are depicted in FIG. 6. In some embodiments, the linker peptide may predominantly include the following amino acid residues: Gly, Ser, Ala, or Thr. The linker peptide should have a length that is adequate to link two molecules in such a way that they assume the correct conformation relative to one another so that they retain the desired activity. In one embodiment, the linker is from about 1 to 50 amino acids in length, preferably about 1 to 30 amino acids in length. In one embodiment, linkers of 1 to 20 amino acids in length may be used, with from about 5 to about 10 amino acids finding use in some embodiments. Useful linkers include glycine-serine polymers, including for example (GS)_n, (GSGGS)_n (SEQ ID NO: 3), (GGGGS)_n (SEQ ID NO: 627), and (GGGS)_n (SEQ ID NO: 4), where n is an integer of at least one (and generally from 3 to 4), glycine-alanine polymers, alanine-serine polymers, and other flexible linkers, some of which are shown in FIG. 5 and FIG. 6.

Alternatively, a variety of nonproteinaceous polymers, including but not limited to polyethylene glycol (PEG), polypropylene glycol, polyoxyalkylenes, or copolymers of polyethylene glycol and polypropylene glycol, may find use as linkers.

(133) Other linker sequences may include any sequence of any length of CL/CH1 domain but not all residues of CL/CH1 domain; for example the first 5-12 amino acid residues of the CL/CH1 domains. Linkers can be derived from immunoglobulin light chain, for example C_κ or C_λ. Linkers can be derived from immunoglobulin heavy chains of any isotype, including for example C_γ1, C_γ2, C_γ3, C_γ4, C_α1, C_α2, C_δ, C_ε, and C_μ. Linker sequences may also be derived from other proteins such as Ig-like proteins (e.g. TCR, FcR, KIR), hinge region-derived sequences, and other natural sequences from other proteins.

(134) In some embodiments, the linker is a “domain linker”, used to link any two domains as outlined herein together. For example, in FIG. 21B, there may be a domain linker that attaches the C-terminus of the CH1 domain of the Fab to the N-terminus of the scFv, with another optional domain linker attaching the C-terminus of the scFv to the CH2 domain (although in many embodiments the hinge is used as this domain linker). While any suitable linker can be used, many embodiments utilize a glycine-serine polymer as the domain linker, including for example (GS)_n, (GSGGS)_n (SEQ ID NO: 3), (GGGGS)_n (SEQ ID NO: 627), and (GGGS)_n (SEQ ID NO: 4), where n is an integer of at least one (and generally from 3 to 4 to 5) as well as any peptide sequence that allows for recombinant attachment of the two domains with sufficient length and flexibility to allow each domain to retain its biological function. In some cases, and with attention being paid to “strandedness”, as outlined below, charged domain linkers, as used in some embodiments of scFv linkers can be used. Exemplary useful domain linkers are depicted in FIG. 6.

(135) With particular reference to the domain linker used to attach the scFv domain to the Fc domain in the “2+1” format, there are several domain linkers that find particular use, including “full hinge C220S variant,” “flex half hinge,” “charged half hinge 1,” and “charged half hinge 2” as shown in FIG. 6.

(136) In some embodiments, the linker is a “scFv linker”, used to covalently attach the VH and VL domains as discussed herein. In many cases, the scFv linker is a charged scFv linker, a number of which are shown in FIG. 5. Accordingly, in some embodiments, the antibodies described herein further provide charged scFv linkers, to facilitate the separation in pI between a first and a second monomer. That is, by incorporating a charged scFv linker, either positive or negative (or both, in the case of scaffolds that use scFvs on different monomers), this allows the monomer comprising the charged linker to alter the pI without making further changes in the Fc domains. These charged linkers can be substituted into any scFv containing standard linkers. Again, as will be appreciated by those in the art, charged scFv linkers are used on the correct “strand” or monomer, according to

the desired changes in pI. For example, as discussed herein, to make 1+1 Fab-scFv-Fc format heterodimeric antibody, the original pI of the Fv region for each of the desired antigen binding domains are calculated, and one is chosen to make an scFv, and depending on the pI, either positive or negative linkers are chosen.

(137) Charged domain linkers can also be used to increase the pI separation of the monomers of the antibodies described herein as well, and thus those included in FIG. 5 can be used in any embodiment herein where a linker is utilized.

(138) In particular, the formats depicted in FIGS. 21A and 21B are antibodies, usually referred to as “heterodimeric antibodies”, meaning that the protein has at least two associated Fc sequences self-assembled into a heterodimeric Fc domain and at least two Fv regions, whether as Fabs or as scFvs.

(139) The PSMA binding domains provided can be included in any useful antibody format including, for example, canonical immunoglobulin, as well as the 1+1 Fab-scFv-Fc and 2+1 Fab2-scFv-Fv formats provided herein. Other useful antibody formats include, but are not limited to, “mAb-Fv,” “mAb-scFv,” “central-Fv,” “one armed scFv-mAb,” “scFv-mAb,” “dual scFv,” and “trident” format antibodies, as disclosed in FIGS. 50A-50K.

(140) In some embodiments, the subject antibody includes one or more of the PSMA ABDs provided herein. In some embodiments, the antibody includes one PSMA ABD. In other embodiments, the antibody includes two PSMA ABDs. In exemplary embodiments, the PSMA ABD includes the variable heavy domain and variable light domain of one of the following PSMA ABDs: PSMA-H H1_L1 and PSMA-H H1_L1.1-L1.84 (FIGS. 17-19). In some embodiments, the PSMA ABD is one of the following PSMA ABDs: PSMA-H H1_L1 and PSMA-H H1_L1.1-L1.84 (FIGS. 17-19).

(141) In an exemplary embodiment, the antibody is a bispecific antibody that includes one or two PSMA ABDs, including any of the PSMA ABDs provided herein. Bispecific antibody that include such PSMA ABDs include, for example, 1+1 Fab-scFv-Fc and 2+1 Fab.sub.2-scFv-Fc bispecifics format antibodies. In exemplary embodiments, the PSMA ABD is one of the following PSMA-H H1_L1 and PSMA-H H1_L1.1-L1.84 (FIGS. 17-19). In exemplary embodiments the PSMA binding domains is a Fab. In some embodiments, such bispecific antibodies are heterodimeric bispecific antibodies that include any of the heterodimerization skew variants, pI variants and/or ablation variants described herein.

(142) A. Chimeric and Humanized Antibodies

(143) In certain embodiments, the antibodies described herein comprise a heavy chain variable region from a particular germline heavy chain immunoglobulin gene and/or a light chain variable region from a particular germline light chain immunoglobulin gene. For example, such antibodies may comprise or consist of a human antibody comprising heavy or light chain variable regions that are “the product of” or “derived from” a particular germline sequence. A human antibody that is “the product of” or “derived from” a human germline immunoglobulin sequence can be identified as such by comparing the amino acid sequence of the human antibody to the amino acid sequences of human germline immunoglobulins and selecting the human germline immunoglobulin sequence that is closest in sequence (i.e., greatest % identity) to the sequence of the human antibody (using the methods outlined herein). A human antibody that is “the product of” or “derived from” a particular human germline immunoglobulin sequence may contain amino acid differences as compared to the germline sequence, due to, for example, naturally-occurring somatic mutations or intentional introduction of site-directed mutation. However, a humanized antibody typically is at least 90% identical in amino acids sequence to an amino acid sequence encoded by a human germline immunoglobulin gene and contains amino acid residues that identify the antibody as being derived from human sequences when compared to the germline immunoglobulin amino acid sequences of other species (e.g., murine germline sequences). In certain cases, a humanized antibody may be at least 95, 96, 97, 98 or 99%, or even at least 96%, 97%, 98%, or 99% identical in amino acid sequence to the amino acid sequence encoded by the germline immunoglobulin gene.

Typically, a humanized antibody derived from a particular human germline sequence will display no more than 10-20 amino acid differences from the amino acid sequence encoded by the human germline immunoglobulin gene (prior to the introduction of any skew, pI and ablation variants herein; that is, the number of variants is generally low, prior to the introduction of the variants described herein). In certain cases, the humanized antibody may display no more than 5, or even no more than 4, 3, 2, or 1 amino acid difference from the amino acid sequence encoded by the germline immunoglobulin gene (again, prior to the introduction of any skew, pI and ablation variants herein; that is, the number of variants is generally low, prior to the introduction of the variants described herein).

(144) In one embodiment, the parent antibody has been affinity matured, as is known in the art. Structure-based methods may be employed for humanization and affinity maturation, for example as described in U.S. Ser. No. 11/004,590. Selection based methods may be employed to humanize and/or affinity mature antibody variable regions, including but not limited to methods described in Wu et al., 1999, J. Mol. Biol. 294:151-162; Baca et al., 1997, J. Biol. Chem. 272(16):10678-10684; Rosok et al., 1996, J. Biol. Chem. 271(37): 22611-22618; Rader et al., 1998, Proc. Natl. Acad. Sci. USA 95: 8910-8915; Krauss et al., 2003, Protein Engineering 16(10):753-759, all entirely incorporated by reference. Other humanization methods may involve the grafting of only parts of the CDRs, including but not limited to methods described in U.S. Ser. No. 09/810,510; Tan et al., 2002, J. Immunol. 169:1119-1125; De Pascalis et al., 2002, J. Immunol. 169:3076-3084, all entirely incorporated by reference.

(145) B. Heterodimeric Antibodies

(146) In exemplary embodiments, the bispecific antibodies provided herein are heterodimeric bispecific antibodies that include two variant Fc domain sequences. Such variant Fc domains include amino acid modifications to facilitate the self-assembly and/or purification of the heterodimeric antibodies.

(147) An ongoing problem in antibody technologies is the desire for “bispecific” antibodies that bind to two different antigens simultaneously, in general thus allowing the different antigens to be brought into proximity and resulting in new functionalities and new therapies. In general, these antibodies are made by including genes for each heavy and light chain into the host cells. This generally results in the formation of the desired heterodimer (A-B), as well as the two homodimers (A-A and B-B (not including the light chain heterodimeric issues)). However, a major obstacle in the formation of bispecific antibodies is the difficulty in biasing the formation of the desired heterodimeric antibody over the formation of the homodimers and/or purifying the heterodimeric antibody away from the homodimers.

(148) There are a number of mechanisms that can be used to generate the subject heterodimeric antibodies. In addition, as will be appreciated by those in the art, these different mechanisms can be combined to ensure high heterodimerization. Amino acid modifications that facilitate the production and purification of heterodimers are collectively referred to generally as “heterodimerization variants.” As discussed below, heterodimerization variants include “skew” variants (e.g., the “knobs and holes” and the “charge pairs” variants described below) as well as “pI variants,” which allow purification of heterodimers from homodimers. As is generally described in U.S. Pat. No. 9,605,084, hereby incorporated by reference in its entirety and specifically as below for the discussion of heterodimerization variants, useful mechanisms for heterodimerization include “knobs and holes” (“KIH”) as described in U.S. Pat. No. 9,605,084, “electrostatic steering” or “charge pairs” as described in U.S. Pat. No. 9,605,084, pI variants as described in U.S. Pat. No. 9,605,084, and general additional Fc variants as outlined in U.S. Pat. No. 9,605,084 and below.

(149) Heterodimerization variants that are useful for the formation and purification of the subject heterodimeric antibody (e.g., bispecific antibodies) are further discussed in detailed below.

(150) 1. Skew Variants

(151) In some embodiments, the heterodimeric antibody includes skew variants which are one or

more amino acid modifications in a first Fc domain (A) and/or a second Fc domain (B) that favor the formation of Fc heterodimers (Fc dimers that include the first and the second Fc domain; (A-B) over Fc homodimers (Fc dimers that include two of the first Fc domain or two of the second Fc domain; A-A or B-B). Suitable skew variants are included in the FIG. 29 of US Publ. App. No. 2016/0355608, hereby incorporated by reference in its entirety and specifically for its disclosure of skew variants, as well as in FIGS. 1A-1E and FIG. 4.

(152) One mechanism is generally referred to in the art as “knobs and holes”, referring to amino acid engineering that creates steric influences to favor heterodimeric formation and disfavor homodimeric formation can also optionally be used; this is sometimes referred to as “knobs and holes”, as described in U.S. Ser. No. 61/596,846, Ridgway et al., Protein Engineering 9(7):617 (1996); Atwell et al., J. Mol. Biol. 1997 270:26; U.S. Pat. No. 8,216,805, all of which are hereby incorporated by reference in their entirety. The Figures identify a number of “monomer A-monomer B” pairs that rely on “knobs and holes”. In addition, as described in Merchant et al., Nature Biotech. 16:677 (1998), these “knobs and hole” mutations can be combined with disulfide bonds to skew formation to heterodimerization.

(153) An additional mechanism that finds use in the generation of heterodimers is sometimes referred to as “electrostatic steering” as described in Gunasekaran et al., J. Biol. Chem. 285(25):19637 (2010), hereby incorporated by reference in its entirety. This is sometimes referred to herein as “charge pairs”. In this embodiment, electrostatics are used to skew the formation towards heterodimerization. As those in the art will appreciate, these may also have an effect on pI, and thus on purification, and thus could in some cases also be considered pI variants. However, as these were generated to force heterodimerization and were not used as purification tools, they are classified as “steric variants”. These include, but are not limited to, D221E/P228E/L368E paired with D221R/P228R/K409R (e.g. these are “monomer corresponding sets) and C220E/P228E/368E paired with C220R/E224R/P228R/K409R.

(154) In some embodiments, the skew variants advantageously and simultaneously favor heterodimerization based on both the “knobs and holes” mechanism as well as the “electrostatic steering” mechanism. In some embodiments, the heterodimeric antibody includes one or more sets of such heterodimerization skew variants. These variants come in “pairs” of “sets”. That is, one set of the pair is incorporated into the first monomer and the other set of the pair is incorporated into the second monomer. It should be noted that these sets do not necessarily behave as “knobs in holes” variants, with a one-to-one correspondence between a residue on one monomer and a residue on the other. That is, these pairs of sets may instead form an interface between the two monomers that encourages heterodimer formation and discourages homodimer formation, allowing the percentage of heterodimers that spontaneously form under biological conditions to be over 90%, rather than the expected 50% (25% homodimer A/A:50% heterodimer A/B:25% homodimer B/B). Exemplary heterodimerization “skew” variants are depicted in FIG. 4. In exemplary embodiments, the heterodimeric antibody includes a S364K/E357Q:L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L; K370S:S364K/E357Q; or a T366S/L368A/Y407V:T366W (optionally including a bridging disulfide, T366S/L368A/Y407V/Y349C:T366W/S354C) “skew” variant amino acid substitution set. In an exemplary embodiment, the heterodimeric antibody includes a “S364K/E357Q:L368D/K370S” amino acid substitution set. In terms of nomenclature, the pair “S364K/E357Q:L368D/K370S” means that one of the monomers includes an Fc domain that includes the amino acid substitutions S364K and E357Q and the other monomer includes an Fc domain that includes the amino acid substitutions L368D and K370S; as above, the “strandedness” of these pairs depends on the starting pI.

(155) In some embodiments, the skew variants provided herein can be optionally and independently incorporated with any other modifications, including, but not limited to, other skew variants (see, e.g., in FIG. 37 of US Publ. App. No. 2012/0149876, herein incorporated by

reference, particularly for its disclosure of skew variants), pI variants, isotypic variants, FcRn variants, ablation variants, etc. into one or both of the first and second Fc domains of the heterodimeric antibody. Further, individual modifications can also independently and optionally be included or excluded from the subject the heterodimeric antibody.

(156) Additional monomer A and monomer B variants that can be combined with other variants, optionally and independently in any amount, such as pI variants outlined herein or other steric variants that are shown in FIG. 37 of US 2012/0149876, the figure and legend and SEQ ID NOs of which are incorporated expressly by reference herein.

(157) In some embodiments, the steric variants outlined herein can be optionally and independently incorporated with any pI variant (or other variants such as Fc variants, FcRn variants, etc.) into one or both monomers, and can be independently and optionally included or excluded from the proteins of the antibodies described herein.

(158) A list of suitable skew variants is found in FIGS. 1A-IE, with FIG. 4 showing some pairs of particular utility in many embodiments. Of particular use in many embodiments are the pairs of sets including, but not limited to, S364K/E357Q:L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L and K370S:S364K/E357Q. In terms of nomenclature, the pair “S364K/E357Q:L368D/K370S” means that one of the monomers has the double variant set S364K/E357Q and the other has the double variant set L368D/K370S.

(159) 2. pI (Isoelectric Point) Variants for Heterodimers

(160) In some embodiments, the heterodimeric antibody includes purification variants that advantageously allow for the separation of heterodimeric antibody (e.g., anti-PSMA×anti-CD3 bispecific antibody) from homodimeric proteins.

(161) There are several basic mechanisms that can lead to ease of purifying heterodimeric antibodies. For example, modifications to one or both of the antibody heavy chain monomers A and B such that each monomer has a different pI allows for the isoelectric purification of heterodimeric A-B antibody from monomeric A-A and B-B proteins. Alternatively, some scaffold formats, such as the “1+1 Fab-scFv-Fc” format and the “2+1 Fab.sub.2-scFv-Fc” format, also allows separation on the basis of size. As described above, it is also possible to “skew” the formation of heterodimers over homodimers using skew variants. Thus, a combination of heterodimerization skew variants and pI variants find particular use in the heterodimeric antibodies provided herein.

(162) Additionally, as more fully outlined below, depending on the format of the heterodimeric antibody, pI variants either contained within the constant region and/or Fc domains of a monomer, and/or domain linkers can be used. In some embodiments, the heterodimeric antibody includes additional modifications for alternative functionalities that can also create pI changes, such as Fc, FcRn and KO variants.

(163) In some embodiments, the subject heterodimeric antibodies provided herein include at least one monomer with one or more modifications that alter the pI of the monomer (i.e., a “pI variant”). In general, as will be appreciated by those in the art, there are two general categories of pI variants: those that increase the pI of the protein (basic changes) and those that decrease the pI of the protein (acidic changes). As described herein, all combinations of these variants can be done: one monomer may be wild type, or a variant that does not display a significantly different pI from wild-type, and the other can be either more basic or more acidic. Alternatively, each monomer is changed, one to more basic and one to more acidic.

(164) Depending on the format of the heterodimer antibody, pI variants can be either contained within the constant and/or Fc domains of a monomer, or charged linkers, either domain linkers or scFv linkers, can be used. That is, antibody formats that utilize scFv(s) such as “1+1 Fab-scFv-Fc”, format can include charged scFv linkers (either positive or negative), that give a further pI boost for purification purposes. As will be appreciated by those in the art, some 1+1 Fab-scFv-Fc formats are useful with just charged scFv linkers and no additional pI adjustments, although the antibodies

described herein do provide pI variants that are on one or both of the monomers, and/or charged domain linkers as well. In addition, additional amino acid engineering for alternative functionalities may also confer pI changes, such as Fc, FcRn and KO variants.

(165) In subject heterodimeric antibodies that utilizes pI as a separation mechanism to allow the purification of heterodimeric proteins, amino acid variants are introduced into one or both of the monomer polypeptides. That is, the pI of one of the monomers (referred to herein for simplicity as “monomer A”) can be engineered away from monomer B, or both monomer A and B change be changed, with the pI of monomer A increasing and the pI of monomer B decreasing. As is outlined more fully below, the pI changes of either or both monomers can be done by removing or adding a charged residue (e.g., a neutral amino acid is replaced by a positively or negatively charged amino acid residue, e.g., glycine to glutamic acid), changing a charged residue from positive or negative to the opposite charge (aspartic acid to lysine) or changing a charged residue to a neutral residue (e.g., loss of a charge; lysine to serine). A number of these variants are shown in the FIGS. 3 and 4.

(166) Thus, in some embodiments, the subject heterodimeric antibody includes amino acid modifications in the constant regions that alter the isoelectric point (pI) of at least one, if not both, of the monomers of a dimeric protein to form “pI antibodies”) by incorporating amino acid substitutions (“pI variants” or “pI substitutions”) into one or both of the monomers. As shown herein, the separation of the heterodimers from the two homodimers can be accomplished if the pIs of the two monomers differ by as little as 0.1 pH unit, with 0.2, 0.3, 0.4 and 0.5 or greater all finding use in the antibodies described herein.

(167) As will be appreciated by those in the art, the number of pI variants to be included on each or both monomer(s) to get good separation will depend in part on the starting pI of the components, for example in the 1+1 Fab-scFv-Fc and 2+1 Fab.sub.2-scFv-Fc formats, the starting pI of the scFv and Fab(s) of interest. That is, to determine which monomer to engineer or in which “direction” (e.g., more positive or more negative), the Fv sequences of the two target antigens are calculated and a decision is made from there. As is known in the art, different Fvs will have different starting pIs which are exploited in the antibodies described herein. In general, as outlined herein, the pIs are engineered to result in a total pI difference of each monomer of at least about 0.1 logs, with 0.2 to 0.5 being preferred as outlined herein.

(168) In the case where pI variants are used to achieve heterodimerization, by using the constant region(s) of the heavy chain(s), a more modular approach to designing and purifying bispecific proteins, including antibodies, is provided. Thus, in some embodiments, heterodimerization variants (including skew and pI heterodimerization variants) are not included in the variable regions, such that each individual antibody must be engineered. In addition, in some embodiments, the possibility of immunogenicity resulting from the pI variants is significantly reduced by importing pI variants from different IgG isotypes such that pI is changed without introducing significant immunogenicity. Thus, an additional problem to be solved is the elucidation of low pI constant domains with high human sequence content, e.g., the minimization or avoidance of non-human residues at any particular position. Alternatively or in addition to isotypic substitutions, the possibility of immunogenicity resulting from the pI variants is significantly reduced by utilizing isosteric substitutions (e.g. Asn to Asp; and Gln to Glu).

(169) As discussed below, a side benefit that can occur with this pI engineering is also the extension of serum half-life and increased FcRn binding. That is, as described in US Publ. App. No. US 2012/0028304 (incorporated by reference in its entirety), lowering the pI of antibody constant domains (including those found in antibodies and Fc fusions) can lead to longer serum retention in vivo. These pI variants for increased serum half-life also facilitate pI changes for purification.

(170) In addition, it should be noted that the pI variants give an additional benefit for the analytics and quality control process of bispecific antibodies, as the ability to either eliminate, minimize and distinguish when homodimers are present is significant. Similarly, the ability to reliably test the

reproducibility of the heterodimeric antibody production is important.

(171) In general, embodiments of particular use rely on sets of variants that include skew variants, which encourage heterodimerization formation over homodimerization formation, coupled with pI variants, which increase the pI difference between the two monomers to facilitate purification of heterodimers away from homodimers.

(172) Exemplary combinations of pI variants are shown in FIGS. 4 and 5, and FIG. 30 of US Publ. App. No. 2016/0355608, all of which are herein incorporated by reference in its entirety and specifically for the disclosure of pI variants. Preferred combinations of pI variants are shown in FIGS. 1 and 2. As outlined herein and shown in the figures, these changes are shown relative to IgG1, but all isotypes can be altered this way, as well as isotype hybrids. In the case where the heavy chain constant domain is from IgG2-4, R133E and R133Q can also be used.

(173) In one embodiment, a preferred combination of pI variants has one monomer (the negative Fab side) comprising 208D/295E/384D/418E/421D variants

(N208D/Q295E/N384D/Q418E/N421D when relative to human IgG1) and a second monomer (the positive scFv side) comprising a positively charged scFv linker, including (GKPGS).sub.4 (SEQ ID NO: 1). However, as will be appreciated by those in the art, the first monomer includes a CH1 domain, including position 208. Accordingly, in constructs that do not include a CH1 domain (for example for antibodies that do not utilize a CH1 domain on one of the domains), a preferred negative pI variant Fc set includes 295E/384D/418E/421D variants (Q295E/N384D/Q418E/N421D when relative to human IgG1).

(174) Accordingly, in some embodiments, one monomer has a set of substitutions from FIG. 2 and the other monomer has a charged linker (either in the form of a charged scFv linker because that monomer comprises an scFv or a charged domain linker, as the format dictates, which can be selected from those depicted in FIG. 5).

(175) In some embodiments, modifications are made in the hinge of the Fc domain, including positions 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, and 230 based on EU numbering. Thus, pI mutations and particularly substitutions can be made in one or more of positions 216-230, with 1, 2, 3, 4 or 5 mutations finding use. Again, all possible combinations are contemplated, alone or with other pI variants in other domains.

(176) Specific substitutions that find use in lowering the pI of hinge domains include, but are not limited to, a deletion at position 221, a non-native valine or threonine at position 222, a deletion at position 223, a non-native glutamic acid at position 224, a deletion at position 225, a deletion at position 235 and a deletion or a non-native alanine at position 236. In some cases, only pI substitutions are done in the hinge domain, and in others, these substitution(s) are added to other pI variants in other domains in any combination.

(177) In some embodiments, mutations can be made in the CH2 region, including positions 233, 234, 235, 236, 274, 296, 300, 309, 320, 322, 326, 327, 334 and 339, based on EU numbering. It should be noted that changes in 233-236 can be made to increase effector function (along with 327A) in the IgG2 backbone. Again, all possible combinations of these 14 positions can be made; e.g., may include a variant Fc domain with 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 CH2 pI substitutions.

(178) Specific substitutions that find use in lowering the pI of CH2 domains include, but are not limited to, a non-native glutamine or glutamic acid at position 274, a non-native phenylalanine at position 296, a non-native phenylalanine at position 300, a non-native valine at position 309, a non-native glutamic acid at position 320, a non-native glutamic acid at position 322, a non-native glutamic acid at position 326, a non-native glycine at position 327, a non-native glutamic acid at position 334, a non-native threonine at position 339, and all possible combinations within CH2 and with other domains.

(179) In this embodiment, the modifications can be independently and optionally selected from position 355, 359, 362, 384, 389, 392, 397, 418, 419, 444 and 447 (EU numbering) of the CH3 region. Specific substitutions that find use in lowering the pI of CH3 domains include, but are not

limited to, a non-native glutamine or glutamic acid at position 355, a non-native serine at position 384, a non-native asparagine or glutamic acid at position 392, a non-native methionine at position 397, a non-native glutamic acid at position 419, a non-native glutamic acid at position 359, a non-native glutamic acid at position 362, a non-native glutamic acid at position 389, a non-native glutamic acid at position 418, a non-native glutamic acid at position 444, and a deletion or non-native aspartic acid at position 447.

(180) In general, as will be appreciated by those in the art, there are two general categories of pI variants: those that increase the pI of the protein (basic changes) and those that decrease the pI of the protein (acidic changes). As described herein, all combinations of these variants can be done: one monomer may be wild type, or a variant that does not display a significantly different pI from wild-type, and the other can be either more basic or more acidic. Alternatively, each monomer is changed, one to more basic and one to more acidic.

(181) Preferred combinations of pI variants are shown in FIG. 4. As outlined herein and shown in the figures, these changes are shown relative to IgG1, but all isotypes can be altered this way, as well as isotype hybrids. In the case where the heavy chain constant domain is from IgG2-4, R133E and R133Q can also be used.

(182) In one embodiment, for example in the FIGS. 21A and 21B formats, a preferred combination of pI variants has one monomer (the negative Fab side) comprising 208D/295E/384D/418E/421D variants (N208D/Q295E/N384D/Q418E/N421D when relative to human IgG1) and a second monomer (the positive scFv side) comprising a positively charged scFv linker, including (GKPGS).sub.4 (SEQ ID NO: 1). However, as will be appreciated by those in the art, the first monomer includes a CH1 domain, including position 208. Accordingly, in constructs that do not include a CH1 domain (for example for antibodies that do not utilize a CH1 domain on one of the domains, for example in a dual scFv format or a “one armed” format such as those depicted in FIG. 42B, C or D), a preferred negative pI variant Fc set includes 295E/384D/418E/421D variants (Q295E/N384D/Q418E/N421D when relative to human IgG1).

(183) Accordingly, in some embodiments, one monomer has a set of substitutions from FIG. 4 and the other monomer has a charged linker (either in the form of a charged scFv linker because that monomer comprises an scFv or a charged domain linker, as the format dictates, which can be selected from those depicted in FIG. 5).

(184) 3. Isotypic Variants

(185) In addition, many embodiments of the antibodies described herein rely on the “importation” of pI amino acids at particular positions from one IgG isotype into another, thus reducing or eliminating the possibility of unwanted immunogenicity being introduced into the variants. A number of these are shown in FIG. 21 of US Publ. 2014/0370013, hereby incorporated by reference. That is, IgG1 is a common isotype for therapeutic antibodies for a variety of reasons, including high effector function. However, the heavy constant region of IgG1 has a higher pI than that of IgG2 (8.10 versus 7.31). By introducing IgG2 residues at particular positions into the IgG1 backbone, the pI of the resulting monomer is lowered (or increased) and additionally exhibits longer serum half-life. For example, IgG1 has a glycine (pI 5.97) at position 137, and IgG2 has a glutamic acid (pI 3.22); importing the glutamic acid will affect the pI of the resulting protein. As is described below, a number of amino acid substitutions are generally required to significantly affect the pI of the variant antibody. However, it should be noted as discussed below that even changes in IgG2 molecules allow for increased serum half-life.

(186) In other embodiments, non-isotypic amino acid changes are made, either to reduce the overall charge state of the resulting protein (e.g. by changing a higher pI amino acid to a lower pI amino acid), or to allow accommodations in structure for stability, etc. as is more further described below.

(187) In addition, by pI engineering both the heavy and light constant domains, significant changes in each monomer of the heterodimer can be seen. As discussed herein, having the pIs of the two monomers differ by at least 0.5 can allow separation by ion exchange chromatography or

isoelectric focusing, or other methods sensitive to isoelectric point.

(188) 4. Calculating pI

(189) The pI of each monomer can depend on the pI of the variant heavy chain constant domain and the pI of the total monomer, including the variant heavy chain constant domain and the fusion partner. Thus, in some embodiments, the change in pI is calculated on the basis of the variant heavy chain constant domain, using the chart in the FIG. 19 of US Pub. 2014/0370013. As discussed herein, which monomer to engineer is generally decided by the inherent pI of the Fv and scaffold regions. Alternatively, the pI of each monomer can be compared.

(190) 5. pI Variants that Also Confer Better FcRn In Vivo Binding

(191) In the case where the pI variant decreases the pI of the monomer, they can have the added benefit of improving serum retention in vivo.

(192) Although still under examination, Fc regions are believed to have longer half-lives in vivo, because binding to FcRn at pH 6 in an endosome sequesters the Fc (Ghetie and Ward, 1997 Immunol Today. 18(12): 592-598, entirely incorporated by reference). The endosomal compartment then recycles the Fc to the cell surface. Once the compartment opens to the extracellular space, the higher pH, ~7.4, induces the release of Fc back into the blood. In mice, Dall'Acqua et al. showed that Fc mutants with increased FcRn binding at pH 6 and pH 7.4 actually had reduced serum concentrations and the same half-life as wild-type Fc (Dall'Acqua et al. 2002, J. Immunol. 169:5171-5180, entirely incorporated by reference). The increased affinity of Fc for FcRn at pH 7.4 is thought to forbid the release of the Fc back into the blood. Therefore, the Fc mutations that will increase Fc's half-life in vivo will ideally increase FcRn binding at the lower pH while still allowing release of Fc at higher pH. The amino acid histidine changes its charge state in the pH range of 6.0 to 7.4. Therefore, it is not surprising to find His residues at important positions in the Fc/FcRn complex.

(193) Recently it has been suggested that antibodies with variable regions that have lower isoelectric points may also have longer serum half-lives (Igawa et al., 2010 PEDS. 23(5): 385-392, entirely incorporated by reference). However, the mechanism of this is still poorly understood. Moreover, variable regions differ from antibody to antibody. Constant region variants with reduced pI and extended half-life would provide a more modular approach to improving the pharmacokinetic properties of antibodies, as described herein.

(194) C. Additional Fc Variants for Additional Functionality

(195) In addition to the heterodimerization variants discussed above, there are a number of useful Fc amino acid modifications that can be made for a variety of reasons, including, but not limited to, altering binding to one or more FcγR receptors, altered binding to FcRn receptors, etc, as discussed below.

(196) Accordingly, the antibodies provided herein (heterodimeric, as well as homodimeric) can include such amino acid modifications with or without the heterodimerization variants outlined herein (e.g., the pI variants and steric variants). Each set of variants can be independently and optionally included or excluded from any particular heterodimeric protein.

(197) 1. FcγR Variants

(198) Accordingly, there are a number of useful Fc substitutions that can be made to alter binding to one or more of the FcγR receptors. In certain embodiments, the subject antibody includes modifications that alter the binding to one or more FcγR receptors (i.e., "FcγR variants"). Substitutions that result in increased binding as well as decreased binding can be useful. For example, it is known that increased binding to FcγRIIIa generally results in increased ADCC (antibody dependent cell-mediated cytotoxicity; the cell-mediated reaction wherein nonspecific cytotoxic cells that express FcγRs recognize bound antibody on a target cell and subsequently cause lysis of the target cell). Similarly, decreased binding to FcγRIIb (an inhibitory receptor) can be beneficial as well in some circumstances. Amino acid substitutions that find use in the antibodies described herein include those listed in U.S. Pat. No. 8,188,321 (particularly FIG. 41)

and U.S. Pat. No. 8,084,582, and US Publ. App. Nos. 20060235208 and 20070148170, all of which are expressly incorporated herein by reference in their entirety and specifically for the variants disclosed therein. Particular variants that find use include, but are not limited to, 236A, 239D, 239E, 332E, 332D, 239D/332E, 267D, 267E, 328F, 267E/328F, 236A/332E, 239D/332E/330Y, 239D/332E/330L, 243A, 243L, 264A, 264V and 299T.

(199) In addition, there are additional Fc substitutions that find use in increased binding to the FcRn receptor and increased serum half-life, as specifically disclosed in U.S. Ser. No. 12/341,769, hereby incorporated by reference in its entirety, including, but not limited to, 434S, 434A, 428L, 308F, 259I, 428L/434S, 259I/308F, 436I/428L, 436I or V/434S, 436V/428L and 259I/308F/428L. Such modification may be included in one or both Fc domains of the subject antibody.

(200) 2. Ablation Variants

(201) Similarly, another category of functional variants are “FcγR ablation variants” or “Fc knock out (FcKO or KO)” variants. In these embodiments, for some therapeutic applications, it is desirable to reduce or remove the normal binding of the Fc domain to one or more or all of the Fcγ receptors (e.g. FcγR1, FcγRIIa, FcγRIIb, FcγRIIIa, etc.) to avoid additional mechanisms of action. That is, for example, in many embodiments, particularly in the use of bispecific antibodies that bind CD3 monovalently it is generally desirable to ablate FcγRIIIa binding to eliminate or significantly reduce ADCC activity. wherein one of the Fc domains comprises one or more Fcγ receptor ablation variants. These ablation variants are depicted in FIG. 14, and each can be independently and optionally included or excluded, with preferred aspects utilizing ablation variants selected from the group consisting of G236R/L328R, E233P/L234V/L235A/G236del/S239K, E233P/L234V/L235A/G236del/S267K, E233P/L234V/L235A/G236del/S239K/A327G, E233P/L234V/L235A/G236del/S267K/A327G and E233P/L234V/L235A/G236del. It should be noted that the ablation variants referenced herein ablate FcγR binding but generally not FcRn binding.

(202) As is known in the art, the Fc domain of human IgG1 has the highest binding to the Fcγ receptors, and thus ablation variants can be used when the constant domain (or Fc domain) in the backbone of the heterodimeric antibody is IgG1. Alternatively, or in addition to ablation variants in an IgG1 background, mutations at the glycosylation position 297 (generally to A or S) can significantly ablate binding to FcγRIIIa, for example. Human IgG2 and IgG4 have naturally reduced binding to the Fc receptors, and thus those backbones can be used with or without the ablation variants.

(203) D. Combination of Heterodimeric and Fc Variants

(204) As will be appreciated by those in the art, all of the recited heterodimerization variants (including skew and/or pI variants) can be optionally and independently combined in any way, as long as they retain their “strandedness” or “monomer partition”. In some embodiments, the heterodimeric antibodies provided herein include the combination of heterodimerization skew variants, isosteric pI substitutions and FcKO variants as depicted in FIG. 4. In addition, all of these variants can be combined into any of the heterodimerization formats.

(205) In the case of pI variants, while embodiments finding particular use are shown in the Figures, other combinations can be generated, following the basic rule of altering the pI difference between two monomers to facilitate purification.

(206) In addition, any of the heterodimerization variants, skew and pI, are also independently and optionally combined with Fc ablation variants, Fc variants, FcRn variants, as generally outlined herein.

(207) Exemplary combination of variants that are included in some embodiments of the heterodimeric 1+1 Fab-scFv-Fc and 2+1 Fab.sub.2-scFv-Fc format antibodies are included in FIG. 4. In certain embodiments, the antibody is a heterodimeric 1+1 Fab-scFv-Fc or 2+1 Fab.sub.2-scFv-Fc format antibody as shown in FIGS. 21A and 21B.

(208) E. Anti-PSMA×Anti-CD3 Bispecific Antibodies

(209) In another aspect, provided herein are anti-PSMA $\times$ anti-CD3 (also referred to herein as “ α PSMA $\times$ α CD3”) bispecific antibodies. Such antibodies include at least one PSMA binding domain and at least one CD3 binding domain. In some embodiments, bispecific α PSMA $\times$ α CD3 provided herein immune responses selectively in tumor sites that express PSMA.

(210) Note that unless specified herein, the order of the antigen list in the name does not confer structure; that is a PSMA $\times$ CD3 1+1 Fab-scFv-Fc antibody can have the scFv bind to PSMA or CD3, although in some cases, the order specifies structure as indicated.

(211) As is more fully outlined herein, these combinations of ABDs can be in a variety of formats, as outlined below, generally in combinations where one ABD is in a Fab format and the other is in an scFv format. Exemplary formats that are used in the bispecific antibodies provided herein include the 1+1 Fab-scFv-Fc and 2+1 Fab2-scFv-Fv formats (see, e.g., FIGS. 15A and 15B). Other useful antibody formats include, but are not limited to, “mAb-Fv,” “mAb-scFv,” “central-Fv”, “one armed scFv-mAb,” “scFv-mAb,” “dual scFv,” and “trident” format antibodies, as disclosed in FIG. 50A-50K.

(212) In addition, in general, one of the ABDs comprises a scFv as outlined herein, in an orientation from N- to C-terminus of VH-scFv linker-VL or VL-scFv linker-VH. One or both of the other ABDs, according to the format, generally is a Fab, comprising a VH domain on one protein chain (generally as a component of a heavy chain) and a VL on another protein chain (generally as a component of a light chain).

(213) As will be appreciated by those in the art, any set of 6 CDRs or VH and VL domains can be in the scFv format or in the Fab format, which is then added to the heavy and light constant domains, where the heavy constant domains comprise variants (including within the CH1 domain as well as the Fc domain). The scFv sequences contained in the sequence listing utilize a particular charged linker, but as outlined herein, uncharged or other charged linkers can be used, including those depicted in FIG. 5 and FIG. 6.

(214) In addition, as discussed above, the numbering used in the Sequence Listing for the identification of the CDRs is Kabat, however, different numbering can be used, which will change the amino acid sequences of the CDRs as shown in Table 2.

(215) For all of the variable heavy and light domains listed herein, further variants can be made. As outlined herein, in some embodiments the set of 6 CDRs can have from 0, 1, 2, 3, 4 or 5 amino acid modifications (with amino acid substitutions finding particular use), as well as changes in the framework regions of the variable heavy and light domains, as long as the frameworks (excluding the CDRs) retain at least about 80, 85 or 90% identity to a human germline sequence selected from those listed in FIG. 1 of U.S. Pat. No. 7,657,380, which Figure and Legend is incorporated by reference in its entirety herein. Thus, for example, the identical CDRs as described herein can be combined with different framework sequences from human germline sequences, as long as the framework regions retain at least 80, 85 or 90% identity to a human germline sequence selected from those listed in FIG. 1 of U.S. Pat. No. 7,657,380. Alternatively, the CDRs can have amino acid modifications (e.g., from 1, 2, 3, 4 or 5 amino acid modifications in the set of CDRs (that is, the CDRs can be modified as long as the total number of changes in the set of 6 CDRs is less than 6 amino acid modifications, with any combination of CDRs being changed; e.g., there may be one change in vlCDR1, two in vhCDR2, none in vhCDR3, etc.)), as well as having framework region changes, as long as the framework regions retain at least 80, 85 or 90% identity to a human germline sequence selected from those listed in FIG. 1 of U.S. Pat. No. 7,657,380.

(216) The anti-PSMA $\times$ anti-CD3 bispecific antibody can include any suitable CD3 ABD, including those described herein (see, e.g., FIGS. 10A-10F). In some embodiments, the CD3 ABD of the anti-PSMA $\times$ anti-CD3 bispecific antibody includes the variable heavy domain and variable light domain of a CD3 ABD provided herein, including those described in FIGS. 10A-10F and the sequence listing. In some embodiments, the CD3 ABD includes the variable heavy domain and variable light domain of one of the following CD3 ABDs: H1.30_L1.47, H1.32_L1.47,

H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. **10A-10F**). In exemplary embodiments, the CD3 ABD is one of the following CD3 ABDs: H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. **10A-10F**) or a variant thereof. The anti-PSMA×anti-CD3 bispecific antibody can include any suitable PSMA ABD, including those described herein (see, e.g., FIGS. **12**, **13A-13B**, and **14A-14I**). In some embodiments, the PSMA ABD of the anti-PSMA×anti-CD3 bispecific antibody includes the variable heavy domain and variable light domain of a PSMA ABD provided herein, including those described in FIGS. **12**, **13A-13B**, and **14A-14I** and the sequence listing. In some embodiments, the PSMA ABD includes the variable heavy domain and variable light domain of one of the following PSMA ABDs: PSMA-H L1 and L1.1-L1.84 (FIGS. **17** and **18A-18E**). In exemplary embodiments, the PSMA ABD is one of the following PSMA ABDs: PSMA ABDs: PSMA-H L1 and L1.1-L1.84 (FIGS. **17** and **18A-18E**) or variants thereof.

(217) F. Useful Formats

(218) As will be appreciated by those in the art and discussed more fully below, the heterodimeric bispecific antibodies provided herein can take on a wide variety of configurations, as are generally depicted in FIG. **1**. Some figures depict “single ended” configurations, where there is one type of specificity on one “arm” of the molecule and a different specificity on the other “arm”. Other figures depict “dual ended” configurations, where there is at least one type of specificity at the “top” of the molecule and one or more different specificities at the “bottom” of the molecule. Thus, in some embodiments, the antibodies described herein are directed to novel immunoglobulin compositions that co-engage a different first and a second antigen.

(219) As will be appreciated by those in the art, the heterodimeric formats of the antibodies described herein can have different valencies as well as be bispecific. That is, heterodimeric antibodies of the antibodies described herein can be bivalent and bispecific, wherein one target tumor antigen (e.g. CD3) is bound by one binding domain and the other target tumor antigen (e.g. PSMA) is bound by a second binding domain. The heterodimeric antibodies can also be trivalent and bispecific, wherein the first antigen is bound by two binding domains and the second antigen by a second binding domain. As is outlined herein, when CD3 is one of the target antigens, it is preferable that the CD3 is bound only monovalently, to reduce potential side effects.

(220) The antibodies described herein utilize anti-CD3 antigen binding domains in combination with anti-PSMA binding domains. As will be appreciated by those in the art, any collection of anti-CD3 CDRs, anti-CD3 variable light and variable heavy domains, Fabs and scFvs as depicted in any of the Figures can be used. Similarly, any of the anti-PSMA antigen binding domains can be used, whether CDRs, variable light and variable heavy domains, Fabs and scFvs as depicted in any of the Figures (e.g., FIGS. **17**, **18A-E** and **19A-X**) can be used, optionally and independently combined in any combination.

(221) 1. 1+1 Fab-scFv-Fc Format

(222) One heterodimeric scaffold that finds particular use in the antibodies described herein is the “1+1 Fab-scFv-Fc” or “bottle-opener” format as shown in FIG. **21A** with an exemplary combination of a CD3 binding domain and a tumor target antigen (PSMA) binding domain. In this embodiment, one heavy chain monomer of the antibody contains a single chain Fv (“scFv”, as defined below) and an Fc domain. The scFv includes a variable heavy domain (VH1) and a variable light domain (VL1), wherein the VH1 is attached to the VL1 using an scFv linker that can be charged (see, e.g., FIG. **5**). The scFv is attached to the heavy chain using a domain linker (see, e.g., FIG. **6**). The other heavy chain monomer is a “regular” heavy chain (VH-CH1-hinge-CH2-CH3). The 1+1 Fab-scFv-Fc also includes a light chain that interacts with the VH-CH1 to form a Fab. This structure is sometimes referred to herein as the “bottle-opener” format, due to a rough visual similarity to a bottle-opener. The two heavy chain monomers are brought together by the use

of amino acid variants (e.g., heterodimerization variants, discussed above) in the constant regions (e.g., the Fc domain, the CH1 domain and/or the hinge region) that promote the formation of heterodimeric antibodies as is described more fully below.

(223) There are several distinct advantages to the present “1+1 Fab-scFv-Fc” format. As is known in the art, antibody analogs relying on two scFv constructs often have stability and aggregation problems, which can be alleviated in the antibodies described herein by the addition of a “regular” heavy and light chain pairing. In addition, as opposed to formats that rely on two heavy chains and two light chains, there is no issue with the incorrect pairing of heavy and light chains (e.g. heavy 1 pairing with light 2, etc.).

(224) Many of the embodiments outlined herein rely in general on the 1+1 Fab-scFv-Fc or “bottle opener” format antibody that comprises a first monomer comprising an scFv, comprising a variable heavy and a variable light domain, covalently attached using an scFv linker (charged, in many but not all instances), where the scFv is covalently attached to the N-terminus of a first Fc domain usually through a domain linker (i.e., from N- to C-terminus scFv-linker-CH2-C3). In some embodiments, the variable light domain of the scFv is attached to the first Fc domain. In other embodiments, the variable heavy domain of the scFv is attached to the first Fc domain. The domain linker can be either charged or uncharged and exogenous or endogenous (e.g., all or part of the native hinge domain). Any suitable linker can be used to attach the scFv to the N-terminus of the first Fc domain. In some embodiments, the domain linker is chosen from the domain linkers in FIG. 6. The second monomer of the 1+1 Fab-scFv-Fc format or “bottle opener” format is a heavy chain (i.e., from N- to C-terminus VH-CH1-hinge-CH2-CH3), and the composition further comprises a light chain.

(225) In addition, the Fc domains of the antibodies described herein generally include skew variants (e.g. a set of amino acid substitutions as shown in FIGS. 1 and 4, with particularly useful skew variants being selected from the group consisting of S364K/E357Q:L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L; K370S:S364K/E357Q; T366S/L368A/Y407V:T366W and T366S/L368A/Y407V/Y349C:T366W/S354C), optionally ablation variants (including those shown in FIG. 3), optionally charged scFv linkers (including those shown in FIG. 5) and the heavy chain comprises pI variants (including those shown in FIG. 2).

(226) In general, in many preferred embodiments, the scFv is the domain that binds to the CD3, and the Fab forms a PSMA binding domain. An exemplary anti-PSMA×anti-CD3 bispecific antibody in the 1+1 Fab-scFv-Fc format is depicted in FIG. 21A. Exemplary anti-PSMA×anti-CD3 bispecific antibody in the 1+1 Fab-scFv-Fc format is depicted in FIGS. 22-25. Exemplary variable heavy and light domains of the scFv that binds to CD3 are included in FIG. 10A-10F. Exemplary variable heavy and light domains of the Fv that binds to PSMA are included in FIGS. 17 and 18. In an exemplary embodiment, the PSMA binding domain of the 1+1 Fab-scFv-Fc PSMA×CD3 bispecific antibody includes the VH of PSMA-H H1 (FIG. 17) and VL of one of the following PSMA binding domains: PSMA-H L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E). In one embodiment, the CD3 binding domain of the 1+1 Fab-scFv-Fc PSMA×CD3 bispecific antibody includes the VH and VL of one of the following CD3 binding domains: H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. 10A-10F). Particularly useful PSMA and CD3 combinations for use in the 1+1 Fab-scFv-Fc PSMA×CD3 bispecific antibody format are disclosed in FIGS. 22-25 and include: a) CD3H1.30_L1.47×PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; or PSMA-H H1_L1.13; b) CD3H1.31_L1.47, CD3H1.32_L1.47, or CD3H1.33_L1.47×PSMA-H HILL.

(227) In certain embodiments, the 1+1 Fab-scFv-Fc scaffold format includes a first monomer that

includes a scFv-domain linker-CH2-CH3 monomer, a second monomer that includes a first variable heavy domain-CH1-hinge-CH2-CH3 monomer and a third monomer that includes a first variable light domain. In some embodiments, the CH2-CH3 of the first monomer is a first variant Fc domain and the CH2-CH3 of the second monomer is a second variant Fc domain. In some embodiments, the scFv includes a scFv variable heavy domain and a scFv variable light domain that form a CD3 binding moiety. In certain embodiments, the scFv variable heavy domain and scFv variable light domain are covalently attached using an scFv linker (charged, in many but not all instances. See, e.g., FIG. 5). In some embodiments, the first variable heavy domain and first variable light domain form a PSMA binding domain. CD3 binding domain sequences finding particular use in these embodiments include, but are not limited to, H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (see FIGS. **10A-10F**). PSMA binding domain sequences that are of particular use in these embodiments include, but are not limited to, PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; and PSMA-H H1_L1.13. Particularly useful PSMA and CD3 combinations for use in the 1+1 Fab-scFv-Fc PSMA×CD3 bispecific antibody format are disclosed in FIGS. **22-25** and include: a) CD3H1.30_L1.47×PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; or PSMA-H H1_L1.13; b) CD3H1.31_L1.47, CD3H1.32_L1.47, or CD3H1.33_L1.47×PSMA-H H1L1.

(228) In some embodiments, the 1+1 Fab-scFv-Fc format includes skew variants, pI variants, and ablation variants. Accordingly, some embodiments include 1+1 Fab-scFv-Fc formats that comprise: a) a first monomer (the “scFv monomer”) that comprises a charged scFv linker (with the +H sequence of FIG. 5 being preferred in some embodiments), the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, and an scFv that binds to CD3 as outlined herein; b) a second monomer (the “Fab monomer”) that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain; and c) a light chain that includes a variable light domain light domain (VL) and a constant light domain (CL), wherein numbering is according to EU numbering. The variable heavy domain and variable light domain make up a PSMA binding moiety. CD3 binding domain sequences finding particular use in these embodiments include, but are not limited to, H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (see FIGS. **10A-10F**). PSMA binding domain sequences that are of particular use in these embodiments include, but are not limited to, PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; and PSMA-H H1_L1.13. Particularly useful PSMA and CD3 combinations for use in the 1+1 Fab-scFv-Fc PSMA×CD3 bispecific antibody format are disclosed in FIGS. **22-25** and include: a) CD3H1.30_L1.47×PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; or PSMA-H H1_L1.13; b) CD3H1.31_L1.47, CD3H1.32_L1.47, or CD3H1.33_L1.47×PSMA-H H1L1.

(229) In some embodiments, the 1+1 Fab-scFv-Fc format includes skew variants, pI variants, ablation variants and FcRn variants. Accordingly, some embodiments include 1+1 Fab-scFv-Fc formats that comprise: a) a first monomer (the “scFv monomer”) that comprises a charged scFv linker (with the +H sequence of FIG. 6 being preferred in some embodiments), the skew variants

S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and an scFv that binds to CD3 as outlined herein; b) a second monomer (the “Fab monomer”) that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S, and a variable heavy domain; and c) a light chain that includes a variable light domain (VL) and a constant light domain (CL), wherein numbering is according to EU numbering. The variable heavy domain and variable light domain make up a PSMA binding domain. CD3 binding domain sequences finding particular use in these embodiments include, but are not limited to, H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (see FIGS. 10A-10F). PSMA binding domain sequences that are of particular use in these embodiments include, but are not limited to, PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; and PSMA-H H1_L1.13. Particularly useful PSMA and CD3 combinations for use in the 1+1 Fab-scFv-Fc PSMA×CD3 bispecific antibody format are disclosed in FIGS. 22-25 and include: a) CD3H1.30_L1.47×PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; or PSMA-H H1_L1.13; b) CD3H1.31_L1.47, CD3H1.32_L1.47, or CD3H1.33_L1.47×PSMA-H H1L1.

(230) FIGS. 7A-7D show some exemplary Fc domain sequences that are useful in the 1+1 Fab-scFv-Fc format antibodies. The “monomer 1” sequences depicted in FIGS. 7A-7D typically refer to the Fc domain of the “Fab-Fc heavy chain” and the “monomer 2” sequences refer to the Fc domain of the “scFv-Fc heavy chain.” Further, FIG. 9 provides useful CL sequences that can be used with this format. In some embodiments, any of the VH and VL sequences depicted herein (including all VH and VL sequences depicted in the Figures and Sequence Listings, including those directed to PSMA) can be added to the bottle opener backbone formats of FIG. 7A-7D as the “Fab side”, using any of the anti-CD3 scFv sequences shown in the Figures and Sequence Listings. For bottle opener backbone 1 from FIG. 7A, (optionally including the 428L/434S variants), CD binding domain sequences finding particular use in these embodiments include, but are not limited to, CD3 binding domain anti-CD3H1.30_L1.47, anti-CD3H1.32_L1.47, anti-CD3H1.89_L1.47, anti-CD3H1.90_L1.47, anti-CD3H1.33_L1.47 and anti-CD3H1.31_L1.47, as well as those depicted in FIG. 10A-10F, attached as the scFv side of the backbones shown in FIGS. 7A-7D. Particularly useful PSMA and CD3 sequence combinations (optionally including the 428L/434S variants) and exemplary anti-CD3× anti-PSMA antibodies in the 1+1 Fab-scFv-Fc format are depicted in FIGS. 22-25.

(231) 2. mAb-Fv

(232) One heterodimeric scaffold that finds particular use in the antibodies described herein is the mAb-Fv format (FIG. 50G). In this embodiment, the format relies on the use of a C-terminal attachment of an “extra” variable heavy domain to one monomer and the C-terminal attachment of an “extra” variable light domain to the other monomer, thus forming a third antigen binding domain, wherein the Fab portions of the two monomers bind a PSMA and the “extra” scFv domain binds CD3.

(233) In this embodiment, the first monomer comprises a first heavy chain, comprising a first variable heavy domain and a first constant heavy domain comprising a first Fc domain, with a first variable light domain covalently attached to the C-terminus of the first Fc domain using a domain linker (VH1-CH1-hinge-CH2-CH3-[optional linker]-VL2). The second monomer comprises a second variable heavy domain of the second constant heavy domain comprising a second Fc domain, and a third variable heavy domain covalently attached to the C-terminus of the second Fc

domain using a domain linker (vh1-CH1-hinge-CH2-CH3-[optional linker]-VH2. The two C-terminally attached variable domains make up a Fv that binds CD3 (as it is less preferred to have bivalent CD3 binding). This embodiment further utilizes a common light chain comprising a variable light domain and a constant light domain that associates with the heavy chains to form two identical Fabs that bind a PSMA. As for many of the embodiments herein, these constructs include skew variants, pI variants, ablation variants, additional Fc variants, etc. as desired and described herein.

(234) The antibodies described herein provide mAb-Fv formats where the CD3 binding domain sequences are as shown in FIG. 10A-10F. The antibodies described herein provide mAb-Fv formats wherein the PSMA binding domain sequences are as shown in FIGS. 17 and 18.

(235) In addition, the Fc domains of the mAb-Fv format comprise skew variants (e.g. a set of amino acid substitutions as shown in FIGS. 1 and 4, with particularly useful skew variants being selected from the group consisting of S364K/E357Q:L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L, K370S:S364K/E357Q, T366S/L368A/Y407V:T366W and T366S/L368A/Y407V/Y349C:T366W/S354C), optionally ablation variants (including those shown in FIG. 3), optionally charged scFv linkers (including those shown in FIG. 5) and the heavy chain comprises pI variants (including those shown in FIG. 2).

(236) In some embodiments, the mAb-Fv format includes skew variants, pI variants, and ablation variants. Accordingly, some embodiments include mAb-Fv formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, and a first variable heavy domain that, with the first variable light domain of the light chain, makes up an Fv that binds to PSMA, and a second variable heavy domain; b) a second monomer that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, and a first variable heavy domain that, with the first variable light domain, makes up the Fv that binds to PSMA as outlined herein, and a second variable light chain, that together with the second variable heavy domain forms an Fv (ABD) that binds to CD3; and c) a light chain comprising a first variable light domain and a constant light domain.

(237) In some embodiments, the mAb-Fv format includes skew variants, pI variants, ablation variants and FcRn variants. Accordingly, some embodiments include mAb-Fv formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a first variable heavy domain that, with the first variable light domain of the light chain, makes up an Fv that binds to PSMA, and a second variable heavy domain; b) a second monomer that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a first variable heavy domain that, with the first variable light domain, makes up the Fv that binds to PSMA as outlined herein, and a second variable light chain, that together with the second variable heavy domain of the first monomer forms an Fv (ABD) that binds CD3; and c) a light chain comprising a first variable light domain and a constant light domain.

(238) 3. mAb-scFv

(239) One heterodimeric scaffold that finds particular use in the antibodies described herein is the mAb-scFv format (FIG. 50H). In this embodiment, the format relies on the use of a C-terminal attachment of a scFv to one of the monomers, thus forming a third antigen binding domain, wherein the Fab portions of the two monomers bind PSMA and the “extra” scFv domain binds CD3. Thus, the first monomer comprises a first heavy chain (comprising a variable heavy domain and a constant domain), with a C-terminally covalently attached scFv comprising a scFv variable light domain, an scFv linker and a scFv variable heavy domain in either orientation (VH1-CH1-

hinge-CH2-CH3-[optional linker]-VH2-scFv linker-VL2 or VH1-CH1-hinge-CH2-CH3-[optional linker]-VL2-scFv linker-VH2). This embodiment further utilizes a common light chain comprising a variable light domain and a constant light domain, that associates with the heavy chains to form two identical Fabs that bind PSMA. As for many of the embodiments herein, these constructs include skew variants, pI variants, ablation variants, additional Fc variants, etc. as desired and described herein.

(240) The antibodies described herein provide mAb-scFv formats where the CD binding domain sequences are as shown in FIGS. **10A-10F** and the PSMA binding domain sequences are as shown in FIGS. **17** and **18A-18E**.

(241) In addition, the Fc domains of the mAb-scFv format comprise skew variants (e.g. a set of amino acid substitutions as shown in FIG. **1**, with particularly useful skew variants being selected from the group consisting of S364K/E357Q:L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L, K370S:S364K/E357Q, T366S/L368A/Y407V:T366W and T366S/L368A/Y407V/Y349C:T366W/S354C), optionally ablation variants (including those shown in FIG. **3**), optionally charged scFv linkers (including those shown in FIG. **5**) and the heavy chain comprises pI variants (including those shown in FIG. **2**).

(242) In some embodiments, the mAb-scFv format includes skew variants, pI variants, and ablation variants. Accordingly, some embodiments include mAb-scFv formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to PSMA as outlined herein, and a scFv domain that binds to CD3; b) a second monomer that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to PSMA as outlined herein; and c) a common light chain comprising a variable light domain and a constant light domain.

(243) In some embodiments, the mAb-scFv format includes skew variants, pI variants, ablation variants and FcRn variants. Accordingly, some embodiments include mAb-scFv formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to PSMA as outlined herein, and a scFv domain that binds to CD3; b) a second monomer that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to PSMA as outlined herein; and c) a common light chain comprising a variable light domain and a constant light domain.

(244) 4. 2+1 Fab.sub.2-scFv-Fc Format

(245) One heterodimeric scaffold that finds particular use in the antibodies described herein is the “2+1 Fab.sub.2-scFv-Fc” format (also referred to in previous related filings as “central-scFv format”) shown in FIG. **21B** with an exemplary combination of a CD3 binding domain and two tumor target antigen (PSMA) binding domains. In this embodiment, the format relies on the use of an inserted scFv domain thus forming a third antigen binding domain, wherein the Fab portions of the two monomers bind PSMA and the “extra” scFv domain binds CD3. The scFv domain is inserted between the Fc domain and the CH1-Fv region of one of the monomers, thus providing a third antigen binding domain. As described, PSMA×CD3 bispecific antibodies having the 2+1 Fab.sub.2-scFv-Fc format are potent in inducing redirected T cell cytotoxicity in cellular environments that express low levels of PSMA. Moreover, as shown in the examples, PSMA×CD3 bispecific antibodies having the 2+1 Fab.sub.2-scFv-Fc format allow for the “fine tuning” of

immune responses as such antibodies exhibit a wide variety of different properties, depending on the PSMA and/or CD3 binding domains used. For example, such antibodies exhibit differences in selectivity for cells with different PSMA expression, potencies for PSMA expressing cells, ability to elicit cytokine release, and sensitivity to soluble PSMA. These PSMA antibodies find use, for example, in the treatment of PSMA associated cancers.

(246) In this embodiment, one monomer comprises a first heavy chain comprising a first variable heavy domain, a CH1 domain (and optional hinge) and Fc domain, with a scFv comprising a scFv variable light domain, an scFv linker and a scFv variable heavy domain. The scFv is covalently attached between the C-terminus of the CH1 domain of the heavy constant domain and the N-terminus of the first Fc domain using optional domain linkers (N- to C-terminus: VH1-CH1-[optional linker]-VH2-scFv linker-VL2-[optional linker including the hinge]-CH2-CH3, or the opposite orientation for the scFv, N- to C-terminus: VH1-CH1-[optional linker]-VL2-scFv linker-VH2-[optional linker including the hinge]-CH2-CH3). The optional linkers can be any suitable peptide linkers, including, for example, the domain linkers included in FIG. 6. In some embodiments, the optional linker is a hinge or a fragment thereof. The other monomer is a standard Fab side (i.e., VH1-CH1-hinge-CH2-CH3). This embodiment further utilizes a common light chain comprising a variable light domain and a constant light domain that associates with the heavy chains to form two identical Fabs that bind PSMA. As for many of the embodiments herein, these constructs include skew variants, pI variants, ablation variants, additional Fc variants, etc. as desired and described herein.

(247) In one embodiment, the 2+1 Fab.sub.2-scFv-Fc format antibody includes an scFv with the VH and VL of a CD3 binding domain sequence depicted in FIG. 10A-10F or the Sequence Listing. In one embodiment, the 2+1 Fab.sub.2-scFv-Fc format antibody includes two Fabs having the VH and VL of a PSMA binding domain as shown in FIGS. 17 and 18A-18E and the Sequence Listing. In an exemplary embodiment, the PSMA binding domain of the 2+1 Fab.sub.2-scFv-Fc PSMA×CD3 bispecific antibody includes the VH of PSMA-H H1 (FIG. 17) and VL of one of the following PSMA binding domains: PSMA-H L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E). In one embodiment, the CD3 binding domain of the 2+1 Fab.sub.2-scFv-Fc format antibody includes the VH and VL of one of the following CD3 binding domains: H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. 10A-10F). Particularly useful PSMA and CD3 combinations for use in the 2+1 Fab.sub.2-scFv-Fc format antibody format are disclosed in FIGS. 26-32 and include: a) CD3H1.30_L1.47, CD3H1.32_L1.47, CD3 L1.47_H1.32, CD3H1.89_L1.47, CD3 L1.47_H1.89, CD3H1.33_L1.47, CD3 L1.47_H1.32×b) PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84

(248) In addition, the Fc domains of the 2+1 Fab.sub.2-scFv-Fc format comprise skew variants (e.g. a set of amino acid substitutions as shown in FIG. 1, with particularly useful skew variants being selected from the group consisting of S364K/E357Q:L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L, K370S:S364K/E357Q, T366S/L368A/Y407V:T366W and T366S/L368A/Y407V/Y349C:T366W/S354C), optionally ablation variants (including those shown in FIG. 3), optionally charged scFv linkers (including those shown in FIG. 5) and the heavy chain comprises pI variants (including those shown in FIG. 2).

(249) In some embodiments, the 2+1 Fab.sub.2-scFv-Fc format antibody includes skew variants, pI variants, and ablation variants. Accordingly, some embodiments include 2+1 Fab.sub.2-scFv-Fc formats that comprise: a) a first monomer (the Fab-scFv-Fc side) that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds

to PSMA as outlined herein, and an scFv domain that binds to CD3; b) a second monomer (the Fab-Fc side) that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with variable light domain of the common light chain, makes up an Fv that binds to PSMA as outlined herein; and c) a common light chain comprising the variable light domain and a constant light domain, where numbering is according to EU numbering. In some embodiments, the common light chain and variable heavy domains on each monomer for PSMA binding domains. CD3 binding domain sequences finding particular use in these embodiments include, but are not limited to, H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 as well as those depicted in FIGS. **10A-10F**. PSMA binding domain sequences that are of particular use in these embodiments include, but are not limited to, PSMA-H H1_L1, and PSMA-H H1_L1.1-L1.84 as depicted in FIGS. **17-19**.

(250) In some embodiments, the 2+1 Fab.sub.2-scFv-Fc format antibody includes skew variants, pI variants, ablation variants and FcRn variants. Accordingly, some embodiments include 2+1 Fab.sub.2-scFv-Fc formats that comprise: a) a first monomer (the Fab-scFv-Fc side) that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to PSMA as outlined herein, and an scFv domain that binds to CD3; b) a second monomer (the Fab-Fc side) that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with variable light domain of the common light chain, makes up an Fv that binds to PSMA as outlined herein; and c) a common light chain comprising a variable light domain and a constant light domain, where numbering is according to EU numbering.

(251) In some embodiments, the common light chain and variable heavy domains on each monomer for PSMA binding domains. CD3 binding domain sequences finding particular use in these embodiments include, but are not limited to, H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 as well as those depicted in FIGS. **10A-10F**. PSMA binding domain sequences that are of particular use in these embodiments include, but are not limited to, PSMA-H H1_L1, and PSMA-H H1_L1.1-L1.84 as depicted in FIGS. **17-19**.

(252) FIGS. **8A-8C** shows some exemplary Fc domain sequences that are useful with the 2+1 Fab.sub.2-scFv-Fc format. The “monomer 1” sequences depicted in FIGS. **8A-8C** typically refer to the Fc domain of the “Fab-Fc heavy chain” and the “monomer 2” sequences refer to the Fc domain of the “Fab-scFv-Fc heavy chain.” Further, FIG. **9** provides useful CL sequences that can be used with this format.

(253) Exemplary anti-CD3× anti-PSMA antibodies in the 2+1 Fab.sub.2-scFv-Fc format are depicted in FIGS. **26-32**.

(254) 5. Central-Fv

(255) One heterodimeric scaffold that finds particular use in the antibodies described herein is the central-Fv format (FIG. **50I**). In this embodiment, the format relies on the use of an inserted Fv domain (i.e., the central Fv domain) thus forming an “extra” third antigen binding domain, wherein the Fab portions of the two monomers bind a PSMA and the “extra” central Fv domain binds CD3. The “extra” central Fv domain is inserted between the Fc domain and the CH1-Fv region of the monomers, thus providing a third antigen binding domain (i.e., the “extra” central Fv domain), wherein each monomer contains a component of the “extra” central Fv domain (i.e., one monomer comprises the variable heavy domain and the other a variable light domain of the “extra” central Fv

domain).

(256) In this embodiment, one monomer comprises a first heavy chain comprising a first variable heavy domain, a CH1 domain, and Fc domain and an additional variable light domain. The light domain is covalently attached between the C-terminus of the CH1 domain of the heavy constant domain and the N-terminus of the first Fc domain using domain linkers (VH1-CH1-[optional linker]-VL2-hinge-CH2-CH3). The other monomer comprises a first heavy chain comprising a first variable heavy domain, a CH1 domain and Fc domain and an additional variable heavy domain (VH1-CH1-[optional linker]-VH2-hinge-CH2-CH3). The light domain is covalently attached between the C-terminus of the CH1 domain of the heavy constant domain and the N-terminus of the first Fc domain using domain linkers.

(257) This embodiment further utilizes a common light chain comprising a variable light domain and a constant light domain that associates with the heavy chains to form two identical Fabs that each bind a PSMA. As for many of the embodiments herein, these constructs include skew variants, pI variants, ablation variants, additional Fc variants, etc. as desired and described herein.

(258) The antibodies described herein provide central-Fv formats where the CD3 binding domain sequences are as shown in **10A-10F** and the PSMA binding domain sequences (VH, VL and CDRs) are as shown in **FIGS. 17 and 18**.

(259) 6. One Armed Central-scFv

(260) One heterodimeric scaffold that finds particular use in the antibodies described herein is the one armed central-scFv format (**FIG. 50C**). In this embodiment, one monomer comprises just an Fc domain, while the other monomer includes a Fab domain (a first antigen binding domain), a scFv domain (a second antigen binding domain) and an Fc domain, where the scFv domain is inserted between the Fc domain and the Fc domain. In this format, the Fab portion binds one receptor target and the scFv binds another. In this format, either the Fab portion binds a PSMA and the scFv binds CD3 or vice versa.

(261) In this embodiment, one monomer comprises a first heavy chain comprising a first variable heavy domain, a CH1 domain and Fc domain, with a scFv comprising a scFv variable light domain, an scFv linker and a scFv variable heavy domain. The scFv is covalently attached between the C-terminus of the CH1 domain of the heavy constant domain and the N-terminus of the first Fc domain using domain linkers, in either orientation, VH1-CH1-[optional domain linker]-VH2-scFv linker-VL2-[optional domain linker]-CH2-CH3 or VH1-CH1-[optional domain linker]-VL2-scFv linker-VH2-[optional domain linker]-CH2-CH3. The second monomer comprises an Fc domain (CH2-CH3). This embodiment further utilizes a light chain comprising a variable light domain and a constant light domain that associates with the heavy chain to form a Fab.

(262) As for many of the embodiments herein, these constructs include skew variants, pI variants, ablation variants, additional Fc variants, etc. as desired and described herein.

(263) The antibodies described herein provide central-Fv formats where the CD3 binding domain sequences are as shown in **FIGS. 10A-10F** and the PSMA binding domain sequences (VH, VL and CDRs) are as shown in **FIGS. 17 and 18**.

(264) In addition, the Fc domains of the one armed central-scFv format generally include skew variants (e.g. a set of amino acid substitutions as shown in **FIG. 1**, with particularly useful skew variants being selected from the group consisting of S364K/E357Q:L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L, K370S:S364K/E357Q, T366S/L368A/Y407V:T366W and T366S/L368A/Y407V/Y349C:T366W/S354C), optionally ablation variants (including those shown in **FIG. 3**), optionally charged scFv linkers (including those shown in **FIG. 5**) and the heavy chain comprises pI variants (including those shown in **FIG. 2**).

(265) In some embodiments, the one armed central-scFv format includes skew variants, pI variants, and ablation variants. Accordingly, some embodiments of the one armed central-scFv formats comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants

E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with the variable light domain of the light chain, makes up an Fv that binds to PSMA as outlined herein, and a scFv domain that binds to CD3; b) a second monomer that includes an Fc domain having the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K; and c) a light chain comprising a variable light domain and a constant light domain.

(266) In some embodiments, the one armed central-scFv format includes skew variants, pI variants, ablation variants and FcRn variants. Accordingly, some embodiments of the one armed central-scFv formats comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with the variable light domain of the light chain, makes up an Fv that binds to PSMA as outlined herein, and a scFv domain that binds to CD3; b) a second monomer that includes an Fc domain having the skew variants L368D/K370S, the pI variants

N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, and the FcRn variants M428L/N434S; and c) a light chain comprising a variable light domain and a constant light domain.

(267) 7. One Armed scFv-mAb

(268) One heterodimeric scaffold that finds particular use in the antibodies described herein is the one armed scFv-mAb format (FIG. 50D). In this embodiment, one monomer comprises just an Fc domain, while the other monomer uses a scFv domain attached at the N-terminus of the heavy chain, generally through the use of a linker: VH-scFv linker-VL-[optional domain linker]-CH1-hinge-CH2-CH3 or (in the opposite orientation) VL-scFv linker-VH-[optional domain linker]-CH1-hinge-CH2-CH3. In this format, the Fab portions each bind PSMA and the scFv binds CD3. This embodiment further utilizes a light chain comprising a variable light domain and a constant light domain that associates with the heavy chain to form a Fab. As for many of the embodiments herein, these constructs include skew variants, pI variants, ablation variants, additional Fc variants, etc. as desired and described herein.

(269) The antibodies described herein provide one armed scFv-mAb formats where the CD3 binding domain sequences are as shown in **10A-10F** and wherein the PSMA binding domain sequences (VH, VL and CDRs) are as shown in FIGS. **17** and **18**.

(270) In addition, the Fc domains of the one armed scFv-mAb format generally include skew variants (e.g. a set of amino acid substitutions as shown in FIG. **1**, with particularly useful skew variants being selected from the group consisting of S364K/E357Q:L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L, K370S:S364K/E357Q, T366S/L368A/Y407V:T366W and T366S/L368A/Y407V/Y349C:T366W/S354C), optionally ablation variants (including those shown in FIG. **3**), optionally charged scFv linkers (including those shown in FIG. **5**) and the heavy chain comprises pI variants (including those shown in FIG. **2**).

(271) In some embodiments, the one armed scFv-mAb format includes skew variants, pI variants, and ablation variants. Accordingly, some embodiments of the one armed scFv-mAb formats comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with the variable light domain of the light chain, makes up an Fv that binds to PSMA as outlined herein, and a scFv domain that binds to CD3; b) a second monomer that includes an Fc domain having the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K; and c) a light chain comprising a variable light domain and a constant light domain.

(272) In some embodiments, the one armed scFv-mAb format includes skew variants, pI variants, ablation variants and FcRn variants. Accordingly, some embodiments one armed scFv-mAb formats comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation

variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with the variable light domain of the light chain, makes up an Fv that binds to PSMA as outlined herein, and a scFv domain that binds to CD3; b) a second monomer that includes an Fc domain having the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, and the FcRn variants M428L/N434S; and c) a light chain comprising a variable light domain and a constant light domain.

(273) 8. scFv-mAb

(274) One heterodimeric scaffold that finds particular use in the antibodies described herein is the mAb-scFv format (FIG. 50E). In this embodiment, the format relies on the use of a N-terminal attachment of a scFv to one of the monomers, thus forming a third antigen binding domain, wherein the Fab portions of the two monomers bind PSMA and the “extra” scFv domain binds CD3.

(275) In this embodiment, the first monomer comprises a first heavy chain (comprising a variable heavy domain and a constant domain), with a N-terminally covalently attached scFv comprising a scFv variable light domain, an scFv linker and a scFv variable heavy domain in either orientation ((VH1-scFv linker-VL1-[optional domain linker]-VH2-CH1-hinge-CH2-CH3) or (with the scFv in the opposite orientation) ((VL1-scFv linker-VH1-[optional domain linker]-VH2-CH1-hinge-CH2-CH3)). This embodiment further utilizes a common light chain comprising a variable light domain and a constant light domain that associates with the heavy chains to form two identical Fabs that bind PSMA. As for many of the embodiments herein, these constructs include skew variants, pI variants, ablation variants, additional Fc variants, etc. as desired and described herein.

(276) The antibodies described herein provide scFv-mAb formats where the CD3 binding domain sequences are as shown in **10A-10F** and wherein the PSMA binding domain sequences (VH, VL and CDRs) are as shown in FIGS. **17** and **18**.

(277) In addition, the Fc domains of the scFv-mAb format generally include skew variants (e.g. a set of amino acid substitutions as shown in FIG. **1**, with particularly useful skew variants being selected from the group consisting of S364K/E357Q:L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L, K370S:S364K/E357Q, T366S/L368A/Y407V:T366W and T366S/L368A/Y407V/Y349C:T366W/S354C), optionally ablation variants (including those shown in FIG. **3**), optionally charged scFv linkers (including those shown in FIG. **5**) and the heavy chain comprises pI variants (including those shown in FIG. **2**).

(278) In some embodiments, the scFv-mAb format includes skew variants, pI variants, and ablation variants. Accordingly, some embodiments include scFv-mAb formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to PSMA as outlined herein, and a scFv domain that binds to CD3; b) a second monomer that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to PSMA as outlined herein; and c) a common light chain comprising a variable light domain and a constant light domain.

(279) In some embodiments, the scFv-mAb format includes skew variants, pI variants, ablation variants and FcRn variants. Accordingly, some embodiments include scFv-mAb formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to PSMA as outlined herein, and a scFv domain that binds to CD3; b) a second monomer that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D,

the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to PSMA as outlined herein; and c) a common light chain comprising a variable light domain and a constant light domain.

(280) 9. Dual scFv Formats

(281) The antibodies described herein also provide dual scFv formats (FIG. 50B) as are known in the art. In this embodiment, the PSMA×CD3 heterodimeric bispecific antibody is made up of two scFv-Fc monomers (both in either (VH-scFv linker-VL-[optional domain linker]-CH2-CH3) format or (VL-scFv linker-VH-[optional domain linker]-CH2-CH3) format, or with one monomer in one orientation and the other in the other orientation.

(282) The antibodies described herein provide dual scFv formats where the CD3 binding domain sequences are as shown in FIGS. 10A-10F and wherein the PSMA binding domain sequences (VH, VL and CDRs) are as shown in FIGS. 17 and 18. In some embodiments, the dual scFv format includes skew variants, pI variants, and ablation variants. Accordingly, some embodiments include dual scFv formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, and a first scFv that binds either CD3 or PSMA; and b) a second monomer that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, and a second scFv that binds either CD3 or PSMA. In some embodiments, the dual scFv format includes skew variants, pI variants, ablation variants and FcRn variants. In some embodiments, the dual scFv format includes skew variants, pI variants, and ablation variants. Accordingly, some embodiments include dual scFv formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a first scFv that binds either CD3 or PSMA; and b) a second monomer that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a second scFv that binds either CD3 or PSMA.

(283) 10. Non-Heterodimeric Bispecific Antibodies

(284) As will be appreciated by those in the art, the PSMA and CD3 Fv sequences outlined herein can also be used in both monospecific antibodies (e.g., “traditional monoclonal antibodies”) or non-heterodimeric bispecific formats.

(285) CD3 binding domain sequences finding particular use include, but are not limited to H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 (FIGS. 10A-10F).

(286) PSMA binding domain sequences that are of particular use include, but are not limited to: PSMA-H L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E).

(287) Suitable non-heterodimeric bispecific formats are known in the art, and include a number of different formats as generally depicted in Spiess et al., *Molecular Immunology* (67):95-106 (2015) and Kontermann, *mAbs* 4:2, 182-197 (2012), both of which are expressly incorporated by reference and in particular for the figures, legends and citations to the formats therein.

(288) 11. Trident Format

(289) In some embodiments, the bispecific antibodies described herein are in the “Trident” format (FIG. 50K) as generally described in WO2015/184203, hereby expressly incorporated by reference in its entirety and in particular for the Figures, Legends, definitions and sequences of “Heterodimer-Promoting Domains” or “HPDs”, including “K-coil” and “E-coil” sequences. Tridents rely on using two different HPDs that associate to form a heterodimeric structure as a component of the structure, see FIG. 1K. In this embodiment, the Trident format include a “traditional” heavy and light chain (e.g., VH1-CH1-hinge-CH2-CH3 and VL1-CL), a third chain

comprising a first “diabody-type binding domain” or “DART®”, VH2-(linker)-VL3-HPD1 and a fourth chain comprising a second DART®, VH3-(linker)-(linker)-VL2-HPD2. The VH1 and VL1 form a first ABD, the VH2 and VL2 form a second ABD, and the VH3 and VL3 form a third ABD. In some cases, as is shown in FIG. 1K, the second and third ABDs bind the same antigen, in this instance generally PSMA, e.g., bivalently, with the first ABD binding a CD3 monovalently.

(290) 12. Monospecific, Monoclonal Antibodies

(291) As will be appreciated by those in the art, the novel Fv sequences outlined herein can also be used in both monospecific antibodies (e.g., “traditional monoclonal antibodies”) or non-heterodimeric bispecific formats. Accordingly, in some embodiments, the antibodies described herein provide monoclonal (monospecific) antibodies comprising the 6 CDRs and/or the vh and vl sequences from the figures, generally with IgG1, IgG2, IgG3 or IgG4 constant regions, with IgG1, IgG2 and IgG4 (including IgG4 constant regions comprising a S228P amino acid substitution) finding particular use in some embodiments. That is, any sequence herein with a “H_L” designation can be linked to the constant region of a human IgG1 antibody.

(292) In some embodiments, the monospecific antibody is a PSMA monospecific antibody. In certain embodiments, the monospecific anti-PSMA antibody includes the 6 CDRs of any of the anti-PSMA binding domains selected from: PSMA-H L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E). In certain embodiments, the monospecific anti-PSMA antibody includes the variable heavy domain (VH) and variable light domain (VL) of any of the anti-PSMA binding domains selected from: PSMA-H L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E).

(293) G. Antigen Binding Domains

(294) As discussed herein, the subject heterodimeric antibodies include two antigen binding domains (ABDs), each of which bind to PSMA or CD3. As outlined herein, these heterodimeric antibodies can be bispecific and bivalent (each antigen is bound by a single ABD, for example, in the format depicted in FIG. 21A), or bispecific and trivalent (one antigen is bound by a single ABD and the other is bound by two ABDs, for example as depicted in FIG. 21B).

(295) In addition, in general, one of the ABDs comprises a scFv as outlined herein, in an orientation from N- to C-terminus of VH-scFv linker-VL or VL-scFv linker-VH. One or both of the other ABDs, according to the format, generally is a Fab, comprising a VH domain on one protein chain (generally as a component of a heavy chain) and a VL on another protein chain (generally as a component of a light chain).

(296) The disclosure provides a number of ABDs that bind to a number of different checkpoint proteins, as outlined below. As will be appreciated by those in the art, any set of 6 CDRs or VH and VL domains can be in the scFv format or in the Fab format, which is then added to the heavy and light constant domains, where the heavy constant domains comprise variants (including within the CH1 domain as well as the Fc domain). The scFv sequences contained in the sequence listing utilize a particular charged linker, but as outlined herein, uncharged or other charged linkers can be used, including those depicted in FIG. 6.

(297) In addition, as discussed above, the numbering used in the Sequence Listing for the identification of the CDRs is Kabat, however, different numbering can be used, which will change the amino acid sequences of the CDRs as shown in Table 2.

(298) For all of the variable heavy and light domains listed herein, further variants can be made. As outlined herein, in some embodiments the set of 6 CDRs can have from 0, 1, 2, 3, 4 or 5 amino acid modifications (with amino acid substitutions finding particular use), as well as changes in the framework regions of the variable heavy and light domains, as long as the frameworks (excluding the CDRs) retain at least about 80, 85 or 90% identity to a human germline sequence selected from those listed in FIG. 1 of U.S. Pat. No. 7,657,380, which Figure and Legend is incorporated by reference in its entirety herein. Thus, for example, the identical CDRs as described herein can be combined with different framework sequences from human germline sequences, as long as the framework regions retain at least 80, 85 or 90% identity to a human germline sequence selected

from those listed in FIG. 1 of U.S. Pat. No. 7,657,380. Alternatively, the CDRs can have amino acid modifications (e.g. from 1, 2, 3, 4 or 5 amino acid modifications in the set of CDRs (that is, the CDRs can be modified as long as the total number of changes in the set of 6 CDRs is less than 6 amino acid modifications, with any combination of CDRs being changed; e.g. there may be one change in VLCDR1, two in VHCDR2, none in VHCDR3, etc.)), as well as having framework region changes, as long as the framework regions retain at least 80, 85 or 90% identity to a human germline sequence selected from those listed in FIG. 1 of U.S. Pat. No. 7,657,380.

(299) 1. PSMA Antigen Binding Domains

(300) Provided herein are PSMA antigen binding domain and antibodies that include such binding domains. Suitable sets of 6 CDRs and/or VH and VL domains included in PSMA binding domains are depicted in FIG. 17 (anti-PSMA vhCDRs and VH) and FIGS. 17 and 18A-E (anti-PSMA vlCDRs and VL). In some embodiments, the heterodimeric antibody is a 1+1 Fab-scFv-Fc or 2+1 Fab.sub.2-scFv-Fv format antibody (see, e.g., FIGS. 21A and 21B).

(301) In one embodiment, the PSMA antigen binding domain includes the 6 CDRs (i.e., vhCDR1-3 and vlCDR1-3) of a PSMA ABD described herein, including the figures and sequence listing. In certain embodiments the PSMA antigen binding domain includes a variable heavy domain that includes vhCDR1-3 of PSMA-H H1 (FIG. 17) and a variable light domain that includes the vlCDR1-3 of any of PSMA-H L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E). In exemplary embodiments, the PSMA antigen binding domain includes the 6 CDRs of one of the following PSMA antigen binding domains: PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; and PSMA-H H1_L1.13.

(302) In one embodiment, the PSMA antigen binding domain includes the variable heavy and variable light domain of a PSMA ABD described herein, including the figures and sequence listing. In certain embodiments the PSMA antigen binding domain includes a variable heavy domain that is PSMA-H H1 (FIG. 17) and a variable light domain selected from PSMA-H L1 and L1.1-L1.84 (FIGS. 17 and 18A-18E). In exemplary embodiments, the PSMA antigen binding domain includes the variable heavy domain and variable light domain of one of the following PSMA antigen binding domains: PSMA-H H1_L1, PSMA-H H1_L1.58; PSMA-H H1_L1.11; PSMA-H H1_L1.24; PSMA-H H1_L1.26; PSMA-H H1_L1.75; PSMA-H H1_L1.68; PSMA-H H1_L1.29; PSMA-H H1_L1.52; PSMA-H H1_L1.78; PSMA-H H1_L1.81; PSMA-H H1_L1.84; and PSMA-H H1_L1.13.

(303) As will be appreciated by those in the art, suitable PSMA binding domains can comprise a set of 6 CDRs as depicted in the Figures, either as they are underlined or, in the case where a different numbering scheme is used as described herein and as shown in Table 2, as the CDRs that are identified using other alignments within the VH and VL sequences of those depicted in FIGS. 17 and 18A-E. Suitable ABDs can also include the entire VH and VL sequences as depicted in these sequences and Figures, used as scFvs or as Fabs. In many of the embodiments herein that contain an Fv to PSMA, it is the Fab monomer that binds PSMA.

(304) In addition to the parental CDR sets disclosed in the figures and sequence listing that form an ABD to PSMA, the disclosure provides variant CDR sets. In one embodiment, a set of 6 CDRs can have 1, 2, 3, 4 or 5 amino acid changes from the parental CDRs, as long as the PSMA ABD is still able to bind to the target antigen, as measured by at least one of a Biacore, surface plasmon resonance (SPR) and/or BLI (biolayer interferometry, e.g., Octet assay) assay, with the latter finding particular use in many embodiments.

(305) In addition to the parental variable heavy and variable light domains disclosed herein that form an ABD to PSMA, the disclosure provides variant VH and VL domains. In one embodiment, the variant VH and VL domains each can have from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid changes from the parental VH and VL domain, as long as the ABD is still able to bind to the target

antigen, as measured at least one of a Biacore, surface plasmon resonance (SPR) and/or BLI (biolayer interferometry, e.g. Octet assay) assay, with the latter finding particular use in many embodiments. In another embodiment, the variant VH and VL are at least 90, 95, 97, 98 or 99% identical to the respective parental VH or VL, as long as the ABD is still able to bind to the target antigen, as measured by at least one of a Biacore, surface plasmon resonance (SPR) and/or BLI (biolayer interferometry, e.g. Octet assay) assay, with the latter finding particular use in many embodiments.

(306) 2. CD3 Antigen Binding Domains

(307) In some embodiments, one of the ABDs binds CD3. Suitable sets of 6 CDRs and/or VH and VL domains, as well as scFv sequences, are depicted in FIGS. **10A-10F** and the Sequence Listing. CD3 binding domain sequences that are of particular use include, but are not limited to, anti-CD3H1.30_L1.47, anti-CD3H1.32, anti-CD3 L1.47, anti-CD3H1.89_L1.47, anti-CD3H1.90_L1.47, anti-CD3H1.33_L1.47, anti-CD3H1.31_L1.47, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.32, anti-CD3 L1.47_H1.89, anti-CD3 L1.47_H1.90, anti-CD3 L1.47_H1.33, and anti-CD3 L1.47_H1.31 as depicted in FIGS. **10A-10F**.

(308) As will be appreciated by those in the art, suitable CD3 binding domains can comprise a set of 6 CDRs as depicted in FIGS. **10A-10F**, either as they are underlined or, in the case where a different numbering scheme is used as described herein and as shown in Table 2, as the CDRs that are identified using other alignments within the VH and VL sequences of those depicted in FIGS. **10A-10F**. Suitable ABDs can also include the entire VH and VL sequences as depicted in these sequences and Figures, used as scFvs or as Fabs. In many of the embodiments herein that contain an Fv to CD3, it is the scFv monomer that binds CD3.

(309) In addition to the parental CDR sets disclosed in the figures and sequence listing that form an ABD to CD3, the disclosure provides variant CDR sets. In one embodiment, a set of 6 CDRs can have 1, 2, 3, 4 or 5 amino acid changes from the parental CDRs, as long as the CD3 ABD is still able to bind to the target antigen, as measured by at least one of a Biacore, surface plasmon resonance (SPR) and/or BLI (biolayer interferometry, e.g. Octet assay) assay, with the latter finding particular use in many embodiments.

(310) In addition to the parental variable heavy and variable light domains disclosed herein that form an ABD to CD3, the disclosure provides variant VH and VL domains. In one embodiment, the variant VH and VL domains each can have from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid changes from the parental VH and VL domain, as long as the ABD is still able to bind to the target antigen, as measured at least one of a Biacore, surface plasmon resonance (SPR) and/or BLI (biolayer interferometry, e.g. Octet assay) assay, with the latter finding particular use in many embodiments. In another embodiment, the variant VH and VL are at least 90, 95, 97, 98 or 99% identical to the respective parental VH or VL, as long as the ABD is still able to bind to the target antigen, as measured by at least one of a Biacore, surface plasmon resonance (SPR) and/or BLI (biolayer interferometry, e.g. Octet assay) assay, with the latter finding particular use in many embodiments.

VI. Nucleic Acids

(311) The disclosure further provides nucleic acid compositions encoding the anti-PSMA antibodies provided herein, including, but not limited to, anti-PSMA $\times$ anti-CD3 bispecific antibodies and anti-PSMA monospecific antibodies.

(312) As will be appreciated by those in the art, the nucleic acid compositions will depend on the format and scaffold of the heterodimeric protein. Thus, for example, when the format requires three amino acid sequences, such as for the 1+1 Fab-scFv-Fc format (e.g. a first amino acid monomer comprising an Fc domain and a scFv, a second amino acid monomer comprising a heavy chain and a light chain), three nucleic acid sequences can be incorporated into one or more expression vectors for expression. Similarly, some formats (e.g. dual scFv formats such as disclosed in FIG. **1**) only two nucleic acids are needed; again, they can be put into one or two expression vectors.

(313) As is known in the art, the nucleic acids encoding the components of the antibodies described

herein can be incorporated into expression vectors as is known in the art, and depending on the host cells used to produce the heterodimeric antibodies described herein. Generally, the nucleic acids are operably linked to any number of regulatory elements (promoters, origin of replication, selectable markers, ribosomal binding sites, inducers, etc.). The expression vectors can be extra-chromosomal or integrating vectors.

(314) The nucleic acids and/or expression vectors of the antibodies described herein are then transformed into any number of different types of host cells as is well known in the art, including mammalian, bacterial, yeast, insect and/or fungal cells, with mammalian cells (e.g. CHO cells), finding use in many embodiments.

(315) In some embodiments, nucleic acids encoding each monomer and the optional nucleic acid encoding a light chain, as applicable depending on the format, are each contained within a single expression vector, generally under different or the same promoter controls. In embodiments of particular use in the antibodies described herein, each of these two or three nucleic acids are contained on a different expression vector. As shown herein and in 62/025,931, hereby incorporated by reference, different vector ratios can be used to drive heterodimer formation. That is, surprisingly, while the proteins comprise first monomer:second monomer:light chains (in the case of many of the embodiments herein that have three polypeptides comprising the heterodimeric antibody) in a 1:1:2 ratio, these are not the ratios that give the best results.

(316) The heterodimeric antibodies described herein are made by culturing host cells comprising the expression vector(s) as is well known in the art. Once produced, traditional antibody purification steps are done, including an ion exchange chromatography step. As discussed herein, having the pIs of the two monomers differ by at least 0.5 can allow separation by ion exchange chromatography or isoelectric focusing, or other methods sensitive to isoelectric point. That is, the inclusion of pI substitutions that alter the isoelectric point (pI) of each monomer so that each monomer has a different pI and the heterodimer also has a distinct pI, thus facilitating isoelectric purification of the “1+1 Fab-scFv-Fc” and “2+1” heterodimers (e.g., anionic exchange columns, cationic exchange columns). These substitutions also aid in the determination and monitoring of any contaminating dual scFv-Fc and mAb homodimers post-purification (e.g., IEF gels, cIEF, and analytical IEX columns).

VII. Biological and Biochemical Functionality of the Heterodimeric Bispecific Antibodies

(317) Generally the bispecific PSMA×CD3 antibodies described herein are administered to patients with cancer, and efficacy is assessed, in a number of ways as described herein. Thus, while standard assays of efficacy can be run, such as cancer load, size of tumor, evaluation of presence or extent of metastasis, etc., immuno-oncology treatments can be assessed on the basis of immune status evaluations as well. This can be done in a number of ways, including both in vitro and in vivo assays.

VIII. Treatments

(318) Once made, the compositions of the antibodies described herein find use in a number of applications. PSMA is highly expressed in prostate cancer. Accordingly, the heterodimeric compositions of the antibodies described herein find use in the treatment of such PSMA positive cancers.

IX. Antibody Compositions for In Vivo Administration

(319) Formulations of the antibodies used in accordance with the antibodies described herein are prepared for storage by mixing an antibody having the desired degree of purity with optional pharmaceutically acceptable carriers, excipients or stabilizers (Remington's Pharmaceutical Sciences 16th edition, Osol, A. Ed. [1980]), in the form of lyophilized formulations or aqueous solutions. Acceptable carriers, excipients, or stabilizers are nontoxic to recipients at the dosages and concentrations employed, and include buffers such as phosphate, citrate, and other organic acids; antioxidants including ascorbic acid and methionine; preservatives (such as octadecyldimethylbenzyl ammonium chloride; hexamethonium chloride; benzalkonium chloride,

benzethonium chloride; phenol, butyl or benzyl alcohol; alkyl parabens such as methyl or propyl paraben; catechol; resorcinol; cyclohexanol; 3-pentanol; and m-cresol); low molecular weight (less than about 10 residues) polypeptides; proteins, such as serum albumin, gelatin, or immunoglobulins; hydrophilic polymers such as polyvinylpyrrolidone; amino acids such as glycine, glutamine, asparagine, histidine, arginine, or lysine; monosaccharides, disaccharides, and other carbohydrates including glucose, mannose, or dextrans; chelating agents such as EDTA; sugars such as sucrose, mannitol, trehalose or sorbitol; salt-forming counter-ions such as sodium; metal complexes (e.g. Zn-protein complexes); and/or non-ionic surfactants such as TWEEN™ PLURONICS™ or polyethylene glycol (PEG).

X. Administrative Modalities

(320) The antibodies and chemotherapeutic agents described herein are administered to a subject, in accord with known methods, such as intravenous administration as a bolus or by continuous infusion over a period of time.

XI. Treatment Modalities

(321) In the methods described herein, therapy is used to provide a positive therapeutic response with respect to a disease or condition. By “positive therapeutic response” is intended an improvement in the disease or condition, and/or an improvement in the symptoms associated with the disease or condition. For example, a positive therapeutic response would refer to one or more of the following improvements in the disease: (1) a reduction in the number of neoplastic cells; (2) an increase in neoplastic cell death; (3) inhibition of neoplastic cell survival; (5) inhibition (i.e., slowing to some extent, preferably halting) of tumor growth; (6) an increased patient survival rate; and (7) some relief from one or more symptoms associated with the disease or condition.

(322) Positive therapeutic responses in any given disease or condition can be determined by standardized response criteria specific to that disease or condition. Tumor response can be assessed for changes in tumor morphology (i.e., overall tumor burden, tumor size, and the like) using screening techniques such as magnetic resonance imaging (MRI) scan, x-radiographic imaging, computed tomographic (CT) scan, bone scan imaging, endoscopy, and tumor biopsy sampling including bone marrow aspiration (BMA) and counting of tumor cells in the circulation.

(323) In addition to these positive therapeutic responses, the subject undergoing therapy may experience the beneficial effect of an improvement in the symptoms associated with the disease.

(324) Treatment according to the disclosure includes a “therapeutically effective amount” of the medicaments used. A “therapeutically effective amount” refers to an amount effective, at dosages and for periods of time necessary, to achieve a desired therapeutic result.

(325) A therapeutically effective amount may vary according to factors such as the disease state, age, sex, and weight of the individual, and the ability of the medicaments to elicit a desired response in the individual. A therapeutically effective amount is also one in which any toxic or detrimental effects of the antibody or antibody portion are outweighed by the therapeutically beneficial effects.

(326) A “therapeutically effective amount” for tumor therapy may also be measured by its ability to stabilize the progression of disease. The ability of a compound to inhibit cancer may be evaluated in an animal model system predictive of efficacy in human tumors.

(327) Alternatively, this property of a composition may be evaluated by examining the ability of the compound to inhibit cell growth or to induce apoptosis by in vitro assays known to the skilled practitioner. A therapeutically effective amount of a therapeutic compound may decrease tumor size, or otherwise ameliorate symptoms in a subject. One of ordinary skill in the art would be able to determine such amounts based on such factors as the subject's size, the severity of the subject's symptoms, and the particular composition or route of administration selected.

(328) Dosage regimens are adjusted to provide the optimum desired response (e.g., a therapeutic response). For example, a single bolus may be administered, several divided doses may be administered over time or the dose may be proportionally reduced or increased as indicated by the

exigencies of the therapeutic situation. Parenteral compositions may be formulated in dosage unit form for ease of administration and uniformity of dosage. Dosage unit form as used herein refers to physically discrete units suited as unitary dosages for the subjects to be treated; each unit contains a predetermined quantity of active compound calculated to produce the desired therapeutic effect in association with the required pharmaceutical carrier.

(329) The specification for the dosage unit forms of the disclosure are dictated by and directly dependent on (a) the unique characteristics of the active compound and the particular therapeutic effect to be achieved, and (b) the limitations inherent in the art of compounding such an active compound for the treatment of sensitivity in individuals.

(330) The efficient dosages and the dosage regimens for the bispecific antibodies described herein depend on the disease or condition to be treated and may be determined by the persons skilled in the art.

(331) An exemplary, non-limiting range for a therapeutically effective amount of an bispecific antibody used in the antibodies described herein is about 0.1-100 mg/kg.

(332) All cited references are herein expressly incorporated by reference in their entirety.

(333) Whereas particular embodiments of the disclosure have been described above for purposes of illustration, it will be appreciated by those skilled in the art that numerous variations of the details may be made without departing from the invention as described in the appended claims.

EXAMPLES

(334) Examples are provided below to illustrate the antibodies described herein. These examples are not meant to constrain the antibodies described herein to any particular application or theory of operation. For all constant region positions discussed in the antibodies described herein, numbering is according to the EU index as in Kabat (Kabat et al., 1991, Sequences of Proteins of Immunological Interest, 5th Ed., United States Public Health Service, National Institutes of Health, Bethesda, entirely incorporated by reference). Those skilled in the art of antibodies will appreciate that this convention consists of nonsequential numbering in specific regions of an immunoglobulin sequence, enabling a normalized reference to conserved positions in immunoglobulin families. Accordingly, the positions of any given immunoglobulin as defined by the EU index will not necessarily correspond to its sequential sequence.

(335) General and specific scientific techniques are outlined in US Publications 2015/0307629, 2014/0288275 and WO2014/145806, all of which are expressly incorporated by reference in their entirety and particularly for the techniques outlined therein.

Example 1: Generating Biologically Relevant Surrogates for PSMA-Expressing Tumor Cells

(336) To ensure cell lines with biologically valid PSMA antigen densities were used to inform the development of the novel anti-PSMA $\times$ anti-CD3 bispecific antibodies of the invention, IHC was conducted on paraffin embedded arrays of 160 biopsy cores of prostate cancer, 16 tumor adjacent cores, 16 normal prostate cores, and several cancer cell lines. Illustrative IHC of biopsy cores are depicted in FIG. 12, and samples were qualitatively scored in-house on a scale of 0-3 with 0 representing little to no PSMA expression and 3 representing high PSMA expression (herein referred to as IHC score; breakdown of score for each core is depicted in FIG. 13). Based on the results, it was determined that the novel bispecific antibodies of the invention should target cell lines having IHC scores of 3, 2, and 1 as they represent 97% of the patient segment and should not target score 0 cell lines as they resemble normal tissue. Upon matching the staining intensity between the sample types, cell lines were identified that could serve as surrogates of tumor and normal tissues. LNCaP cancer cells were found to stain as intensely as cancer tumors expressing high amounts of PSMA; 22Rv1 cancer cells were found to stain similar to some tumors expressing moderate amounts of PSMA as well as some normal prostate cores; Huh-7 cancer cells were found to express low levels of PSMA; and A549, ASPC-1, HT29, SKOV3, and PC3 cancer cells were found to express little to no PSMA.

(337) PC3 cell lines expressing varying PSMA antigen densities were also generated in order to

expand the selection of surrogate cell lines having varying antigen density levels to tune the antibodies of the invention. Cell-surface PSMA antigen density levels on the above cancer cell lines and the PSMA-transfected PC3 cell lines were estimated by FACS using fluorescently-labeled beads as advised by the QuickCal protocol (Bangs Laboratories, Inc., Fishers, IN). 50,000 cells per well and MESF beads were combined with A647-conjugated anti-PSMA mAb (J591) for 30 minutes at 4° C. Cells were then washed and fixed in 1% PFA. Flow cytometry was performed to determine antibody binding, and antigen density calculations were carried out using QuickCal® V.2.3. software (Bangs Laboratories, Inc., Fishers, IN). Data depicting PSMA density on the various cell lines are depicted in FIGS. **14-16**. PSMA-transfected PC3 cell lines are hereon referred to by their PSMA density (i.e. PC3 (~100 k) has a MESF score of 100,000).

Example 2: Antigen Binding Domains

(338) 2A: CD3 Binding Domains

(339) Sequences for CD3 binding domains having different CD3 binding affinities are depicted in FIG. **10**.

(340) 2B: PSMA Binding Domains

(341) The variable regions of a mouse anti-human PSMA binding domain were humanized using string content optimization (see, e.g., U.S. Pat. No. 7,657,380, issued Feb. 2, 2010). Sequences for the humanized PSMA binding domain, hereon referred to as PSMA-H, are depicted in FIG. **17**.

(342) 2B(a): Tuning PSMA-H Binding Affinity for PSMA

(343) Variants of PSMA-H were engineered by introducing point substitutions into the variable light region (VL). In a first round, 75 variant VLs were engineered designated as L1.1-L1.75, sequences for which are depicted in FIG. **18**. The variant VLs were paired with the wild-type humanized variable heavy region (VH) of PSMA-H and produced as bivalent mAbs (sequences for which are depicted in FIG. **19**). Binding affinity of the variants for human PSMA was screened using Octet, a BioLayer Interferometry (BLI)-based method. Experimental steps for Octet generally include the following: Immobilization (capture of ligand to a biosensor); Association (dipping of ligand-coated biosensors into wells containing the analyte); and Dissociation (returning of biosensors to well containing buffer). In particular, anti-human Fc sensors were used to capture the bivalent mAbs and dipped into human PSMA antigen. The resulting apparent dissociation constant (KDapp due to avidity concerns), association rate (ka), dissociation rate (kd), as well as sensorgram response are depicted in FIG. **20**.

(344) Based on the above, useful variant VLs were identified on the basis of change in binding affinity for PSMA, reversion of CDRs to human germline sequence (to reduce immunogenic potential), and introduction of negative charge (to aid in longer serum half-life by reducing non-specific clearance). In some instances, substitutions from suitable variant VLs were combined to generate additional VL variants (sequences for which are depicted in FIG. **18** as L1.76-L1.84). The variant VLs were paired with the wild-type humanized VH and produced as anti-PSMA×anti-CD3 bsAbs (sequences for which are depicted in FIG. **22**) and screened for binding affinity for human and cynomolgus PSMA using Octet as generally described above. The resulting dissociation constant (K.sub.D), association rate (k.sub.a), and dissociation rate (k.sub.d) are depicted in FIGS. **34-35**. A range of affinities from 6.55 nM to 116 nM were obtained. Notably, affinity for human and cynomolgus PSMA did not track across all the variants. For example, XENP33756 bound human PSMA with K.sub.D of 18 nM while binding cynomolgus PSMA with K.sub.D of 88 nM. For ease of clinical development, it is advantageous for the PSMA binding domain to bind human and cynomolgus PSMA with similar affinity.

(345) 2C: Engineering Anti-PSMA×Anti-CD3 Bispecific Antibodies (bsAbs)

(346) A number of formats for αPSMA×αCD3 bispecific antibodies (bsAbs) were conceived, illustrative formats for which are outlined below and in FIG. **21**.

(347) One such format is the 1+1 Fab-scFv-Fc format which comprises a single-chain Fv (“scFv”) covalently attached to a first heterodimeric Fc domain, a heavy chain variable region (VH)

covalently attached to a complementary second heterodimeric Fc domain, and a light chain (LC) transfected separately so that a Fab domain is formed with the variable heavy domain.

(348) Another format is the 2+1 Fab2-scFv-Fc format which comprises a VH domain covalently attached to a CH1 domain covalently attached to an scFv covalently attached to a first heterodimeric Fc domain (VH-CH1-scFv-Fc), a VH domain covalently attached to a complementary second heterodimeric Fc domain, and a LC transfected separately so that Fab domains are formed with the VH domains.

(349) DNA encoding chains of the α PSMA $\times\alpha$ CD3 bsAbs were generated by standard gene synthesis followed by isothermal cloning (Gibson assembly) or subcloning into a pTT5 expression vector containing fusion partners (e.g. domain linkers as depicted in FIG. 6 and/or backbones as depicted in FIGS. 7-9). DNA was transfected into HEK293E cells for expression. Sequences for illustrative α PSMA $\times\alpha$ CD3 bsAbs (based on CD3 binding domains as described in Example 2A and PSMA binding domains as described in Example 2B) in the 1+1 Fab-scFv-Fc format and in the 2+1 Fab2-scFv-Fc format are depicted respectively in FIGS. 22-32.

Example 3: Prototypic 1+1 Anti-PSMA $\times$ Anti-CD3 bsAbs Indiscriminately Kill Cell Lines Expressing Low Levels of PSMA

(350) To benchmark the activity of the novel anti-PSMA $\times$ anti-CD3 bsAbs of the invention, the activity prototypic 1+1 anti-PSMA $\times$ anti-CD3 bsAbs XENP14484, XENP34282, and XENP34283 (sequences for which are respectively depicted in FIGS. 22 and 33) were first investigated in redirected T-cell cytotoxicity (RTCC) assays.

(351) PC3 cell lines expressing various PSMA densities (as described in Example 1) were transduced to constitutively express luciferase. Luciferase released from dead cancer cells rapidly degrade in assay media, so live target cells can be quantified based on luminescence readout. The cancer cells were incubated with freshly enriched CD3.sup.+ T cells at an effector to target ratio of 1:1 for 24 hours. Next, the bispecific antibodies were added to the cells at the indicated concentrations. In a first experiment, 48 hours after addition of the bsAbs, Bio-Glo Luciferase reagent (Promega, Madison, WI.) was added and plates were read with the Envision Reader on luminescence setting. % RTCC was calculated by $1 - (\text{Raw Value}/\text{PBS AVG}) \times 100$. In a second experiment, 72 hours after addition of the bsAbs, cells were assayed via flow cytometry for Ki67 expression on T cells (as an indicator of proliferation).

(352) Data showing cell kill are depicted in FIG. 36, and data showing T cell proliferation are depicted in FIG. 37. The data show that the two prototypic 1+1 anti-PSMA $\times$ anti-CD3 induced RTCC on cell lines expressing high and low PSMA levels, including PC3 (~3K) which represent normal tissues. In fact, in an in vivo study (data not shown), cynomolgus monkeys treated with XENP14484 experienced dose-limiting toxicity (DLT; as indicated by high IL-6 release). Accordingly, activity on the PC3 (~3K) cell line may be a surrogate for DLT, and novel bispecific antibodies of the invention should be designed to avoid activity on PC3 (~3K) cells.

(353) In another experiment, CD107a T cell degranulation was investigated as an indicator of activity by the prototypic bispecific antibodies. PC3 cell lines expressing various PSMA densities (~100K vs ~50K) were incubated with freshly enriched CD3.sup.- T cells at an effector to target ratio of 10:1 for 24 hours. Next, the bispecific antibodies were added to the cells at the indicated concentrations. 18 hours after addition of the bsAbs, cells were assayed via flow cytometry for CD107a degranulation on T cells, data for which are shown in FIG. 46. The data show that each of the prototypic bispecific antibodies induced T cell degranulation with very similar potency in the presence of both cells having higher and lower PSMA densities.

Example 4: Tuning Anti-PSMA $\times$ Anti-CD3 bsAbs to Enhance Selectivity and Therapeutic Index

(354) 4A: Tuning PSMA Binding Valency and Binding Affinity

(355) In order to encourage avid binding and strong activity on high PSMA expressing cells (e.g. tumors) while minimizing reactivity on low expressing cells (e.g. normal tissues), the anti-PSMA $\times$ anti-CD3 bsAbs were tuned for PSMA binding valency and affinity as well as CD3 binding

affinity. Towards this, anti-PSMA×anti-CD3 bsAbs were engineered in the 2+1 Fab.sub.2-scFv-Fc format with a range of PSMA binding affinities and reduced CD3 binding affinities and the following illustrative bsAbs were investigated in cell binding and RTCC assays: XENP31855 (having 1 nM K.sub.D PSMA-H_H1L1 and CD3 High-Int #1[VLVH]), XENP32218 (having 7 nM PSMA-H_H1_L1.58 and CD3 High-Int #1[VLVH]), XENP32220 (having 38 nM PSMA-H_H1_L1.24 and CD3 High-Int #1[VLVH]), and XENP32224 (having 83 nM PSMA-H_H1_L1.29 and CD3 High-Int #1[VLVH]).

(356) First, the effect of reduced monovalent PSMA binding affinity on cell binding was investigated. PSMA-transfected PC3 (~32K) cancer cells were treated with the indicated concentrations of the indicated test articles. Binding was detected using anti-human Fc mAb, data for which are shown in FIG. 38. The data show that in comparison the monovalent 1+1 Fab-scFv-Fc format, the bivalent 2+1 Fab2-scFv-Fc format retains cell binding despite reduced monovalent PSMA binding affinity due to avidity.

(357) Next, cell lines expressing various PSMA densities were transduced to constitutively express luciferase. The cells were incubated with freshly enriched CD3.sup.+ T cells at an effector to target ratio of 1:1 for 24 hours. Next, the bispecific antibodies were added to the cells at the indicated concentrations. 18 hours after addition of the bsAbs, cells were assayed via flow cytometry for CD107a degranulation on T cells (as an indicator of T cell activity). 48 hours after addition of the bsAbs, Bio-Glo Luciferase reagent (Promega, Madison, WI.) was added and plates were read with the Envision Reader on luminescence setting. % RTCC was calculated by $1 - (\text{Raw Value}/\text{PBS AVG}) \times 100$. 72 hours after addition of the bsAbs, cells were assayed via flow cytometry for Ki67 expression on T cells (as an indicator of proliferation). Data showing cell kill are depicted in FIG. 39, data showing T cell proliferation are depicted in FIG. 40, and data showing CD107a degranulation are depicted in FIG. 41. Collectively, the data show that as PSMA binding affinity is reduced, selectivity for higher antigen density cell lines over lower antigen density cell lines is improved. Notably as shown in FIGS. 42 and 39, with low PSMA binding affinity (e.g. 38 nM and 83 nM) it is possible to achieve efficacious killing of cell lines exhibiting IHC Score 2 and Score 1 while inducing little to no killing of cell line exhibiting IHC Score 0.

(358) 4B: Further Tuning CD3 Binding Affinity

(359) Another approach explored for optimizing the anti-PSMA×anti-CD3 bispecific antibodies was tuning CD3 binding affinity. According, anti-PSMA×anti-CD3 bsAbs in the 2+1 Fab.sub.2-scFv-Fc format having CD3 High-Int #2[VLVH] binding domain and either 1 nM KD or 7 nM KD PSMA binding domains, respectively XENP31856 and XENP33063, were investigated in assays as generally described above. Data as depicted in FIGS. 43 and 44 show that reducing the CD3 binding affinity may also convey selective targeting to high PSMA expressing cell lines (even with higher affinity PSMA binding).

(360) 4C: Tuned Anti-PSMA×Anti-CD3 Bispecific Antibodies are Able to Kill Clinically Relevant PSMA+ Prostate Cancer Cells

(361) To confirm that the tuned anti-PSMA×anti-CD3 bispecific antibodies of the invention are able to kill clinically relevant PSMA.sup.+ prostate cancer cells, their ability to induce RTCC on LNCaP (human prostate adenocarcinoma cells having IHC score 3 and ~140K PSMA density) and 22Rv1 (human prostate carcinoma cells having IHC score 2 and ~115K PSMA density).

(362) Cancer cells were incubated with freshly enriched CD3.sup.+ T cells at an effector to target ratio of 10:1 for 24 hours. Next, the bispecific antibodies were added to the cells at the indicated concentrations. 72 hours after addition of the bsAbs, Bio-Glo Luciferase reagent (Promega, Madison, WI.) was added and plates were read with the Envision Reader on luminescence setting. % RTCC was calculated by $1 - (\text{Raw Value}/\text{PBS AVG}) \times 100$. Data depicted in FIG. 45 show that tuned anti-PSMA×anti-CD3 bispecific mAb XENP32220 was able to induce cell kill on both LNCaP and 22Rv1 cancer cells.

Example 5: Tuned PSMA×CD3 Bispecific Antibodies are Active In Vivo

(363) Next, the in vivo anti-tumor effect of the tune anti-PSMA×anti-CD3 bispecific antibodies of the invention was investigated. NOD-SCID gamma (NSG) mice were engrafted intradermally with 1×10^6 PC3 (~100K) cells in the right flank on Day -16. On Day -1, mice were engrafted intraperitoneally with 5×10^6 human PBMCs. Mice (n=10) were then treated on Days 0, 8, 15, and 22 with 3 mg/kg XENP32218, XENP32220, or XENP32224. Controls (N=10) used were PBS and 3 mg/kg anti-PD-1 mAb (a checkpoint inhibitor which enhances anti-tumor activity by de-repressing the engrafted human T cells). Tumor volumes were monitored by caliper measurements, data for which are shown in FIG. 47 for Day 19 and change over time in FIG. 48. The data show that each of the tuned PSMA×CD3 bispecific antibodies significantly enhanced ($p < 0.05$ vs. PBS or α PD-1 mAb) anti-tumor activity (as indicated by tumor volume; statistics performed on baseline corrected data using unpaired t-test) despite their reduced PSMA binding affinity.

Example 6: Tuned PSMA×CD3 bsAbs Demonstrate Favorable Tolerability and Pharmacokinetics in Cynomolgus

(364) In a cynomolgus study, each healthy male cynomolgus (n=1) was administered by IV either a 1× dose, 10× dose, or 60× dose of tuned PSMA×CD3 bsAbs XENP32218, XENP32220, or XENP32224, or the analogs of these three molecules that additionally contain the Xtend mutation (M428L/N434S for enhanced serum half-life) in the Fc domain-XENP34626, XENP34627, or XENP34628, respectively. All XENPs were generally well tolerated (i.e., no dose limiting side effects) up to the highest 60× dose (data not shown). As seen in FIG. 49, the variants with the Xtend mutations resulted in improved pharmacokinetics, particularly at the 1× and 10× doses. For example, the terminal serum half-lives of XENP32220 were 1.59, 3.01, or 7.95 days at each relative dose level, while its Xtend analog XENP34627 had serum half-lives of 4.5, 8.6, and 10.8 days across each dose level. Additionally, the lower PSMA affinity XENP32224 had half-lives of 2.53 or 3.34 days at the 1× and 10× doses, while its Xtend analog XENP34628 had half-lives of 5.9 or 9.3 days at the 1× and 10× dose levels. These half life measurements of the Xtend analog were a significant improvement over half life data for a comparator PSMA×CD3 bsAb molecule at a comparable dose (as reported in literature).

(365) At the lowest 1× dose level, serum clearance was PSMA affinity-dependent. XENP32218, having the highest PSMA binding affinity, demonstrated the fastest clearance, and XENP32224, having the lowest PSMA binding affinity, demonstrated the slowest clearance. Notably, the 60× dose was high enough to clear the sink effect so that half lives at that dose were no longer affinity-dependent.

Claims

1. A composition comprising a Prostate Specific Membrane Antigen (PSMA) binding domain comprising: a) a variable heavy domain comprising the variable heavy complementarity determining regions vhCDR1, vhCDR2, and vhCDR3 of a variable heavy domain having an amino acid sequence of SEQ ID NO: 217; and b) a variable light domain comprising the variable light complementarity determining regions vlCDR1, vlCDR2, and vlCDR3 of a variable light domain having an amino acid sequence of one of the following: SEQ ID NO: 252, SEQ ID NO: 257, and SEQ ID NO: 286.
2. A composition comprising a Prostate Specific Membrane Antigen (PSMA) binding domain comprising: a) a variable heavy domain having an amino acid sequence of SEQ ID NO: 217; and b) a variable light domain having an amino acid sequence of one of the following: SEQ ID NO: 252, SEQ ID NO: 257, and SEQ ID NO: 286.
3. A nucleic acid composition comprising: a) a first nucleic acid encoding the variable heavy domain according claim 2; and b) a second nucleic acid encoding the variable light domain according to claim 2.

4. An expression vector composition comprising: a) a first expression vector comprising the first nucleic acid of claim 3; and b) a second expression vector comprising the second nucleic acid of claim 3.
 5. A host cell comprising the expression vector composition according to claim 4.
 6. A method of making a Prostate Specific Membrane Antigen (PSMA) binding domain comprising culturing the host cell according to claim 5 under conditions wherein the PSMA binding domain is expressed, and recovering the PSMA binding domain.
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